

Curriculum Vitae

Name: Samia J. Khoury, MD, FAAN, FRCP
Office Address: Nehme and Therese Tohme Multiple Sclerosis Center
AUBMC
Work Phone: +961-1-350000 ext 7420
Work E-Mail: sk88@aub.edu.lb
Work FAX: +961-1-366143
ORCID: 0000-0003-3198-6063
Scopus Author ID: 7102393760

Place of Birth: Beirut, Lebanon

Education

1980	B.S. with honors	Biology/Chemistry	American University of Beirut, Lebanon
1984	M.D.	Medicine	American University of Beirut, Lebanon

Postdoctoral Training

1984-1985	Intern	Internal Medicine	Cleveland Metropolitan General Hospital, Cleveland, Ohio
1985-1988	Resident	Neurology	Case Western Reserve University Hospitals, Cleveland Ohio
1988-1991	Research Fellow	Medicine	Harvard Medical School, Boston, Massachusetts

Certificates

1989 American Board of Psychiatry and Neurology (Currently valid)

Faculty Academic Appointments

1991-1994	Instructor in Medicine	Harvard Medical School
1994-1998	Assistant Professor of Neurology	Harvard Medical School
1998-2007	Associate Professor of Neurology	Harvard Medical School
2007-2009	Professor of Neurology	Harvard Medical School
2009-2011	Visiting Professor of Neurology	American University of Beirut
2009-2013	Jack, Sadie, and David Breakstone Professor of Neurology	Harvard Medical School

2011-	Professor of Neurology and Immunology	American University of Beirut
2011-	Director, Multiple Sclerosis Center	American University of Beirut
2011-	Director, Abu Haidar Neuroscience Institute	American University of Beirut
2012-2021	Associate Dean for Clinical and Transl. research	American University of Beirut
2012-	Member, Academic Committee	American University of Beirut
2012-2021	Chair, Research Committee	American University of Beirut
2013-2015	Visiting Professor of Neurology	Harvard Medical School
2015-2024	Senior Lecturer	Harvard Medical School
2017	Tenured Professor of Neurology and Immunology	American University of Beirut

Appointments at Hospitals/Affiliated Institutions

- Past*

1991-2000	Attending	Inpatient and Consult Neurology Service	West Roxbury VA Medical Center
1993-1994	Attending	Outpatient Resident Neurology Clinic	West Roxbury VA Medical Center
1989-2010	Attending	Outpatient Neurology Clinic	Brigham and Women's Hospital
1991-2011	Associate Physician of Neurology		Department of Medicine Brigham and Women's Hospital
1998-2011	Director	Clinical Immunology Laboratory Department of Neurology	Center for Neurologic Diseases Brigham and Women's Hospital
2001-2011	Co-Director	Partners Multiple Sclerosis Center	Partners Healthcare Systems, Inc., Brigham and Women's Hospital
2007-2011	Senior Scientist	Clinical Immunology Laboratory Division of Multiple Sclerosis	Department of Neurology Brigham and Women's Hospital

Other Professional Positions

1995-1997	Physician Advisory Board	Biogen Idec Pharmaceuticals
2000-2004	Scientific Program Subcommittee, Chair	American Association of Neurologists
2002	Scientific Advisory Board	Biogen Idec Pharmaceuticals
2004	Tolerance Assay Group	Immune Tolerance Network
2004	Scientific Advisory Board	TcLand Expression
2007	Consultant	Daiichi-Suntory
2007	Consultant	Wyeth Pharmaceuticals
2008	Scientific Advisory Board	EpiVax
2011	Scientific Advisory Board	Novartis
2011	Consultant	Kirin
2023-	Chair, Independent Data Monitoring Committee	Biogen

Major Administrative Leadership Positions

- *Local Positions*

1998-2011	Director, Clinical Immunology Laboratory	Center for Neurologic Diseases Brigham and Women's Hospital
1999-2008	Organizer, Annual Harvard Autoimmunity Symposium	Harvard Medical School Brigham and Women's Hospital Center for Neurologic Diseases
2001-2011	Co-Director, Partners Multiple Sclerosis Center	Brigham and Women's hospital Partners HealthCare Systems, Inc.
2004-2011	Member, Finance Committee	Department of Neurology Brigham and Women's Hospital
2009-2011	Member, Research Advisory Council	Center for Women's Health and Gender Biology, Division of Women's Health Brigham and Women's Hospital
2011-	Director, Abu Haidar Neuroscience Institute	American University of Beirut
2011-	Director, Multiple Sclerosis Center	American University of Beirut
2012-2021	Associate Dean for Clinical and Transl. research	American University of Beirut

- *National Positions*

1999-2001	Organizer, Beyond the Basics: MS Insights	National Multiple Sclerosis Society
2000-2004	Chair, MS and Related Disease Topics Scientific Program Subcommittee	American Association of Neurologist
2006	Organizer, Scientific Highlights in Multiple Sclerosis	American Academy of Neurology
2007-2013	Science Committee	American Academy of Neurology
2007-2010	Neuroscience Research Prize Subcommittee	American Academy of Neurology
2014-2023	MS Expert Review Advisory Panel	American Academy of Neurology

- *International Positions*

1998-	Lecturer, Various topics in Neurology and MS	American University Hospital of Beirut
2009	Visiting Professor in Neurology	American University Hospital of Beirut
2011	Symposium Organizer "Update on MS"	American University Hospital of Beirut
2011-2012	Chair, Scientific Organizing Committee	International Society of Neuroimmunology
2015	Member, Evaluation Committee	Kuwait Prize-Basic Sciences

Committee Service

- *Local*

1999-2002	Scientific Review Committee for the GCRC	Brigham and Women's Hospital
2001-2011	Clinical Trials Review Committee Chair Partners Multiple Sclerosis Center	Partners HealthCare Systems, Inc.

2002-2007	Physician Organization Board of Trustees	Brigham and Women's Hospital
1995-2010	Board Member	Affirmative Action Representative for Faculty Recruitment, Department of Neurology, Brigham and Women's Hospital
2005-2011	Neurology Department Finance Committee	Brigham and Women's Hospital
2007-2011	Connors Center Research Advisory Committee	Brigham and Women's Hospital
2007-2011	Stem Cell, Regenerative Medicine and Tissue Engineering Center Working Group	Brigham and Women's Hospital
2008-2009	Neurosurgery Chair Search Committee	BI-Deaconess Medical Center
2010-2012	Neurosurgery Chair Search Committee	BI-Deaconess Medical Center
2009-2011	Promotions, Reappointments and Appointments Committee	Harvard Medical School
2009	ad hoc Committee for Professorship Promotion	Harvard Medical School
2010	Neurology Chair Search Committee	Massachusetts General Hospital
2010	ad hoc Committee for Professorship Promotion	Harvard Medical School
2011	ad hoc Committee for Professorship Promotion	Harvard Medical School
2011	Co-Chair, Search committee for the Naef Basile Cancer Center Director	American University of Beirut
2011-2015	Member, Tenure Task Force	American University of Beirut
2012	Co-Chair, IRB	American University of Beirut
2012	Chair, Spine Program Task Force	American University of Beirut
2012-2021	Chair, Research Committee	American University of Beirut
2012-2021	Chair, Farouk Jabr Grant Review Committee	American University of Beirut
2012-	Member, Performance Improvement Committee	American University of Beirut
2012-2013	Chair, Strategic Plan for Research Task Force	American University of Beirut
2012	Member, Scientific Committee of MEMA 2013	American University of Beirut
2013-	Academic Committee	American University of Beirut
2014-2020	Graduate Studies Committee	American University of Beirut
2014-	AUBMC-MUSC fellowship Committee	American University of Beirut
2014	Member, Task force on Professionalism	American University of Beirut
2014-2015	Organizing Committee, Internat. Science Conference Advancement of Science	Lebanese Association for the
2016-2018	Co-Chair, Collaborative Research Stimulus	American University of Beirut
2016-	Chair, Tissue Bank Committee	American University of Beirut
2016-	Faculty Mentoring Committee	American University of Beirut
2016-	MUSC MD/PhD applicant committee	American University of Beirut
2017-	Ambulatory Services Committee	American University of Beirut

2017-	Member of MS Thesis Committee (for JM)	Hariri School of Nursing American University of Beirut
2018-	Lebanese Medical Training Leadership	American University of Beirut
2018-2019	Chair, Promotion Review Committee	American University of Beirut
2018-	Medical Ethics committee	American University of Beirut
2019	Chair, Task Force on Faculty Mentorship	American University of Beirut
2020-2021	Chair, Promotion Review Committee for HZ	American University of Beirut
2020-2021	Member, Promotion Review Committee for NZ	American University of Beirut
2021-2022	Chair, Tenure Review Committee for ER	American University of Beirut
2022	Member, PhD thesis Committee for LA	University of Paris – Sorbonne
2022-2024	Chair, PhD thesis Committee for MD	American University of Beirut
2023-2024	Tenure and Promotion Appeals Committee	American University of Beirut
2023-2024	Chair, Promotion Review Committee for MM	American University of Beirut
2023-2024	Chair, Promotion Review Committee for MA	American University of Beirut
2024-	University Tenure and Promotion Committee	American University of Beirut
2024-2025	Chair, search Committee for Biochem and molecular Genetics Department chair	American University of Beirut
2025-	Member of PhD Thesis Committee	Hariri School of Nursing, AUB
2025	Chair, Post-tenure Review Committee for GA	American University of Beirut
2025	Chair, Post-tenure Review Committee for GD	American University of Beirut

- *National*

1999	NIAID Ad Hoc Council	National Institute of Health
1999-2004	NMSS Scientific Peer Review Committee	National Multiple Sclerosis Society
1999-2005	Autoimmunity Centers of Excellence	NIAID
2000-2004	American Academy of Neurology Scientific Program Chair, Subcommittee, MS and Related Disease Topics	American Academy of Neurology
2006-2013	American Academy of Neurology Science Committee	American Academy of Neurology
2007-2012	American Academy of Neurology Neuroscience Award subcommittee	American Academy of Neurology
2008-2012	Clinical Immunology Society Council	Clinical Immunology Society
2009-2012	Chair, Membership Committee	Clinical Immunology Society

2013	Member, Oversight Committee	Consortium on progenitor cells for the treatment of multiple sclerosis from the State University of New York
2014-2023	Member, Expert Review Advisory Panel	American Academy of Neurology
• <i>International</i>		
2009-	Neuroscience Task-force Committee	American University of Beirut
2009-2012	ISNI 2012 meeting Organizing Committee Chair, ISNI 2012 Meeting	International Society of Neuroimmunol ISNI
2014-2018	International Advisory Board	International Society of Neuroimmunol ISNI
2024-	Chair of IDMC for 257MS201trial	Biogen
2024-	Steering committee for an international Delphi consensus project for management of secondary hypogammaglobulinemia due to B-cell therapies in autoimmune diseases.	

Professional Societies

1986-	American Medical Association
1986-	Ohio State Medical Society
1988-	Massachusetts Medical Society
1989-	American Academy of Neurology
1996-	American Association of Immunologists
2001	Federation of Clinical Immunology Societies
2002-	American Society of Clinical Investigation
2002-	American Neurological Association
2002-	American Association for the Advancement of Science
2002-	International Society of Neuroimmunology
2006-	Clinical Immunology Society
2013	UK Royal College of Physicians-Fellow
2013	American Academy of Neurology-Fellow
2013	American Neurological Association-Fellow

Grant Review Activities

1994, 2000	Ad Hoc Review Committee	NINDS
1996-	Abstract Review Committee	American Academy of Neurology
1996-1999	Ad Hoc Reviewer	
2004-2008	Permanent member	
1998	Review Committee for Seed grants	Brigham and Women's Hospital
1998-2004	Scientific Peer Review Committee	National Multiple Sclerosis Society

1998-	Pilot grants Ad Hoc reviewer	National Multiple Sclerosis Society
2001	Ad Hoc Committee: RFA for the Study of Gender and Autoimmunity	National Institutes of Health
2002	Ad Hoc Committee: RFA for the Statistical and Clinical Coordination Center-Autoimmune Disease Clinical Trials	National Institutes of Health
2002	Hertie Foundation Competition for Molecular Neuroimmunology in Multiple Sclerosis Grants	Hertie Foundation, Germany
2002	Expert Panel--Coordinating Committee Autoimmune Disease Research Plan, Ad Hoc Member	National Institutes of Health
2003	VA Pilot Grants for Multiple Sclerosis	Veterans Administration
2004	BCDN4 Study Section, Ad Hoc Reviewer	National Institutes of Health
2004-2008	CNBT Study Section, Permanent member	National Institutes of Health
2004-2008	Tolerance Assay Group, Permanent member	Immune Tolerance Network
2010	Ad Hoc reviewer	Hilton Foundation
2008, 2010	Ad Hoc reviewer	National MS Society of Canada
2012, '13, '17	Ad hoc Reviewer	Conseil National de la Recherche Scientifique, Lebanon
2012, 2013	Ad Hoc Review Committee	NINDS
2013, 2014, 2015	Ad Hoc Reviewer	Hertie Foundation, Germany
2013	Reviewer	Wellcome Trust, UK
2015	Ad Hoc reviewer: Transformative Research Award	National Institutes of Health
2015	Board of Scientific Counselors	NINDS
2016-2018	Reviewer	Award for Excellence in Research Hamad Medical Corp. Doha, Qatar
2023	Reviewer	Albrecht Stuppler fellowship s Technical University of Munich
2023	Reviewer	Promotion file for Professor King Saud University
2023	Reviewer	European Science Foundation

Editorial Activities

- *Ad hoc Reviewer*
 - Neurology
 - Journal of Experimental Medicine

- Journal of Immunology
- Journal of Neuroimmunology
- Journal Watch Neurology
- Journal of Clinical Investigation
- Immunity
- Blood
- Cellular Immunology
- Annals of Neurology
- Brain
- Circulation
- Nature Medicine
- New England Journal of Medicine
- Lebanese Medical Journal
- Journal of Neuroscience
- Proceedings of the National Academy of Sciences
- Nature Medicine
- MS Journal
- American Journal of Pathology
- Journal of Neuroscience

• *Other Editorial Roles*

2000-2004	Associate Editor	Journal of Immunology
2004	Editorial Board	Current Immunology Reviews
2005-2023	Section Editor	Clinical Immunology
2010-2018	Editorial Board	Multiple Sclerosis Journal
2013-2023	Associate Editor	Current Treatment Options in Neurology
2016-2021	Associate Editor	Annals of Neurology
2022-	Associate Editor	Annals of Clinical and Translational Neurology
2023-	Editor In Chief	Clinical Immunology

Honors and Prizes

1977-1984	Dean's Honor List	American University of Beirut, Lebanon
1980	B.S. Degree with Distinction	American University of Beirut, Lebanon
1984	Alpha Omega Alpha	The Alpha Omega Alpha Honor Medical Society
2002	Elected member	American Society of Clinical Investigation

2002	Elected member	American Neurological Association
2007	Kuwait Prize for Immunology	Kuwait Foundation for the Advancement of Sciences
2009-2012	Jack, Sadie and David Breakstone Chair	Harvard Medical School
2011	Elected Fellow	American Academy of Neurology
2011	Gold Award	Medical Alumni of the American Univ of Beirut
2012	Excellence in Science Award	Lebanese Society for the Advancement of Science
2013	Elected Fellow	Royal College of Physician-England
2013	Elected Fellow	American Neurological Association
2017	Research Excellence Award	Conseil National pour la Recherche Scientifique
2017	Innovative Doctors	Saudi Arabia Embassy
2019	Elected Member	Lebanese Academy of Science
2022	Board Designated Professorship	American University of Beirut, Lebanon
2022	King Salman International Award For disability research	Riyadh- KSA
2023	Mustafa Prize Life and Medical Science and Technology	Isfahan-Iran
2025	Dean's Excellence Award for Research	American University of Beirut, Lebanon

Report of Funded and Unfunded Projects

Funding Information

- *Current*

2024-2026	PI	MENACTRIMS	\$25,000
Development of a Personalized Prognostic Algorithm for Stratifying Multiple Sclerosis Severity using OCT			
2024-2025	Co-PI	MENACTRIMS	\$25,000
Iron Deposition, Microstructural Integrity, and their Implications in Relapsing-Remitting Multiple Sclerosis			
2024-2025	Co-PI	AUB	\$25,000
Predicting Multiple Sclerosis Disease Activity Using Large Language Models			

- *Past*

1989-1991	National Multiple Sclerosis Society Fellowship Grant.		
1994-1996	Co-Investigator	Pharmacia	
<i>Immunologic monitoring of MS patients in the Linomide study</i>			
1995-1996	Co-Investigator	NDCWW	
<i>Multicenter investigation of IFN-gamma production in MS</i>			

1992-1997	Co-Investigator	National Institutes of Health
<i>Induction of transplantation tolerance by the oral route</i>		
1994-1997	Principal Investigator	National Multiple Sclerosis Society
<i>Acquired thymic tolerance in EAE</i>		
1996-1998	PI	Autoimmune
<i>Immunoreactivity to Myloral in relapsing-remitting MS patients</i>		
1998-1999	PI	National Multiple Sclerosis Society
<i>Immunologic Effects of Albuterol therapy in MS</i>		
1996-2001	PI	National Institutes of Health
<i>Immunoregulation in EAE: Role of T Cell Costimulation</i>		
1997-2000	PI	National Multiple Sclerosis Society
<i>Mechanisms of acquired thymic tolerance in EAE</i>		
1998-2001	PI	National Multiple Sclerosis Society
<i>Synergy and Differential Effects of CD40L/CD40 And CD28/B7 Blockade In EAE</i>		
1997-2002	Co-Investigator	National Institutes of Health
<i>Costimulation and Cytokines in Tolerance</i>		
1999-2003	PI	National Institutes of Health
<i>Autoimmunity: Treatment by Costimulatory Signal Blockade</i>		
1999-2004	PI	National Institutes of Health
<i>Costimulatory regulation of Th1/Th2 cytokines in EAE</i>		
2000-2005	Co-Investigator	National Institutes of Health
<i>Investigation of Immunologic Markers as Surrogates of Disease</i>		
The objectives of the work to be performed for this contract are: 1) to determine whether there is a link between immune abnormalities in the peripheral blood of MS patients and the level of disease activity as measured clinically and by MRI imaging; 2) to determine whether there is a link between immune measures and therapeutic efficacy of currently used drugs for MS; 3) to determine whether immune measures are linked to the change of patients from the relapsing remitting to the progressive form of the disease; and 4) to determine whether immune measures are predictive of the future course of the MS patients as measured both clinically and by MRI.		
2002-2005	PI	National Multiple Sclerosis Society
<i>Mechanisms of Immune Regulation in Chronic EAE</i>		
The overall goal of this grant is to dissect the mechanisms of immune regulation of chronic experimental autoimmune encephalomyelitis (EAE), the animal model of multiple sclerosis (MS).		
2003-2005	Co-Investigator	National Institutes of Health
<i>Denver Autoimmune Center of Excellence Rapamune Trial</i>		
The main goal of this project is to determine the safety and tolerability of Rapamune oral therapy in subjects with Multiple Sclerosis who have failed standard INF treatment with interferons and glatiramer acetate by clinical and magnetic resonance imaging (MRI) measure.		

- 2001-2006 Co-Investigator National Institutes of Health
U01 DK 61926 (Hafler)
Role of Regulatory CD4+CD25+T Cells in Diabetes
The main goal of this proposal is to investigate the mechanisms of costimulatory signal blockade by CD4+CD25hi T cells therapy in multiple sclerosis and diabetes.
- 2002-2006 Co-Investigator National Institutes of Health
R01
Immunoregulatory T cells in multiple sclerosis
- 2003-2006 Co-Investigator NIH/University of Colorado
U19 AI046374 (Holers)
Treatment of MS with Copaxone and Albuterol
The main goal of this project is to investigate whether Copaxone and Albuterol treatment induces more prominent changes in immune and MS responses as compared to Copaxone and placebo.
- 2002-2007 Co-Investigator NIH/U. of California San Francisco
N01 AI15416 (Bluestone)
A Phase I Study to Assess the Safety of One Dose of RG2077 (CTLA4-IgG4m) in Patients with Relapsing-Remitting Multiple Sclerosis
The main goal of this project is to evaluate the safety of a single IV infusion of RG2077 in patients with relapsing-remitting multiple sclerosis.
- 2003-2007 PI National Institutes of Health
R01 AI058680
Role and Mechanism of PD1-PDL Pathway in EAE
The main goal of this project is to define the functions and mechanisms of the PD1 pathway in regulating EAE.
- 2003-2007 PI National Multiple Sclerosis Society
RG 3504-A-8
Role of PD1-PDL Pathway in EAE
The main goal of this project is to define the functions and mechanisms of the PD1 pathway in regulating EAE.
- 1985-2008 Co-Investigator National Institutes of Health /NINDS
R01 NS23132
Immunoregulatory T cells in multiple sclerosis
The goals of this project is to analyze immune dysregulation in MS as measured by chemokine receptor expression both in the peripheral blood and the central nervous system; investigate the role of IL-12 and IL-18 and dendritic cells in creating Th1 type immunologic milieu in MS; Investigate immunoregulatory defects in MS related to regulation by IL-10 and TGF-beta and to CD4+CD25+ regulatory cells; determine the degree to which immune dysregulation and the abnormalities we have identified are linked to clinical disease activity and whether the dysregulation is altered by immunomodulatory therapies.
- 1999-2008 PI National Institutes of Health
R01 AI043496
Immune Regulation of Neural Stem Cell Program in EAE
The main goal for this project is to find out what the effects of acute inflammation on the dynamics and multi-potentiality of endogenous neural stem and progenitor cells in vivo.
- 1999-2008 PI National Institutes of Health
U19 AI46130
Treatment of Autoimmune Disease by Costimulatory Signal Blockade
The main goal of this proposal is to investigate the clinical efficacy and mechanisms of costimulatory signal blockade in 3 autoimmune diseases: multiple sclerosis, diabetes, and psoriasis.

2003-2008	Co-Investigator	Millennium Pharmaceuticals, Inc.	
<i>Millennium Registry Collaboration, Multiple Sclerosis Registry</i>			
The aim is to establish a registry of patients diagnosed with multiple sclerosis (MS). The long-term goals of the registry are to determine and validate biomarkers that may correlate with the onset, progression, and severity of MS as well as to determine and validate biomarkers that may correlate with patients' responses to drug therapy. Nine hundred and twenty patients with MS or monosymptomatic disease will be enrolled in the study over a five-year period, as well as one hundred healthy controls. Patients will be followed by serial examinations, MRI and cytokine gene expression profiling in addition to functional immunologic analyses. These biomarkers may serve as future therapeutic targets in MS.			
2006-2008	Co-Investigator	NIH/NIAID/Columbia University	
U19 AI056362 (Diamond)			
<i>Treatment of Multiple Sclerosis with Copaxone and Albuterol</i>			
The main goal of this project is to investigate whether Copaxone and Albuterol treatment induces more prominent changes in immune and MS responses as compared to Copaxone and placebo. This is a continuing support from U19 AI046374.			
2005-2008	PI	National Multiple Sclerosis Society	
RG 3666 A9			
<i>Memory T Cells in EAE</i>			
The major goal of this project is to investigate the role of memory T cells in EAE, and to investigate their susceptibility to tolerance strategies in vivo.			
2004 – 2009	Co-Investigator	National Institute of Health	\$100,000
U19 AI056362			
<i>Autoimmunity Centers of Excellence</i>			
The primary objective of this project is to determine the safety and tolerability of Rapamune oral therapy in subjects with multiple sclerosis who have failed standard IFN- β or Copaxone treatment.			
2008 – 2009	Project PI	National Institutes of Health	Core A: \$82,620
U19 AI056362			
Project 1: \$145,535			
<i>Treatment of Autoimmune Disease by Costimulatory Signal Blockade</i>			
The main goal of this proposal is to investigate the clinical efficacy and mechanisms of costimulatory signal blockade in 3 autoimmune diseases: multiple sclerosis, diabetes, and psoriasis. This is a cost extension.			
2007 – 2009	PI	Fidelity Foundation	\$150,000
<i>Microglia and the Stem Cell Niche</i>			
The goal of this project is to study the interaction of microglia with stem cells.			
2007 – 2010	PI	National Multiple Sclerosis Society	\$346,080
RG 3945 A10			
<i>Impact of IFN-gamma on Neural Stem Cell Repair Potential in EAE</i>			
The goals of this project are to examine the effect of IFN- γ and the transcription factor stat1 on the self-renewal program of NSCs in vivo and on their molecular program in vitro; to examine the effect of IFN- γ on migration of NSCs in vivo and on the molecular program in vitro; and to examine the effect of stat1 on differentiation of NSCs in vivo and on the differentiation molecular program in vitro focusing on neurogenic differentiation.			
2008 – 2010	PI	National Institutes of Health	\$500,000
R56 AI058680			
<i>Role and Mechanism of Negative Costimulatory Pathways in EAE</i>			
The main goal of this project is to define the functions and mechanisms of the PD1 pathway in regulating EAE.			
2009 – 2010	PI	BIOGEN Idec	\$89,201
<i>T cell and dendritic cell function during natalizumab treatment in MS</i>			

The goal of this project is to determine effects of natalizumab treatment on the innate and adaptive immune system in patients with MS.

2007 – 2011	PI	National Institutes of Health R01 AI071448	\$1,000,000
<i>Impact of IFN-gamma on Neural Stem Cell Repair Potential in EAE</i>			
The goals of this project are to examine the effect of IFN-g and the transcription factor stat1 on the self-renewal program of NSCs in vivo and on their molecular program in vitro; to examine the effect of IFN-g on migration of NSCs in vivo and on the molecular program in vitro; and to examine the effect of stat1 on differentiation of NSCs in vivo and on the differentiation molecular program in vitro focusing on neurogenic differentiation.			
2010 – 2011	PI	BIOGEN Idec	\$129,000
<i>T cell and dendritic cell function during natalizumab treatment in MS</i>			
The goal of this project is to determine effects of long-term natalizumab treatment on the innate and adaptive immune system in patients with MS.			
2009 – 2011	PI	National Institutes of Health R56 AI058680-05	\$402,588
<i>Role and Mechanism of Negative Costimulatory Pathways in EAE</i>			
The main goal of this proposal is to explore the immune regulatory mechanisms of PDL1/PDL2 and the B7H3 and B7H4 in regulating EAE. We have unique reagents (monoclonal antibodies and fusion proteins) and animal models (gene knockout and TCR transgenic animals) that will enable us to dissect the role of this pathway in a clinically relevant disease model.			
2010-2012	Co-PI	AUBMC MPP	\$80,000
<i>Inflammatory pattern, vitamin D level and cognitive performance in adults and older adults.</i>			
The main goal of this proposal is to investigate the correlation between serum levels of vitamin D and cytokines with cognitive performance in adults (> 30 years) and older adults (> 60 years).			
2008 – 2013	PI	National Institutes of Health R01 AI067472	\$1,250,000
<i>Memory T Cells in EAE</i>			
The goal of this project is to understand the immunopathogenetic mechanisms of autoimmune disease, and provide the rationale to develop novel strategies to prevent and/or interrupt progression of disease.			
2009 – 2014	Co-Investigator	EMD Serono, Inc. A203325.03 (Weiner)	\$5,412,998
<i>CLIMB Study</i>			
The purpose of this study is to identify clinical and MRI features in MS patients followed longitudinally that characterize disease subtypes, prognosis and response to therapy.			
2008 – 2013	PI	National Institutes of Health R01 AI067472	\$1,250,000
<i>Memory T Cells in EAE</i>			
The goal of this project is to understand the immunopathogenetic mechanisms of autoimmune disease, and provide the rationale to develop novel strategies to prevent and/or interrupt progression of disease.			
2009 – 2014	Co-Investigator	EMD Serono, Inc. A203325.03 (Weiner)	\$5,412,998
<i>CLIMB Study</i>			
The purpose of this study is to identify clinical and MRI features in MS patients followed longitudinally that characterize disease subtypes, prognosis and response to therapy.			
2012 – 2013	PI	National Institutes of Health R56 AI093838	\$247,607

Neural Stem Cells and Regulatory T Cells

The main goal of this proposal is to investigate the mechanisms of how neural stem cells induce regulatory T cells.

2012 –2014 PI AUBMC-MPP \$40,000

Effect of Vitamin D replacement on immune function and cognition in Multiple Sclerosis patients

The main goal of this proposal is to investigate the immunological changes in peripheral blood of MS patients treated with Vit D

2012 –2014 PI CNRS, Lebanon \$26,670

Effect of Vitamin D replacement on immune function and cognition in Multiple Sclerosis patients

This is a supplement to the MPP grant. The main goal of this proposal is to investigate the immunological changes in peripheral blood of MS patients treated with Vit D

2013-2016 PI NMSS \$563,665

Transcriptional Regulation of the Resistance of Memory T Cells in EAE

The goal of this proposal is to investigate the mechanisms of auto-reactive memory CD4+ T cells resistance to suppression focusing on the role of TGF- β 1 signaling pathway

2010 –2016 PI ITN/NIH
N01 AI15416 \$125,086

The Collaborative Network for Clinical Research in Immune Tolerance: Protocol Chair support

This is a phase 2 multicenter trial of Abetacept in RR MS.

2011 –2016 PI National Institutes of Health
R01 AI043496 \$250,000

Immune Regulation of Neural Stem Cell Program in EAE

The main goal for this project is to find out what the effects of acute inflammation on the dynamics and multi-potentiality of endogenous neural stem and progenitor cells in vivo

2016 –2018 PI CNRS, Lebanon \$26,670

EBV Infected B Lymphocyte Exosomes and Dendritic Cells: Differential Cytokine Profile in Multiple

Sclerosis. The main goal of this proposal is to investigate the immunological changes in dendritic cells of MS patients treated with exosomes from EBV infected B cells

2017-2019 PI ONR \$382,278

Microglial activation and neural stem cells cross talk post TBI

The goal of this proposal is to investigate the role of microglia activation after mild traumatic brain injury

2017 –2020 Co-PI AUBMC-MPP \$120,000

Longitudinal effect of Vitamin D3 replacement on cognitive performance and MRI markers in Multiple Sclerosis patients: A Single-Blind Randomized Clinical Trial. The goal of this proposal is to assess the

difference in cognitive performance and brain MRI measures in MS patients between the intervention groups (high versus low dose vitamin D supplementation)

2018-2020	Co-PI	NMSS	\$50,000
<i>Interacting with Nature using virtual reality: A pilot intervention to restore cognitive fatigue in patients with Multiple Sclerosis.</i> The main goal of this proposal is to investigate the effect of 12 weeks of interactive virtual reality, nature compared to the urban environment, on cognitive fatigue.			
2019-2020	PI	MENACTRIMS	\$20,000
<i>Morphometric analysis of the visual pathways and brain volumetry through retinal OCT and MRI to differentiate Multiple Sclerosis from Neuromyelitis Optica Spectrum Disorder patients.</i> The main goal of this proposal is to differentiate MS from NMO by using MRI and OCT.			
2020-2021	PI	MENACTRIMS	\$20,000
<i>Analysis of Exosome Neuronal and Astrocytic Proteins as Prognostic Biomarkers in Multiple Sclerosis.</i> The main goal of this proposal is to investigate by proteomics the content of CSF exosomes and if they can predict prognosis.			
2021-2022	Co-PI	Octave Bioscience	\$20,000
<i>Serum Biomarkers in Multiple Sclerosis.</i> The purpose is to evaluate a comprehensive biomarker marker panel in association with relapses and gadolinium enhancing lesions, with the goal of finalizing a proteomic signature correlating with and predictive of new inflammatory activity in MS patients			
2021-2023	PI	MPP	\$90,000
<i>Proteomic profiles of exosomes as prognostic biomarkers in Multiple Sclerosis.</i> The purpose is to analyze the CSF exosome cargo in MS that has a great potential for finding a relevant biomarker for MS and potentially for identifying biomarkers predictive of disease course			

Clinical trials

ENSEMBLE – MA30143: An Open-Label, Single-arm Study to Evaluate the Effectiveness and Safety of Ocrelizumab in Patients with Early-Stage Relapsing-Remitting Multiple Sclerosis.

Date: 2017-2023

Site PI

CONSONANCE-MN39159: An open-label, single-arm, 4-year study to evaluate the effectiveness and safety of ocrelizumab treatment in patients with progressive multiple sclerosis.

Date: 2019-Ongoing

Site PI

COMB157G23101: A single-arm, prospective, multicentre, open-label study to evaluate ofatumumab treatment effectiveness and patient-reported outcomes in patients with relapsing multiple sclerosis transitioning from dimethyl fumarate or fingolimod therapy.

Date: 2021-Ongoing

Site PI

WA40404 O'Hand: A phase III b multicenter, randomized, double-blind placebo-controlled study to evaluate the efficacy and safety of ocrelizumab in adults with primary progressive multiple sclerosis.

Date: 2022-2024

Site PI

P3-IMU-838-RMS-01 (ENSURE-1): A Multi-Center, Randomized, Double-Blinded Phase 3 Study to Evaluate the Efficacy, Safety, and Tolerability of IMU-838 versus Placebo in Adults with Relapsing Multiple Sclerosis (ENSURE-1)

NCT05134441

Date: Jan 2024-

Site PI

SBS-2018-0498: Interacting with Nature using virtual reality: A pilot intervention to restore cognitive fatigue in patients with Multiple Sclerosis (MS)

Date: 2019-2022

PI: Samia Khoury

BIO-2017-0395: Longitudinal Effect of Vitamin D3 Replacement on Cognitive Performance and MRI Markers in Multiple Sclerosis Patients: A Single-Blind Randomized Clinical Trial

NCT03610139

Date: 2018-Ongoing

PI: Samia Khoury

IM.SK1.01: Comprehensive Multiple Sclerosis Database.

Date: 2012 – Ongoing

PI: Dr. Samia Khoury

109MS311: A Multicenter Extension Study to Determine the Long-Term Safety and Efficacy of BG00012 in Pediatric Subjects With Relapsing-Remitting Multiple Sclerosis.

Date: 2016 – 2018

Site Co-PI

EFC11759: A two year, multicenter, randomized, double-blind, placebo-controlled, parallel group trial to evaluate efficacy, safety, tolerability, and pharmacokinetics of teriflunomide administered orally once daily in pediatric patients with relapsing forms of multiple sclerosis followed by an open-label extension.

NCT02201108

Date: 2014 – 2019

Site Co-PI

MS200136_0035: A Multicenter, prospective, non-interventional study to assess Adherence to treatment for patients with Relapsing Multiple Sclerosis (RMS) who are prescribed subcutaneous (sc) interferon beta-1a.

Date: 2017 – 2021

Site Co-PI

TERIFR07918: Longitudinal assessment of clinical disability, cognition, quality of life, fatigue, brain magnetic resonance imaging measures, and optical coherence tomography findings in multiple sclerosis patients treated with AUBAGIO® (teriflunomide).

Date: 2016 – 2017

Site Co-PI

IM.SK1.04: Effect of Vitamin D Replacement on Immune Function and Cognition in MS Patients.

NCT01952483

Date: 2012-2017

PI: Samia Khoury

109MS202: Open-Label, Multicenter, Multiple-Dose Study of the Effect of BG00012 on MRI Lesions and Pharmacokinetics in Pediatric Subjects With Relapsing-Remitting Multiple Sclerosis Aged 10 to 17 Years.

NCT02410200

Date: 2015 – 2016

Site Co-PI

Effect of Vitamin D Replacement on Immune Function and Cognition in MS Patients

NCT01952483

Date: 2012-2017

PI: Samia Khoury

AMS01: Treatment of Multiple Sclerosis with Copaxone (Glatiramer acetate) and Albuterol

NCT00039988

Date: 2001-2006

PI: Samia Khoury

ITN035AI – A Phase II, Randomized, Double-blind, Parallel-group, Placebo-controlled, Multicenter Study to Evaluate the Safety and Efficacy of Abatacept in Adults with Relapsing-remitting Multiple Sclerosis

NCT01116427

Date: 2010-2015

Report of Local Teaching and Training

Teaching of Students in Courses

1989	Neurologic Physical Diagnosis (Harvard)		
25 Medical Students	Instructor	18 hours preparation, 12 hours contact time	
1998	Human Nervous System (Harvard)		
25-30 Medical Students	Instructor	18 hours preparation, 12 hours contact time	
2012	Human Nervous System (American University of Beirut)		
10-5 Medical Students	Instructor	5 hours preparation, 4 hours contact time	
2012	Multiple Sclerosis (American University of Beirut)		
100 Medical Students (Med II)	Instructor	6 hours preparation, 1 hour contact time	
2013	Multiple Sclerosis (American University of Beirut)		
100 Medical Students (Med II)	Instructor	6 hours preparation, 1 hour contact time	
2014	Multiple Sclerosis (American University of Beirut)		
100 Medical Students (Med II)	Instructor	6 hours preparation, 1 hour contact time	
2015	Brain and cognition (American University of Beirut)		
100 Medical Students (Med II)	Instructor	6 hours preparation, 3 hrs contact time	
2015	Clinical Skills (American University of Beirut)		
20 Medical Students (Med II)	Instructor for neuro exam	3 hours preparation, 6 hrs contact time	
2016	Brain and cognition (American University of Beirut)		
100 Medical Students (Med II)	Instructor	6 hours preparation, 3 hrs contact time	
2016	Clinical Skills (American University of Beirut)		
20 Medical Students (Med II)	Instructor for neuro exam	3 hours preparation, 6 hrs contact time	
2017	Brain and cognition (American University of Beirut)		
100 Medical Students (Med II)	Instructor	6 hours preparation, 3 hrs contact time	
2017	Clinical Skills (American University of Beirut)		
20 Medical Students (Med II)	Instructor for neuro exam	3 hours preparation, 6 hrs contact time	
2018	Brain and cognition (American University of Beirut)		
100 Medical Students (Med II)	Instructor	6 hours preparation, 3 hrs contact time	
2018	Clinical Skills (American University of Beirut)		
20 Medical Students (Med II)	Instructor for neuro exam	6 hours preparation, 6 hrs contact time	
2019	Brain and cognition (American University of Beirut)		
100 Medical Students (Med II)	Instructor	6 hours preparation, 3 hrs contact time	
2019	Clinical Skills (American University of Beirut)		
20 Medical Students (Med II)	Instructor for neuro exam	6 hours preparation, 6 hrs contact time	
2020	Brain and cognition (American University of Beirut)		
100 Medical Students (Med II)	Instructor	6 hours preparation, 3 hrs contact time	
2020	Clinical Skills (American University of Beirut)		
20 Medical Students (Med II)	Instructor for neuro exam	3 hours preparation, 6 hrs contact time	
2021	Brain and cognition (American University of Beirut)		

100 Medical Students (Med II)	Instructor	6 hours preparation, 3 hrs online
2022	Brain and cognition (American University of Beirut)	
100 Medical Students (Med II)	Instructor	6 hours preparation, 3 hrs online
2022	Brain and cognition (American University of Beirut) Interactive session	
50 Medical Students (Med II)	Instructor	6 hours preparation, 1 hrs online
2023	Brain and cognition (American University of Beirut)	
100 Medical Students (Med II)	Instructor	6 hours preparation, 3 hrs online
2023	Brain and cognition (American University of Beirut) Interactive session	
50 Medical Students (Med II)	Instructor	6 hours preparation, 1 hrs online
2024	Brain and cognition (American University of Beirut)	
100 Medical Students (Med II)	Instructor	6 hours preparation, 3 hrs online
2025	Brain and cognition (American University of Beirut)	
100 Medical Students (Med II)	Instructor	6 hours preparation, 3 hrs online

Formal Teaching of Residents, Clinical Fellows and Research Fellows (Post-docs):

1990-	CND Research Conference, Various Topics	
45 Residents, Post-Doctoral Fellows	Brigham and Women's Hospital	
1996	"New Treatments in MS," Neurology Residents Conference	
40 Residents	New England Medical Center	
2001, 2003-2004	"Clinical and Research Updates in MS,"	
10-20 Residents	Brigham and Women's Hospital	
2005	"Neural Stem Cell Interact with Immune Mediators in EAE: Consequences for Repair"	
40 Residents, Post-Docs, Attendings	Massachusetts General Hospital	
2005	"Scientific Basis of MS Therapy," Allergy/ Immunology Seminar	
20 Residents, Post-Docs, Attendings	Massachusetts General Hospital	
2006	"Interaction between neural stem cells and the immune system"	
20 Residents	Brigham and Women's Hospital	
2012-2013	Multiple Sclerosis (American University of Beirut)	
10 Neurology residents	Instructor	5 hours preparation, 4 hours contact time
2013-2014	Multiple Sclerosis (American University of Beirut)-Core curriculum	
10 Neurology residents/interns	Instructor	5 hours preparation, 4 hours contact time
2014-2015	Multiple Sclerosis (American University of Beirut)	
10 Neuro residents/interns	Instructor	5 hours preparation, 4 hours contact time
2014-2015	Multiple Sclerosis (American University of Beirut)	
10 Neuro residents/interns	Instructor	5 hours preparation, 4 hours contact time
2015-2016	Multiple Sclerosis (American University of Beirut)	
10 Neuro residents/interns	Instructor	5 hours preparation, 4 hours contact time
2016-2017	Multiple Sclerosis (American University of Beirut)	

10 Neuro residents/interns	Instructor	5 hours preparation, 4 hours contact time
2017-2018	Multiple Sclerosis (American University of Beirut)	
10 Neuro residents/interns	Instructor	5 hours preparation, 4 hours contact time

Clinical Supervisory and Training Responsibilities

2003-2009	Outpatient Neurology Clinic, preceptor	Multiple Sclerosis Center Brigham and Women's Hospital	One session per week
2011-	Outpatient Clinic	MS Center American University of Beirut	3 sessions per week

Laboratory and Other Research Supervisory and Training Responsibilities

1990-	Research Laboratory Supervisor	Daily mentorship
1995, 1996	Research Internship Program Mentor	80 hours per year

Formally Supervised Trainees

1993-1996	Lorenzo Gallon, M.D.	Faculty at University of Chicago
1993-1996	Wanjun Chen, M.D.	Faculty at the NIH
1996-1997	Shohreh Issazadeh, Ph.D	Faculty at Lund University, Sweden
1996-1998	Manuel Comabella, M.D.	Faculty at Hospital Universitari Vall d'Hebron, Barcelona
1998-2001	Karim Makhoul, M.D.	Faculty at Tunisia University Hospital, Tunisia
1997-2007	Jaime Imitola, M.D.	Faculty at Ohio State University
1998-2003	Tanuja Chitnis, M.D.	Faculty at Brigham and Women's Hospital, Boston
1998-2001	Anukanth Anumanthan, Ph.D	Working at the Harvard Institute of Proteomics
2001-2003	Christina Benou, M.D.	Currently in private practice, Berlin
1999-2002	Khaddir Raddassi, Ph.D	Associate Research Scientist, Yale
1998-2004	Kasia Bourcier, Ph.D	Program Officer, NIH
2002-2010	Bing Zhu, Ph.D	Medical director of clinical development, Biogen
2002-2005	Cristina Gutierrez, M.D.	Assistant Professor at MD Anderson TX
2002-2006	Susan Gauthier, M.D.	Currently on Faculty at New York Hospital, Cornell
2003-2006	Sal Napoli, M.D.	Currently in private practice
2003-2011	Wassim Elyaman, Ph.D.	Faculty at Columbia University, NY
2004-2007	Vissia Viglietta, M.D.	Director of Medical research, Biogen
2004-2006	Maria Houtchens, M.D.	Faculty at Brigham and Women's Hospital, Boston
2004-2010	Yue Wang, M.D. Ph.D.	Residency training
2005- 2011	Pia Kivisakk, Ph.D.	Faculty at Brigham and Women's Hospital, Boston
2004-2007	Stine Rassmussen	Postdoctoral Fellow in Sweden
2005-2008	Yoshio Bando, Ph.D.	Faculty at Asahikawa Medical College, Japan

2006-2008	Eman Nassim Ali, M.D.	Consultant at King Fahad Specialist Hospital Dammam
2007-2009	Jennifer Moodie, M.D.	Faculty at UMass Medical Center
2007-2011	Sarah Christin Starossom	Faculty Charite Hospital Berlin
2008-2009	Amar Elkhalfa, M.D.	Neurologist Cleveland Clinic Dubai
2008-2011	Li Cao, Ph.D.	Professor, Dept Neurobiol, Second Military Medical University, Shanghai, China.
2009-2010	Ribal Bassil, MD	Neuro Critical Care fellow, MGB Boston
2009-2010	Carolina Sandoval, MD	Neurosurgeon, University of Wisconsin
2009-2010	Younna Lahoud, MD	Assistant Professor, Boston University
2010-2011	Channa Kolb, MD	Clinical Fellow at the Multiple Sclerosis Center
2011-2012	Marta Olah, PhD	Instructor at Columbia University, NY
2011-2012	William Orent	PhD Student
2011-2012	Sahar Farhat, MD	Neurology Resident at AUBMC
2012-2013	Yasmina Pappas, MD	Clinical Research Fellow at the AUB MS Center
2012-2014	Raed Ismail, MD	ER Resident at AUBMC
2013-2014	Ralph Maroun, MD	Clinical Research Fellow at the AUB MS Center
2012-2013	Ahmad Kurdi	Post-Doctoral fellow
2012-2016	Lama Fawaz, PhD	Instructor in Immunology, AUB
2013-2016	Tarek Mouhieddine	Residency training
2013-2016	May Mrad, PhD	Post Doctoral fellow, AUB
2013-2014	Mariam Ghantous	Medical student, AUB
2013-2014	Souraya Sayegh	Masters Student, AUB
2014-2016	Stephanie Ghassan, MD	Clinical Research Fellow at the AUB MS Center
2014-2015	Sahar Osman, MD	Residency training
2014-2015	Nabil El-Ayoubi, MD	Faculty AUBMC
2014-2015	Zeinab Abdallah, MD	Neurology Resident, FRRP project supervisor
2015-2017	Marianne Said, MD	Clinical Research Fellow at the AUB MS Center
2016-2018	Ribal Haddad, MD	Clinical Research Fellow at the AUB MS Center
2016	Leslie Saba	Visiting Medical student from University College Dublin
2016-2017	Rami Fakihi, MD	Clinical Research Fellow at the AUB MS Center
2015-2017	Nabil El-Ayoubi, MD	Multiple Sclerosis Clinical Fellow
2017-2019	Marwa Baalbaki, MD	Clinical Research Fellow at the AUB MS Center
2017	Jose Kmeid, MD	Clinical Research Fellow at the AUB MS Center
2017-2020	Esber Saba, PhD	Post Doctoral fellow, AUB
2018-2020	Maysam El Najjar, MD	Clinical Research Fellow at the AUB MS Center

2018-2020	Wassim Assad, MD	Clinical Research Fellow at the AUB MS Center
2019-2020	Hussein Moussa, MD	Clinical Research Fellow at the AUB MS Center
2019-2020	Sarah Bou Reslan, MD	Clinical Research Fellow at the AUB MS Center
2020-2020	Sally El Sammak, MD	Clinical Research Fellow at the AUB MS Center
2019-2021	Georges Saab, MD	Multiple Sclerosis Clinical Fellow
2021-2022	Hadi Al Sabbagh, MD	Clinical Research Fellow at the AUB MS Center
2021-2022	Antoine Younes, MD	Clinical Research Fellow at the AUB MS Center
2021-2022	Nicole Metri, MD	Clinical Research Fellow at the AUB MS Center
2022-2023	Fares Fahd, MD	Clinical Research Fellow at the AUB MS Center
2022-2023	Lama Younes, MD	Clinical Research Fellow at the AUB MS Center
2022-2023	Ali Ezzeddine, MD	Clinical Research Fellow at the AUB MS Center
2023-2024	Jad El Ahdab, MD	Clinical Research Fellow at the AUB MS Center
2023-2024	Georgio Sader, MD	Clinical Research Fellow at the AUB MS Center
2023-2025	Joseph Abboud, MD	Clinical Research Fellow at the AUB MS Center
2024-2025	James Dagher, MD	Clinical Research Fellow at the AUB MS Center
2024-2026	Nour Najjar, MD	Clinical Research Fellow at the AUB MS Center
2025-2026	Medhat Siddik, MD	Clinical Research Fellow at the AUB MS Center

Mentorship

2016-2021	Faculty Mentoring Committee	American University of Beirut
2018- 2019	Participating as a mentor in “The Women Leading in Neurology Program” by the American Academy of Neurology to mentor junior AAN members	
2018-2020	Participating as a mentor in a program by International Women In MS (iWiMS)	
2020-2022	Mentored 2 MED II students in their project for the Scholarly Concentration Track	
2022-2024	Mentored a MED II student in her project for the Scholarly Concentration Track	
2023-2025	Co-mentored a MED II student in his project for the Scholarly Concentration Track	
2023-2025	Mentored a MED II student in her project for the Scholarly Concentration Track	
2024-	Mentored a group of 5 MED II students in a project for the Scholarly Concentration Track	

Formal Teaching of Peers (e.g., CME and other continuing education courses)

1991-2003	New Therapies for Multiple Sclerosis	2-3 hrs prep time, 1-2 hrs contact time
-----------	--------------------------------------	---

Intensive Review of Neurology	Boston, MA	Harvard Medical School
2000-2006	Yearly Harvard Autoimmunity Symposium Boston, MA	20 hrs prep time, 8 hrs contact time Harvard Medical School
2011-2014	Yearly Multiple Sclerosis Symposium Beirut, Lebanon	10 hrs prep time, 8 hrs contact time
2012	International Society of Neuroimmunology Chair, Scientific Program Organizing Committee	100 hrs prep time, 30 hrs contact time
2013	Neuroscience Symposium, AUBMC Neuromodulation in Brain Diseases/Disorders	5 hrs prep time, 8 hrs contact time
2013-2015	Monthly Neuroscience Seminar series AUBMC	10 hrs prep time, 8 hrs contact time
2015, 2016	Neuroscience Symposium, AUBMC	

Local Invited Presentations

1995	New Therapies in Multiple Sclerosis Waltham-Deaconess Hospital	Medical Grand Rounds
1996	Oral Tolerance as Therapy Lahey-Hitchcock Clinic	Medical Grand Rounds
1996	"Longitudinal MRI Studies in MS," Harvard Medical School	Harvard Neurology Research Symposium
1996	Costimulatory Blockade in Autoimmune disease Brigham and Women's Hospital	Medical Grand Rounds
1996	New Therapies in MS Brigham and Women's Hospital	Neurology Grand Rounds
1999	Costimulatory Signal Blockade: From Bench to Bedside Harvard Medical School	Harvard Autoimmunity Symposium
1999	MS Treatment: From Bench to Bedside Brigham and Women's Hospital	Neurology Grand Rounds
1999	MS Treatment: From Bench to Bedside Massachusetts General Hospital	Neurology Grand Rounds
2003	Regulatory function of CD8+CD28- T cells in EAE Harvard Medical School	Harvard Autoimmunity Symposium
2004	MS Treatment: From Bench to Bedside Faulkner Hospital	Medical Grand Rounds,
2004	MS Treatment: From Bench to Bedside Brigham and Women's Hospital	Rheumatology Grand Rounds
2004	MS Therapy: From Bench to Bedside Brigham and Women's Hospital	Neurology Resident Rounds
2006	Interaction between Neural Stem Cells and the Immune System, Brigham and Women's Hospital	Neurology Resident Rounds
2012	"MS immunology: Why does it matter?" Beirut, Lebanon	Medical Grand-Rounds, AUBMC

2013	“What Defines a Good Mentor” Beirut, Lebanon	Invited Speaker, FRRP AUBMC
2014	“MS: what should the Ob/Gyn clinician know” Beirut, Lebanon	Invited speaker, Ob/Gyn Dept AUBMC
2019	“Microglia activation in MS and EAE”	Invited speaker, Research Seminar, AUB
2019	“Progressive MS”	Invited Speaker, Lebanese Soc of Neurol
2019	“NMOSD”	Neurology Grand-Rounds, AUBMC
2020	“ Update on NMO”	Neurology Grand-Rounds, AUBMC
2020	“On mentorship”	Invited Speaker, HSON, AUB
2021	“ Biomarkers for MS”	Neurology Grand-Rounds, AUBMC
2022	“ Progressive MS”	Neurology Grand-Rounds, AUBMC
2023	“Multiple Sclerosis Workshop “	Neurology Interest Group at AUBFM
2024	“ New Treatments for Progressive MS”	Neurology Grand-Rounds, AUBMC
2024	“MS for OB-Gyn”	OB-Gyn Grand-Rounds, AUBMC

Report of Regional, National and International Invited Teaching and Presentations

Regional, National and International Invited Presentations and Courses

- *Regional Contributions*

1993	"Treatment of Multiple Sclerosis" New Hampshire Chapter	Invited Speaker National Multiple Sclerosis Society
1995	"New Therapies in Multiple Sclerosis" Massachusetts Chapter	Invited Speaker National Multiple Sclerosis Society
1999	“Immunology for the Neurologist” Newport, Rhode Island	Invited Speaker, MS Regional Advisory Meeting
2000	“New Therapies for Multiple Sclerosis” Maine	Invited Speaker Maine Neurologic Association
2000	“New Therapies for Multiple Sclerosis” Lenox, Massachusetts	Invited Speaker Arab American Medical Association
2000	“Recent Advances and New Directions in MS Research” Central New England Chapter	Stanley Waterman Guest Lecturer National Multiple Sclerosis Society
2002	Update on MS, Intensive Review of Neurology	Invited Speaker

	Belmont, Massachusetts	McLean Hospital
2004	Future Directions in Multiple Sclerosis Belmont, Massachusetts	Intensive Review of Neurology McLean Hospital
2005	Immunomodulation and Tissue Repair in MS Philadelphia, Pennsylvania	Invited Speaker Thomas Jefferson University
2002	“Role of MRI in MS” Newport, Rhode Island	Invited Speaker Serono Symposium
2002	“Scientific Basis of MS Therapies” Boston, Massachusetts	Invited Speaker Serono Preceptorship
2006	Block Symposium, “Autoimmunity of the CNS” Boston, Massachusetts	Co-Chair American Association of Immunologists
2006	“Neural Stem Cell Interactions with the Inflammatory Milieu” Boston, Massachusetts	Invited Speaker World Transplant Congress
2006	“Balancing Risks and Benefits in Therapeutic Choices” Newport, Rhode Island	Invited Speaker Biogen-Idec Consultants Meeting
2007	“Update in Multiple Sclerosis” Cambridge, Massachusetts	Invited Speaker Pri-Med Updates in Neurology
2008	“Update in Multiple Sclerosis” Cambridge, Massachusetts	Invited Speaker ALD symposium honoring Hugo Moser
2008	“Update in Multiple Sclerosis” Newton, MA	Invited Speaker McCourt Foundation program for patients
2010	“Update on MS Therapies” Shrewsbury, MA	Invited Speaker, MS Immunology
2011	“New Treatments for Multiple Sclerosis” Newton, MA	Invited Speaker UMASS Medical School/NMSS

• *National Contributions*

1996	"Acquired Thymic Tolerance," Immunology Seminar Portland, Oregon	Invited Speaker Oregon Health and Science University
1997	"New Therapies for Multiple Sclerosis: From Bench to Bedside" Scientific Basis of Neurology course	Invited Speaker American Academy of Neurology
1998	“Cyclophosphamide Therapy in MS,” Contemporary Issues in MS Atlanta, Georgia	Invited Speaker National Multiple Sclerosis Society
1998	“Experimental Therapy in MS” New Dimensions in Basic Research and their Clinical Application Symposium on Multiple Sclerosis Philadelphia, Pennsylvania	Invited Speaker
1998	“IL-12, HLA Expression and Disease Progression” International Symposium on Use of Imaging and Other Monitoring Measurements in MS Santa Fe, New Mexico	Invited Speaker
1999	“Costimulatory Signal Blockade: From Bench to Bedside” Miami Beach, Florida	Invited Speaker American Society of Nephrology
2002	“The John N. Whitaker Visiting Professorship”	Visiting Professor

	Little Rock, Arkansas	University of Arkansas for Medical Sciences
2002	"Costimulation" San Francisco, California	Symposium Chair Federation of Clinical Immunology Societies
2004	"Multiple Sclerosis: Bench to Bedside" Garson Lecturer for Clinical Neuroscience Grand Rounds Series Cleveland, Ohio	Visiting Professor Case Western Reserve University Hospitals
2004	"New Approaches in Biomarker Research," MS Biomarker Workshop Washington, D.C.	Invited speaker National Institutes of Health
2005	"Therapeutic Intervention in Multiple Sclerosis," "CTLA4IG Treatment in MS," Translational Medicine in Autoimmunity, Big Sky, Montana	Symposium Chair Keystone Symposia
2005	"Immune Intervention: From Bench to Bedside" Miami Beach, Florida	Faculty member American Academy of Neurology
2005	"Negative Costimulatory Pathways in Autoimmunity" Boston, Massachusetts	Symposium Chair/Invited Speaker Federation of Clinical Immunology Society
2005	"Novel Immunomodulatory Therapy in MS," CME course Philadelphia, Pennsylvania	Invited Speaker Thomas Jefferson University
2005	Roundtable Discussion, "Prognostic Indicators of MS Therapies" Houston, Texas	Invited Speaker Jobson Education Group
2005	"Multiple Sclerosis: Bench to Bedside" New York, New York	Visiting Professor Albert Einstein College of Medicine
2005	Immunology Colloquium Seminar Series, "Negative Costimulatory Pathways in Autoimmunity" Philadelphia, Pennsylvania	University of Pennsylvania
2005	"Immune Regulation of Neural Stem Function: Implications for Therapy in MS" and "Overview of MS," Cellular Therapy for Treatment of Autoimmune Disease Newport Beach, California	Invited Speaker City of Hope meeting
2005	Renal Week: "Negative Costimulatory Pathways in Autoimmunity" Philadelphia, Pennsylvania	Invited Speaker American Society of Nephrology
2006	Update on Multiple Sclerosis AAN Winter Symposium, Las Vegas, Nevada	Invited Speaker American Academy of Neurology
2006	"Multiple Sclerosis: Immune Intervention, Bench to Bedside," Immunotherapy in Autoimmunity Symposium Lebanon, New Hampshire	Invited Speaker Dartmouth Medical School
2006	"Multiple Sclerosis: Bench to Bedside," Dept of Neurology Baltimore, Maryland	Visiting Professor, Grand Rounds University of Maryland
2008	"Update on MS" Department of Neurology, New York, New York	Invited Speaker, Grand-Rounds New York University
2008	"Immune Regulation of Neural Stem Function" Neuroimmunology Program, Ann Arbor, Michigan	Visiting Professor University of Michigan
2008	"NSC Interaction with the Inflammatory Milieu" Dallas-Fort Worth, TX	Invited Speaker International Society of Neuroimmunology Congress
2009	"Neural stem cells interactions with the immune system" Temple University School of Medicine	Invited Speaker Neuroscience Symposium

2009	“Update on MS” New York, New York	Invited Speaker, Grand-Rounds Albert Einstein College of Medicine
2010	“Multiple Sclerosis: From Pathogenesis to Treatment” New Brunswick, New Jersey	Invited Speaker, Grand-Rounds UMDNJ
2010	“Costimulation: Bench to Bedside and Back” New York, New York	Invited Speaker, Lecture Series Weill Cornell Medical College
2012	“Immunology of Multiple Sclerosis” Broumana, Lebanon	Invited Speaker, Lebanese Society of Neurology
2012	“Hormones and pregnancy in Autoimmune Diseases” Beirut, Lebanon	Invited Speaker, Lebanese Society of Obstetrics and Gynecology
2013	“Neural Stem Cell Therapy: Fantasy or reality” Tripoli, Lebanon	Invited Speaker, Lebanese Order of Physicians
2013	“Innate Immunity and CNS Inflammation” Birmingham, AL	Invited Speaker, UAB MS
2023	“Progressive MS”	Neurology Grand-Rounds AUBMC
2023	“Microglia activation in MS and EAE”	Invited speaker, EPIM Dept, AUB
2024	“B Cells in MS” NDS Jbeil Lebanon	Invited Speaker, 1 st MS Day
2025	“MS Treatment”	Neurology Grand-Rounds AUBMC
2026	“MS in older age”	Neurology Grand-Rounds AUBMC

• *International Contributions*

1993	“Oral Tolerance” Bergamo, Italy	Visiting Scientist, Invited Speaker Mario Negri Institute for Pharmacological Research
1995	“New Therapies in Multiple Sclerosis”	Invited Speaker Neurologic Institute of Tunisia
1995	“Immunologic Monitoring in MS: How to Identify Responders in Clinical Trials,” Clinical Trials in MS Paris, France	Invited Speaker, Panel Discussion Charcot Foundation
1995	"Oral Tolerance," Autoimmunity Symposium Sao Paolo, Brazil	Invited Speaker University of Sao Paolo
1995	"Mechanisms of Acquired Thymic Tolerance: Lessons from Autoimmune Models" Angra Dos Reis, Brazil	Invited Speaker Brazilian Congress of Immunology
1996	“Acquired Thymic Tolerance,” Killam Seminar Series Montreal, Canada	Invited Speaker Montreal Neurologic Institute
1997	“T Cell Costimulation in Autoimmune Disease,” and “Mechanisms of Acquired Thymic Tolerance” Rio Dos Pedras, Brazil	Invited Speaker Brazilian Congress of Immunology
1997	“New Therapies for Multiple Sclerosis,” Neurology Grand Rounds Sao Paolo, Brazil	Invited Speaker University of Sao Paolo
1998-	Various Topics in Neurology and Multiple Sclerosis	Lecture/Invited Speaker

	Beirut, Lebanon	American University Hospital of Beirut
2001	“Multiple Sclerosis: Etiological Mechanisms” Edinburgh, Scotland	Symposium Chair International Society of Neuroimmunology
2002	“Multiple Sclerosis” Grand Rounds Emirate of Abu Dhabi, UAE	Visiting Professor Tawam Hospital Neurology Dept.
2003	“Immunotherapy” Paris, France	Symposium Chair Federation of Clinical Immunology Societies
2003	“Multiple Sclerosis: An Immune Mediated Disease” Hungary	Invited Speaker Multiple Sclerosis Symposium
2004	“Costimulation” Montreal, Canada	Symposium Chair Federation of Clinical Immunology Societies
2004	Various Topics Venice, Italy	Invited Speaker Neuroimmunology Congress
2004	“Multiple Sclerosis, bench to bedside” Tokyo, Japan	Invited Speaker Japanese Society of Neuroimmunology
2005	“Mechanisms of T Cell Activation” Singapore	Symposium Chair World Transplant Congress
2005	“Immunopathogenesis of Multiple Sclerosis” Tokyo, Japan	Invited Speaker Research Committee
2006	“NSC Interaction with the Inflammatory Milieu” Nagoya, Japan	Invited Speaker International Society of Neuroimmunology Congress
2008	“Treatments that target Microglia” Montreal, Canada	Invited Speaker ECTRIMS-ACRIMS
2008	“Multiple Sclerosis: Immune Intervention, Bench to Bedside” Marmaris, Turkey	Invited Speaker School of Neuroimmunology
2009	“Stem cell Interaction with the Inflammatory Milieu” Alexandria, Egypt	Invited Speaker International Congress On tissue engineering
2009	“Interaction of NSC with the immune system” Fort-Forth, Texas	Am Society for Neurochemistry
2009	“Outlook for using stem cells in treatment of immune mediated diseases” Beirut, Lebanon	Invited Speaker CAST congress
2010	“CTLA4Ig in Multiple Sclerosis” Beirut, Lebanon	Invited speaker, Research Day American University of Beirut
2010	“Outlook and challenges for using stem cells in treatment of immune mediated diseases” Cairo, Egypt	Invited speaker, CAST congress
2010	“Neutral Stem Cells For Treatment Of Demyelinating Diseases: Fantasy Or Real Prospect” Beirut, Lebanon	Invited speaker, The Naji Sahyoun Memorial lecture, MEMA
2010	" Cell Therapy for Neurological Degenerative and Autoimmune Diseases” Carthage, Tunisia	Invited Speaker, MESOT

2010	“Costimulation: Bench to Bedside and Back” Montreal, Canada	Keynote Speaker Montreal Neurological Institute
2010	“Neural Stem Cell Therapy: Fantasy or reality” Montreal, Canada	Keynote Speaker, Dinner Seminar Montreal Neurological Institute
2011	“Immunology of Multiple Sclerosis” Beirut, Lebanon	Invited speaker, MS symposium
2012	“Immunology of Multiple Sclerosis” Beirut, Lebanon	Invited speaker, MS symposium
2012	“Scientific Basis of MS Therapies” Alexandria, Egypt	Plenary speaker, MS Day
2013	“Beirut to Boston and Back: It’s All About Tolerance” Boston, MA	Keynote Speaker, Harvard Arab Weekend
2013	“MS Immunology: why is it relevant” Beirut, Lebanon	Invited Speaker, Lebanese Society of Neurology
2013	“MS Immunology: why is it relevant” Riadh, Saudi Arabia	Invited Speaker, Saudi Society of Neurology
2014	“Immune basis of new MS therapies” Austin, Texas	Invited Speaker, Texas Neurology Society
2014	“AUBMC-MSKCC collaboration” New York, NY	Invited panelist, C3 Summit
2014	“Immunopathogenesis of MS” Dubai, UAE	Invited Speaker, MENACTRIMS
2014	“MS new drugs or repositioning of old drugs” Mainz, Germany	Invited Speaker, ESNI
2014	“Innate Immunity and CNS Inflammation” Mainz, Germany	Invited Speaker, ISNI
2015	“MS Differential Diagnosis” Doha, Qatar	Invited Speaker, Qatar Neurology Symposium
2015	“CIS” Doha, Qatar	Invited Speaker, Qatar Neurology Symposium
2015	“Biomarkers in MS” Istanbul, Turkey	Invited Speaker, Istanbul MS Days Symposium
2015	“Vitamin D in MS” Beirut, Lebanon	Invited Speaker, Lebanese Society of Neurology
2015	“Breakthroughs in MS” Orlando, Fl	Invited Speaker, Breakthroughs in Neurology conference, AAN
2016	“First line DMTs: when to start/stop”	Invited Speaker, Hamad Neuroscience

	Doha, Qatar	Symposium
2016	“MS Immunology” Amman, Jordan	Invited Speaker, MENACTRIMS plenary session
2017	“Diagnostic Dilemmas and Therapeutic Decisions in Demyelinating Diseases” Phoenix, AZ	Course organizer, Breakthroughs in Neurology conference, AAN
2017	“Biomarkers in MS” Dubai, UAE	Invited Speaker, MENACTRIMS
2017	“Biomarkers of Repair ” Kyrinia, Cyprus	Invited Speaker, Istanbul MS Days Symposium
2018	“Role of microglia in MS” Kyrinia, Cyprus	Invited Speaker, Istanbul MS Days Symposium
2019	“Data Science in Healthcare”	Symposium Chair
2019	“Women Leaders and Health” Beirut, Lebanon	Invited Speaker and Chair Arab International Women’s Forum
2019	“Progressive MS” Beirut, Lebanon	Invited Speaker, Lebanese Society of Neurology
2019	“Free Communications” Stockholm, Sweden	Session Chair, ECTRIMS
2019	“HERV-W-ENV In Sera of MS” Lyon, France	Invited Speaker, 3rd international workshop on HERVs and Disease
2019	“Multiple Sclerosis and Other Demyelinating Diseases” Dubai, UAE	Course organizer, World Congress of Neurology
2019	“Progressive MS” Dubai, UAE	Invited Speaker, World Congress of Neurology
2019	“Biomarkers for MS” Dubai, UAE	Invited Speaker, World Congress of Neurology
2019	“Neurofilaments in MS” Dubai, UAE	Invited Speaker, MENACTRIMS
2019	“Common Imaging for common phenotypes among different neurological disorders” Istanbul, Turkey	Symposium Chair, Istanbul MS Days Symposium
2019	“Chronic MS”	Invited Speaker, Bezmialem Univ Istanbul

2021 “Late Breaking News” (virtual)	Session Chair, ECTRIMS
2021 “Autoimmune demyelinating CNS diseases besides NMOSD and MOGAD”	Session Chair, ECTRIMS
2021 “Biomarkers in the Differential Diagnosis of MS”	Invited speaker, ECTRIMS
2021 “Neuroimmunology of CNS demyelinating disorders (Plenary I)”	Session Chair, ISNI
2021 “Neurofilaments in MS: Update” Dubai, UAE	Invited Speaker, MENACTRIMS
2021 “Biomarkers for MS” George Washington University	Invited Grand-Round Speaker,
2022 “Progressive Multiple Sclerosis: To treat or NOT to treat?”	IVPN-NeuroPsychiatry Journal Club
2022 “Microglia and Ageing” Cappadoccia, Turkey	Invited Speaker, Istanbul MS Days symposium
2022 “Progressive MS: What’s New” Toronto University	Invited Grand-Round Speaker, Barlo MS center speaker series
2022 “Translational Research” Egypt, online	Invited speaker, Harvard Scholarly immersion program
2022 “Microglia: Sexual dimorphism and aging” Elalamein, Egypt	Invited Speaker, IBRO-MENA
2024 “New therapies in MS: CD40L, CART and others” Dubai, UAE	Invited Speaker, MENACTRIMS Workshop
2025 “MS in Older Age: to stop DMTs or not” Dubai, UAE	Invited Speaker, MENACTRIMS
2025 “Microglia in MS” Istanbul, Turkey	Session Chair, Istanbul MS Days symposium

Report of Clinical Activities and Innovations

Current Licensure and Certification

1983	ECFMG
1986	Ohio State Medical License
1988	Massachusetts State Medical License

Practice Activities

- Area of Specialty: Multiple Sclerosis

Attending Outpatient Neurology Clinic

Preceptor Outpatient Neurology Resident clinic

Multiple Sclerosis Center
Brigham and Women's Hospital
Multiple Sclerosis Center
Brigham and Women's Hospital

Report on Clinical Activities

Since Joining the AUB Medical Center in Beirut, I have established the first multidisciplinary MS center in the region with 3 neurologists (MS specialists), 3 infusion nurses, on-site pharmacist, 8 infusion chairs, and collaborating specialists in Urology, Ophthalmology, and Obstetrics-Gynecology. We have established practice guidelines, and initiated patient education programs. I have also started a weekly MS seminar attended by neurologists, residents, students, nurses, and pharmacists. I initiated a series of fund-raising events and activities to support needy patients and local research.

As the Director of Abu Haidar Neuroscience Institute, I have chaired a committee to generate a 5-year strategic plan for the Institute that includes improvements in patient care, recruitment of faculty, and enabling research. We have reorganized the outpatient neurology and neurosurgery services and consolidated these services to a new outpatient facility with multidisciplinary approach to care and a focus on patient education.

Report of Education of Patients and Service to the Community

Activities

2000-2012	Speaker for National MS Society Educational Programs (yearly), Boston
2012	Organizer and speaker, MS Patient Education Day, Beirut, Lebanon
2012	In collaboration with the Lebanese Ministry of Health, we initiated a national awareness campaign about multiple sclerosis
2013	Organizer and speaker, MS Patient Education Day, Zahle, Lebanon
2014	Organizer and speaker, MS Patient Education Day, Metn, Lebanon
2015	MS Awareness Day, AUB Campus
2016	Organizer and speaker, MS Awareness Day, Byblos, Lebanon
2016	Organizer and speaker, MS Patient Education Day, Beirut, Lebanon
2017	Organizer, MS Patient Education Day, Beirut, Lebanon
2018	Organizer and speaker, Women and MS, Beirut, Lebanon
2019	Chair, Scientific Committee for AAHCI regional conference
2021	Organizer and speaker, MS Patient Education Day, Beirut, Lebanon
2022	Organizer and speaker, MS Patient Education Day, Beirut, Lebanon
2023	Organizer and speaker, MS Patient Education Day, Beirut, Lebanon

Report of Scholarship

Publications

- **Peer Reviewed Publications in print or other media**

1. Meisler DM, Tomsak RL, **Khoury S**, Hanson MR, Schwab IR, Ransohoff RM. Anterior uveitis and multiple sclerosis. *Cleve Clin J Med* 1989;56(5):535-538.
2. Santos, LM, Lider O, Audette J, **Khoury SJ**, Weiner HL. Characterization of immunomodulatory properties and accessory cell function of small intestinal epithelial cells. *Cell Immunol* 1990;127(1):26-34.
3. **Khoury SJ**, Lider O, Al-Sabbagh A, Weiner HL. Suppression of experimental autoimmune encephalomyelitis by oral administration of myelin basic protein. III. Synergistic effect of lipopolysaccharide. *Cell Immunol* 1990;131(2):302-310. doi: 10.1016/0008-8749(90)90256-Q
4. Sayegh MH, **Khoury SJ**, Hancock WW, Weiner HL, Carpenter CB. Induction of immunity and oral tolerance with polymorphic class II major histocompatibility complex allopeptides in the rat. *Proc Natl Acad Sci USA* 1992;89(16):7762-7766. PMC49791
5. **Khoury SJ**, Hancock WW, Weiner HL. Oral tolerance to myelin basic protein and natural recovery from experimental autoimmune encephalomyelitis are associated with downregulation of inflammatory cytokines and differential upregulation of transforming growth factor beta, interleukin 4, and prostaglandin E expression in the brain. *J Exp Med* 1992;176(5):1355-1364.
6. Weiner HL, Mackin GA, Matsui M, Orav EJ, **Khoury SJ**, Dawson DM, Hafler DA. Double-blind pilot trial of oral tolerization with myelin antigens in multiple sclerosis. *Science* 1993;259(5099):1321-1324.
7. **Khoury SJ**, Sayegh MH, Hancock WW, Gallon L, Carpenter CB, Weiner HL. Acquired tolerance to experimental autoimmune encephalomyelitis by intrathymic injection of myelin basic protein or its major encephalitogenic peptide. *J Exp Med* 1993;178(2):559-566.
8. Hancock WW, **Khoury SJ**, Carpenter CB, Sayegh MH. Differential effects of oral versus intrathymic administration of polymorphic major histocompatibility complex class II peptides on mononuclear and endothelial cell activation and cytokine expression during a delayed-type hypersensitivity response. *Am J Pathol* 1994;144(6):1149-1158. PMC1887471
9. **Khoury SJ**, Weiner HL. Kappa light chains in spinal fluid for diagnosing multiple sclerosis. *JAMA* 1994;272(3):242-243.
10. **Khoury SJ**, Guttmann CR, Orav EJ, Hohol MJ, Ahn SS, Hsu L, Kikinis R, Mackin GA, Jolesz FA, Weiner HL. Longitudinal MRI in multiple sclerosis: correlation between disability and lesion burden. *Neurology* 1994;44(11):2120-2124.
11. Balashov KE, **Khoury SJ**, Hafler DA, Weiner HL. Inhibition of T cell responses by activated human CD8+ T cells is mediated by interferon-gamma and is defective in chronic progressive multiple sclerosis. *J Clin Invest* 1995;95(6):2711-2719. PMC295954. doi: 10.1172/JCI117973
12. **Khoury SJ**, Gallon L, Chen W, Betres K, Russell ME, Hancock WW, Carpenter CB, Sayegh MH, Weiner HL. Mechanisms of acquired thymic tolerance in experimental autoimmune encephalomyelitis: thymic dendritic-enriched cells induce specific peripheral T cell unresponsiveness in vivo. *J Exp Med* 1995;182(2):357-366.
13. **Khoury SJ**, Akalin E, Chandraker A, Turka LA, Linsley PS, Sayegh MH, Hancock WW. CD28-B7 costimulatory blockade by CTLA4Ig prevents actively induced experimental autoimmune encephalomyelitis and inhibits Th1 but spares Th2 cytokines in the central nervous system. *J Immunol* 1995;155(10):4521-4524.
14. Fukaura H, Kent SC, Pietrusewicz MJ, **Khoury SJ**, Weiner HL, Hafler DA. Induction of circulating

- myelin basic protein and proteolipid protein-specific transforming growth factor-beta1-secreting Th3 T cells by oral administration of myelin in multiple sclerosis patients. *J Clin Invest* 1996;98(1):70-77. PMC507402. doi: 10.1172/JCI118779
15. Chen W, Murphy B, Waaga AM, Willett TA, Russell ME, **Khoury SJ**, and Sayegh MH. Mechanisms of indirect allorecognition in graft rejection: class II MHC allopeptide-specific T cell clones transfer delayed-type hypersensitivity responses in vivo. *Transplantation* 1996;62(6):705-710.
 16. **Khoury SJ**, Gallon L, Verburg RR, Chandraker A, Peach R, Linsley PS, Turka LA, Hancock WW, Sayegh MH. Ex vivo treatment of antigen-presenting cells with CTLA4Ig and encephalitogenic peptide prevents experimental autoimmune encephalomyelitis in the Lewis rat. *J Immunol* 1996;157(8):3700-3705.
 17. Balashov KE, Smith DR, **Khoury SJ**, Hafler DA, Weiner HL. Increased interleukin 12 production in progressive multiple sclerosis: induction by activated CD4⁺ T cells via CD40 ligand. *Proc Natl Acad Sci USA* 1997;94(2):599-603. PMC19559. doi: 10.1073/pnas.94.2.599
 18. Hohol MJ, Guttmann CR, Orav J, Mackin GA, Kikinis R, **Khoury SJ**, Jolesz FA, Weiner HL. Serial neuropsychological assessment and magnetic resonance imaging analysis in multiple sclerosis. *Arch Neurol* 1997;54(8):1018-1025.
 19. Chen W, Issazadeh S, Sayegh MH, **Khoury SJ**. In vivo mechanisms of acquired thymic tolerance. *Cell Immunol* 1997;179(2):165-173. doi: 10.1006/cimm.1997.1165
 20. Smith DR, Balashov KE, Hafler DA, **Khoury SJ**, Weiner HL. Immune deviation following pulse cyclophosphamide/methylprednisolone treatment of multiple sclerosis: increased interleukin-4 production and associated eosinophilia. *Ann Neurol* 1997;42(3):313-318. doi: 10.1002/ana.410420307
 21. Gallon L, Chandraker A, Issazadeh S, Peach R, Linsley PS, Turka LA, Sayegh MH, **Khoury SJ**. Differential effects of B7-1 blockade in the rat experimental autoimmune encephalomyelitis model. *J Immunol* 1997;159(9):4212-4216.
 22. Chen W, Sayegh MH, **Khoury SJ**. Mechanisms of acquired thymic tolerance in vivo: intrathymic injection of antigen induces apoptosis of thymocytes and peripheral T cell anergy. *J Immunol* 1998;160(3):1504-1508.
 23. Parr MJ, Wen PY, Schaub M, **Khoury SJ**, Sayegh MH, Fine HA. Immune parameters affecting adenoviral vector gene therapy in the brain. *J Neurovirol* 1998;4(2):194-203.
 24. Samoilova EB, Horton JL, Zhang H, **Khoury SJ**, Weiner HL, Chen Y. CTLA-4 is required for the induction of high dose oral tolerance. *Int Immunol* 1998;10(4):491-498. doi: 10.1093/intimm/10.4.491
 25. Takashima H, Smith DR, Fukaura H, **Khoury SJ**, Hafler DA, Weiner HL. Pulse cyclophosphamide plus methylprednisolone induces myelin-antigen-specific IL-4-secreting T cells in multiple sclerosis patients. *Clin Immunol Immunopathol* 1998;88(1):28-34.
 26. Issazadeh S, Navikas V, Schaub M, Sayegh MH, **Khoury S**. Kinetics of expression of costimulatory molecules and their ligands in murine relapsing experimental autoimmune encephalomyelitis in vivo. *J Immunol* 1998;161(3):1104-1112.
 27. Comabella M, Balashov K, Issazadeh S, Smith D, Weiner HL, **Khoury SJ**. Elevated interleukin-12 in progressive multiple sclerosis correlates with disease activity and is normalized by pulse cyclophosphamide therapy. *J Clin Invest* 1998;102(4):671-678. PMC508928. doi: 10.1172/JCI3125
 28. Balashov KE, Olek MJ, Smith DR, **Khoury SJ**, Weiner HL. Seasonal variation of interferon-gamma production in progressive multiple sclerosis. *Ann Neurol* 1998;44(5):824-828. doi: 10.1002/ana.410440519
 29. Issazadeh S, Zhang M, Sayegh MH, **Khoury SJ**. Acquired thymic tolerance: role of CTLA4 in the initiation and maintenance of tolerance in a clinically relevant autoimmune disease model. *J Immunol*

- 1999;162(2):761-765.
30. Schaub M, Issazadeh S, Stadlbauer TH, Peach R, Sayegh MH, **Khoury SJ**. Costimulatory signal blockade in murine relapsing experimental autoimmune encephalomyelitis. *J Neuroimmunol* 1999;96(2):158-166.
 31. **Khoury SJ**, Orav EJ, Guttmann CR, Kikinis R, Jolesz FA, Weiner HL. Changes in serum levels of ICAM and TNF-R correlate with disease activity in multiple sclerosis. *Neurology* 1999;53(4):758-764.
 32. Hohol MJ, Olek MJ, Orav EJ, Stazzone L, Hafler DA, **Khoury SJ**, Dawson DM, Weiner HL. Treatment of progressive multiple sclerosis with pulse cyclophosphamide/methylprednisolone: response to therapy is linked to the duration of progressive disease. *Mult Scler* 1999;5:403-409. doi: 10.1177/135245859900500i606
 33. Weiner HL, Guttmann CR, **Khoury SJ**, Orav EJ, Hohol MJ, Kikinis R, Jolesz FA. Serial magnetic resonance imaging in multiple sclerosis: correlation with attacks, disability, and disease stage. *J Neuroimmunol* 2000;104(2):164-173.
 34. Issazadeh S, Abdallah K, Chitnis T, Chandraker A, Wells AD, Turka LA, Sayegh MH, **Khoury SJ**. Role of passive T-cell death in chronic experimental autoimmune encephalomyelitis. *J Clin Invest* 2000;105(8):1109-1116. PMC3000830. doi: 10.1172/JCI8607
 35. Kishimoto K, Dong VM, Issazadeh S, Fedoseyeva EV, Waaga AM, Yamada A, Sho M, Benichou G, Auchincloss H Jr, Grusby MJ, **Khoury SJ**, Sayegh MH. The role of CD154-CD40 versus CD28-B7 costimulatory pathways in regulating allogeneic Th1 and Th2 responses in vivo. *J Clin Invest* 2000;106(1):63-72. PMC314364. doi: 10.1172/JCI9586
 36. Balashov KE, Comabella M, Ohashi T, **Khoury SJ**, Weiner HL. Defective regulation of IFNgamma and IL-12 by endogenous IL-10 in progressive MS. *Neurology* 2000;55(2):192-198.
 37. **Khoury SJ**, Guttmann CR, Orav EJ, Kikinis R, Jolesz FA, Weiner HL. Changes in activated T cells in the blood correlate with disease activity in multiple sclerosis. *Arch Neurol* 2000;57(8):1183-1189.
 38. Frank MH, Denton MD, Alexander SI, **Khoury SJ**, Sayegh MH, Briscoe DM. Specific MDR1 P-glycoprotein blockade inhibits human alloimmune T cell activation in vitro. *J Immunol* 2001;166(4):2451-2459.
 39. Chitnis T, Najafian N, Abdallah KA, Dong V, Yagita H, Sayegh MH, **Khoury SJ**. CD28-independent induction of experimental autoimmune encephalomyelitis. *J Clin Invest* 2001;107(5):575-583. PMC199425. doi: 10.1172/JCI11220
 40. Waaga AM, Gasser M, Kist-van Holthe JE, Najafian N, Muller A, Vella JP, Womer KL, Chandraker A, **Khoury SJ**, Sayegh MH. Regulatory functions of self-restricted MHC class II allopeptide-specific Th2 clones in vivo. *J Clin Invest* 2001;107(7):909-916. PMC199572. doi: 10.1172/JCI11427
 41. Makhlouf K, Comabella M, Imitola J, Weiner HL, **Khoury SJ**. Oral salbutamol decreases IL-12 in patients with secondary progressive multiple sclerosis. *J Neuroimmunol* 2001;117(1-2):156-165.
 42. Makhlouf K, Weiner HL, **Khoury SJ**. Increased percentage of IL-12+ monocytes in the blood correlates with the presence of active MRI lesions in MS. *J Neuroimmunol* 2001;119(1):145-149.
 43. Chitnis T, Najafian N, Benou C, Salama AD, Grusby MJ, Sayegh MH, **Khoury SJ**. Effect of targeted disruption of STAT4 and STAT6 on the induction of experimental autoimmune encephalomyelitis. *J Clin Invest* 2001;108(5):739-747. PMC209380. doi: 10.1172/JCI12563
 44. Karni A, Koldzic DN, Bharanidharan P, **Khoury SJ**, Weiner HL. IL-18 is linked to raised IFN-gamma in multiple sclerosis and is induced by activated CD4(+) T cells via CD40-CD40 ligand interactions. *J Neuroimmunol*. 2002 Apr;125(1-2):134-140.
 45. Comabella M, Imitola J, Weiner HL, **Khoury SJ**. Interferon-beta treatment alters peripheral blood monocytes chemokine production in MS patients. *J Neuroimmunol*. 2002 May;126(1-2):205-212.

46. O'Connor KC, Chitnis T, Griffin DE, Piyasirisilp S, Bar-Or A, **Khoury S**, Wucherpfennig KW, Hafler DA. Myelin basic protein-reactive autoantibodies in the serum and cerebrospinal fluid of multiple sclerosis patients are characterized by low-affinity interactions. *J Neuroimmunol.* 2003 Mar;136(1-2):140-148.
47. Ansari MJ, Salama AD, Chitnis T, Smith RN, Yagita H, Akiba H, Yamazaki T, Azuma M, Iwai H, **Khoury SJ**, Auchincloss H Jr, Sayegh MH. The programmed death-1 (PD-1) pathway regulates autoimmune diabetes in nonobese diabetic (NOD) mice. *J Exp Med.* 2003 Jul 7;198(1):63-69. PMC196083. doi: 10.1084/jem.20022125
48. Salama AD, Chitnis T, Imitola J, Ansari MJ, Akiba H, Tushima F, Azuma M, Yagita H, Sayegh MH, **Khoury SJ**. Critical role of the programmed death-1 (PD-1) pathway in regulation of experimental autoimmune encephalomyelitis. *J Exp Med.* 2003 Jul 7;198(1):71-8. PMC2196082. doi: 10.1084/jem.20022119.
49. Najafian N, Chitnis T, Salama AD, Zhu B, Benou C, Yuan X, Clarkson MR, Sayegh MS, **Khoury SJ**. Regulatory functions CD8+CD28- T cells in an autoimmune disease model. *J Clin Invest.* 2003 Oct;112(7):1037-1048. PMC198520. doi: 10.1172/JCI17935
50. Sheen VL, Ganesh VS, Topcu M, Sebire G, Bodell A, Hill RS, Grant PE, Shugart YY, Imitola J, **Khoury SJ**, Guerrini R, Walsh CA. Mutations in ARFGEF2 implicate vesicle trafficking in neural progenitor proliferation and migration in the human cerebral cortex. *Nat Genet.* 2004 Jan;36(1):69-76. doi: 10.1038/ng1276
51. Karni A, Balashov K, Hancock WW, Bharanidharan P, Abraham M, **Khoury SJ**, Weiner HL. Cyclophosphamide modulates CD4+ T cells into a T helper type 2 phenotype and reverses increased IFN-gamma production of CD8+ T cells in secondary progressive multiple sclerosis. *J Neuroimmunol.* 2004 Jan;146(1-2):189-198.
52. Chitnis T, Salama AD, Grusby MJ, Sayegh MH, **Khoury SJ**. Defining Th1 and Th2 immune responses in a reciprocal cytokine environment in vivo. *J Immunol.* 2004 Apr 1;172(7):4260-4265.
53. Imitola J, Comabella M, Chandraker AK, Dangond F, Sayegh MH, Snyder EY, **Khoury SJ**. Neural stem/progenitors cells express costimulatory molecules that are differentially regulated by inflammatory and apoptotic stimuli. *Am J Pathol.* 2004 May;164(5):1615-1625. PMC1615661. doi: 10.1016/S0002-9440(10)63720-0
54. Magee CC, Denton MD, Womer KL, **Khoury SJ**, Sayegh MH. Assessment by flow cytometry of intracellular cytokine production in the peripheral blood cells of renal transplant recipients. *Clin Transplant.* 2004 Aug;18(4):395-401. doi: 10.1111/j.1399-0012.2004.00179.x
55. Meier DS, Weiner HL, **Khoury SJ**, Guttman CRG. Magnetic resonance imaging surrogates of multiple sclerosis pathology and their relationship to central nervous system atrophy. *J Neuroimaging.* 2004;14(3 Suppl):46S-53S. doi:10.1177/1051228404266268
56. Nieto M, Monuki ES, Tang H, Imitola J, Haubst N, **Khoury SJ**, Cunningham J, Gotz M, Walsh CA. Expression of Cux-1 and Cux-2 in the subventricular zone and upper layers II-IV of the cerebral cortex. *J Comp Neurol.* 2004 Nov 8;479(2):168-180. doi: 10.1002/cne.20322
57. Imitola J, Raddassi K, Park KI, Mueller FJ, Nieto M, Teng YD, Frenkel D, Li J, Sidman RL, Walsh CA, Snyder EY, **Khoury SJ**. Directed migration of neural stem cells to sites of CNS injury by the stromal cell-derived factor 1alpha/CXC chemokine receptor 4 pathway. *Proc Natl Acad Sci USA.* 2004 Dec 28;101(52):18117-18122. PMC536055. doi: 10.1073/pnas.0408258102
58. Kutok JL, Yang X, Folkerth RD, Imitola J, Raddassi K, Yano Y, Salahuddin S, Lawitts J, Imboden H, Chinami M, Shirakawa T, Turner H, **Khoury S**, Sayegh MH, Scadden D, Adra C. The cell cycle associated protein, HTm4, is expressed in differentiating cells from hematopoietic and central nervous system in mice. *J Mol Histol.* 2005 Feb;36(1-2):77-87. doi: 10.1007/s10735-004-3913-8

59. Benou C, Wang Y, Imitola J, VanVlerken L, Chandras C, Karalis KP, **Khoury SJ**. Corticotropin-releasing hormone contributes to the peripheral inflammatory response in experimental autoimmune encephalomyelitis. *J Immunol*. 2005 May 1;174(9):5407-5413.
60. Prat A, Biernacki K, Saroli T, Orav JE, Guttman CR, Weiner HL, **Khoury SJ**, Antel JP. Kinin B1 receptor expression on multiple sclerosis mononuclear cells: correlation with magnetic resonance imaging T2-weighted lesion volume and clinical disability. *Arch Neurol*. 2005 May;62(5):795-800. doi: 10.1001/archneur.62.5.795
61. Camelo S, Iglesias AH, Hwang D, Due B, Ryu H, Smith K, Gray SG, Imitola J, Duran G, Assaf B, Langley B, **Khoury SJ**, Stephanopoulos G, De Girolami U, Ratan, RR, Ferrante, RJ, Dangond F. Transcriptional therapy with the histone deacetylase inhibitor trichostatin A ameliorates experimental autoimmune encephalomyelitis. *J Neuroimmunol*. 2005 Jul;164(1-2):10-21. doi: 10.1016/j.jneuroim.2005.02.022
62. Guleria I, Khosroshahi A, Ansari MJ, Habicht A, Azuma M, Yagita H, Noelle RJ, Coyle A, Mellor AL, **Khoury SJ**, Sayegh MH. A critical role for the programmed death ligand 1 in fetomaternal tolerance. *J Exp Med*. 2005 Jul 18;202(2):231-237. PMC2213002. doi: 10.1084/jem.20050019
63. Zhu C, Anderson AC, Schubart A, Xiong H, Imitola J, **Khoury SJ**, Zheng XX, Strom TB, Kuchroo VK. The Tim-3 ligand galectin-9 negatively regulates T helper type 1 immunity. *Nat Immunol*. 2005 Dec;6(12):1245-1252. doi: 10.1038/ni1271
64. Zhu B, Guleria I, Khosroshahi A, Chitnis T, Imitola J, Azuma M, Yagita H, Sayegh MH, **Khoury SJ**. Differential role of programmed death-1 ligand and programmed death-2 ligand in regulating the susceptibility and chronic progression of experimental autoimmune encephalomyelitis. *J Immunol*. 2006 Mar 15;176(6):3480-3489.
65. Laplaud DA, Berthelot L, Miqueu P, Bourcier K, Moynard J, Oudinet Y, Guillet M, Ruiz C, Oden N, Brouard S, Guttman CR, Weiner HL, **Khoury SJ**, Soullillou JP. Serial blood T cell repertoire alterations in multiple sclerosis patients; correlation with clinical and MRI parameters. *J Neuroimmunol*. 2006 Aug;177(1-2):151-160. doi: 10.1016/j.jneuroim.2006.05.006
66. Karni A, Abraham M, Monsonego A, Cai G, Freeman GJ, Hafler D, **Khoury SJ**, Weiner HL. Innate immunity in multiple sclerosis: myeloid dendritic cells in secondary progressive multiple sclerosis are activated and drive a proinflammatory immune response. *J Immunol*. 2006 Sep 15;177(6):4196-4202.
67. Kaneko S, Wang J, Kaneko M, Yiu G, Hurrell JM, Chitnis T, **Khoury SJ**, He Z. Protecting axonal degeneration by increasing nicotinamide adenine dinucleotide levels in experimental autoimmune encephalomyelitis models. *J Neurosci*. 2006 Sep 20;26(38):9794-9804. doi: 10.1523/JNEUROSCI.2116-06.2006
68. Chitnis T, Imitola J, Wang Y, Elyaman W, Chawla P, Sharuk M, Raddassi K, Bronson RT, **Khoury SJ**. Elevated neuronal expression of CD200 protects Wld^s mice from inflammation-mediated neurodegeneration. *Am J Pathol*. 2007 May;170(5):1695-1712. PMC1854964. doi: 10.2353/ajpath.2007.060677
69. Gauthier SA, Mandel M, Guttman CR, Glanz BI, **Khoury SJ**, Betensky RA, Weiner HL. Predicting short-term disability in multiple sclerosis. *Neurology*. 2007 Jun 12;68(24):2059-2065. doi: 10.1212/01.wnl.0000264890.97479.b1
70. Xiao S, Najafian N, Reddy J, Albin M, Zhu C, Jensen E, Imitola J, Korn T, Anderson AC, Zhang Z, Gutierrez C, Moll T, Sobel RA, Umetsu DT, Yagita H, Akiba H, Strom T, Sayegh MH, DeKruyff RH, **Khoury SJ**, Kuchroo VK. Differential engagement of Tim-1 during activation can positively or negatively costimulate T cell expansion and effector function. *J Exp Med*. 2007 Jul 9;204(7):1691-1702. PMC2118637. doi: 10.1084/jem.20062498
71. Glanz B, Holland C, Gauthier S, Amunwa E, Liptak Z, Houtchens M, Sperling R, **Khoury S**, Guttman C, Weiner H. Cognitive dysfunction in patients with clinically isolated syndromes or newly diagnosed multiple sclerosis. *Mult Scler*. 2007 Sep;13(8):1004-1010. doi: 10.1177/1352458507077943

72. Zhu B, Bando Y, Xiao S, Yang K, Anderson AC, Kuchroo VK, **Khoury SJ**. CD11b+Ly-6Chi Suppressive Monocytes in Experimental Autoimmune Encephalomyelitis. *J Immunol*. 2007 Oct 15;179(8):5228-5237.
73. Rasmussen S, Wang Y, Kivisäkk P, Bronson RT, Meyer M, Imitola J, **Khoury SJ**. Persistent activation of microglia is associated with neuronal dysfunction of callosal projecting pathways and multiple sclerosis-like lesions in relapsing remitting experimental autoimmune encephalomyelitis. *Brain*. 2007 Nov;130(Pt 11):2816-2829. doi: 10.1093/brain/awm219
74. Elyaman W, Bradshaw EM, Wang Y, Oukka M, Kivisäkk P, Chiba S, Yagita H, **Khoury SJ**. JAGGED1 and delta1 differentially regulate the outcome of experimental autoimmune encephalomyelitis. *J Immunol*. 2007 Nov 1;179(9):5990-5998.
75. De Jager PL, Rossin E, Pyne S, Tamayo P, Ottoboni L, Viglietta V, Weiner M, Soler D, Izmailova E, Faron-Yowe L, O'Brien C, Freeman S, Granados S, Parker A, Roubenoff R, Mesirov JP, **Khoury SJ**, Hafler DA, Weiner HL. Cytometric profiling in multiple sclerosis uncovers patient population structure and a reduction of CD8low cells. *Brain*. 2008 Jul;131(Pt 7):1701-1711. PMC2730047. doi: 10.1093/brain/awn118
76. Elyaman W, Kivisäkk P, Reddy J, Chitnis T, Raddassi K, Imitola J, Bradshaw E, Kuchroo VK, Yagita H, Sayegh MH, **Khoury SJ**. Distinct functions of autoreactive memory and effector CD4+ T cells in experimental autoimmune encephalomyelitis. *Am J Pathol*. 2008 Aug;173(2):411-422. PMC2475778. doi: 10.2353/ajpath.2008.080142
77. Liptak Z, Berger AM, Sampat MP, Charil A, Felsovalyi O, Healy BC, Hildenbrand P, **Khoury SJ**, Weiner HL, Bakshi R, Guttmann CR. Medulla oblongata volume: a biomarker of spinal cord damage and disability in multiple sclerosis. *AJNR Am J Neuroradiol*. 2008 Sep;29(8):1465-1470. doi: 10.3174/ajnr.A1162
78. Viglietta V, Bourcier K, Buckle GJ, Healy B, Weiner HL, Hafler DA, Egorova S, Guttmann CR, Rusche JR, **Khoury SJ**. CTLA4Ig treatment in patients with multiple sclerosis: an open-label, phase 1 clinical trial. *Neurology*. 2008 Sep 16;71(12):917-24. doi: 10.1212/01.wnl.0000325915.00112.61
79. Pluchino S, Muzio L, Imitola J, Deleidi M, Alfaro-Cervello C, Salani G, Porcheri C, Brambilla E, Cavasinni F, Bergamaschi A, Garcia-Verdugo JM, Comi G, **Khoury SJ**, Martino G. Persistent inflammation alters the function of the endogenous brain stem cell compartment. *Brain*. 2008 Oct;131(Pt 10):2564-2578. PMC2570715. doi: 10.1093/brain/awn198
80. Wang Y, Imitola J, Rasmussen S, O'Connor KC, **Khoury SJ**. Paradoxical dysregulation of the neural stem cell pathway sonic hedgehog-Gli1 in autoimmune encephalomyelitis and multiple sclerosis. *Ann Neurol*. 2008 Oct;64(4):417-427. PMC2762219. doi: 10.1002/ana.21457
81. Bakshi R, Neema M, Healy BC, Liptak Z, Betensky RA, Buckle GJ, Gauthier SA, Stankiewicz J, Meier D, Egorova S, Arora A, Guss ZD, Glanz B, **Khoury SJ**, Guttmann CR, Weiner HL. Predicting clinical progression in multiple sclerosis with the magnetic resonance disease severity scale. *Arch Neurol*. 2008 Nov;65(11):1449-1453. PMC2762216. doi: 10.1001/archneur.65.11.1449
82. Dardalhon V, Awasthi A, Kwon H, Galileos G, Gao W, Sobel RA, Mitsdoerffer M, Strom TB, Elyaman W, Ho IC, **Khoury S**, Oukka M, Kuchroo VK. IL-4 inhibits TGF-beta-induced Foxp3+ T cells and, together with TGF-beta, generates IL-9+ IL-10+ Foxp3(-) effector T cells. *Nat Immunol*. 2008 Dec;9(12):1347-134755. PMC2999006. doi: 10.1038/ni.1677
83. Quintana FJ, Farez MF, Viglietta V, Iglesias AH, Merbl Y, Izquierdo G, Lucas M, Basso AS, **Khoury SJ**, Lucchinetti CF, Cohen IR, Weiner HL. Antigen microarrays identify unique serum autoantibody signatures in clinical and pathologic subtypes of multiple sclerosis. *Proc Natl Acad Sci USA*. 2008 Dec 2;105(48):18889-18894. PMC2596207. doi: 10.1073/pnas.0806310105
84. Neema M, Arora A, Healy BC, Guss ZD, Brass SD, Duan Y, Buckle GJ, Glanz B, Stazzone L, **Khoury SJ**, Weiner HL, Guttmann CRG, Bakshi R. Deep gray matter involvement on brain MRI scans is

- associated with clinical progression in multiple sclerosis. *J Neuroimaging*. 2009 Jan;19(1):3-8. PMC2762230. doi: 10.1111/j.1552-6569.2008.00296.x
85. Wang Y, Evans JT, Rodriguez F, Fields P, Mueller C, Chitnis T, **Khoury SJ**, Bynoe MS. A tale of two STAT6 knock out mice in the induction of experimental autoimmune encephalomyelitis. *J Neuroimmunol* 2009 Jan 3;206 (1-2):76-85. PMC2762219. doi: 10.1016/j.jneuroim.2008.11.003
 86. Ali EN, Healy BC, Stazzone LA, Brown BA, Weiner HL, **Khoury SJ**. Daclizumab in Treatment of Multiple Sclerosis Patients. *Multiple Sclerosis*. 2009 Feb;15(2):272-274. doi: 10.1177/1352458508097468
 87. Gauthier SA, Berger AM, Liptak Z, Duan Y, Egorova S, Buckle G, Glanz BI, **Khoury SJ**, Bakshi R, Weiner HL, Guttmann CRG. Rate of Brain Atrophy in Benign vs Early Multiple Sclerosis. *Arch Neurol*. 2009 Feb;66(2):234-237. doi: 10.1001/archneurol.2008.567
 88. Kivisäkk P, Imitola J, Rasmussen S, Elyaman W, Zhu B, Ransohoff RM, **Khoury SJ**. Localizing central nervous system immune surveillance: Meningeal antigen-presenting cells activate T cells during experimental autoimmune encephalomyelitis. *Ann Neurol*. 2009 Apr;65(4):457-469. doi: 10.1002/ana.21379
 89. Kivisäkk P, Healy BC, Viglietta V, Quintana FJ, Hootstein MA, Weiner HL, **Khoury SJ**. Natalizumab treatment is associated with peripheral sequestration of proinflammatory T cells. *Neurology*. 2009 Jun 2;72(22):1922-30. PMC2690969. doi: 10.1212/WNL.0b013e3181a8266f
 90. Healy BC, Ali E, Guttmann CRG, Chitnis T, Glanz B, Buckle G, Houtchens M, Stazzone L, Moodie J, Berger AM, Duan Y, Bakshi R, **Khoury SJ**, Weiner HL, Ascherio A. Smoking and disease progression in multiple sclerosis. *Arch Neurol*. 2009 Jul;66(7):858-64. PMC2754172. doi: 10.1001/archneurol.2009.122
 91. Elyaman W, Bradshaw EM, Uyttenhove C, Dardalhon V, Awasthi A, Imitola J, Bettelli E, Oukka M, van Snick J, Renauld JC, Kuchroo VK, **Khoury SJ**. IL-9 induces differentiation of TH17 cells and enhances function of FoxP3+ natural regulatory T cells. *Proc Natl Acad Sci USA*. 2009 Aug 4;106(31):12885-12890. PMC2722314. doi: 10.1073/pnas.0812530106
 92. Gauthier SA, Glanz BI, Mandel M, Tsagkaropoulos A, Neema M, Stankiewicz J, Arora A, Duan Y, Liptak Z, Egorova S, Buckle GJ, Bakshi R, Guttmann CR, **Khoury SJ**, Weiner HL. Incidence and factors associated with treatment failure in the CLIMB multiple sclerosis cohort study. *J Neurol Sci*. 2009 Sep 15;284(1-2):116-119. doi: 10.1016/j.jns.2009.04.020
 93. McLaughlin KA, Chitnis T, Newcombe J, Franz B, Kennedy J, McArdel S, Kuhle J, Kappos L, Rostasy K, Pohl D, Gagne D, Ness JM, Tenenbaum S, O'Connor KC, Viglietta V, Wong SJ, Tavakoli NP, de Seze J, Idrisova Z, **Khoury SJ**, Bar-Or A, Hafler DA, Banwell B, Wucherpfennig KW: Age-dependent B cell autoimmunity to a myelin surface antigen in pediatric multiple sclerosis, *J Immunol* 2009 Sep 15; 183 (6):4067-4076. PMC2795331. doi: 10.4049/jimmunol.0801888
 94. Ilan Y, Zigmond E, Lalazar G, Dembinsky A, Ben Ya'acov A, Hemed N, Kasis I, Axelrod E, Zolotarov L, Klein A, El Haj M, Gandhi R, Baecher-Allan C, Wu H, Murugaiyan G, Kivisäkk P, Farez MF, Quintana FJ, **Khoury SJ**, Weiner HL. Oral Administration of OKT3 Monoclonal Antibody to Human Subjects Induces a Dose-Dependent Immunologic Effect in T Cells and Dendritic Cells. *J Clin Immunol*. 2010 Jan;30(1):167-77. PMC2822125. doi: 10.1007/s10875-009-9323-7
 95. Stankiewicz JM, Glanz BI, Healy BC, Arora A, Neema M, Benedict RH, Guss ZD, Tauhid S, Buckle GJ, Houtchens MK, **Khoury SJ**, Weiner HL, Guttmann CR, Bakshi R. Brain MRI Lesion Load at 1.5T and 3T versus Clinical Status in Multiple Sclerosis. *J Neuroimaging*. 2009 Nov 3; 21(2):e50-6. NIHMSID251387. doi: 10.1111/j.1552-6569.2009.00449.x
 96. Liu Y, Bando Y, Vargas-Lowy D, Elyaman W, **Khoury SJ**, Huang T, Reif K, Chitnis T. CD200R1 Agonist Attenuates Mechanisms of Chronic Disease in a Murine Model of Multiple Sclerosis. *J Neurosci*. 2010 Feb 30 (6): 2025-2038. PMC2837938. doi: 10.1523/JNEUROSCI.4272-09.2010

97. Beriou G, Bradshaw EM, Lozano E, Costantino CM, Hastings WD, Orban T, Elyaman W, **Khoury SJ**, Kuchroo VK, Baecher-Allan C, Hafler DA. TGF- β Induces IL-9 Production from Human Th17 Cells. *J Immunol*. 2010 Jul 1;185(1):46-54. PMC2936106. doi: 10.4049/jimmunol.1000356
98. Healy BC, Liguori M, Tran D, Chitnis T, Glanz B, Wolfish C, Gauthier S, Buckle G, Houtchens M, Stazzone L, **Khoury S**, Hartzmann R, Fernandez-Vina M, Hafler DA, Weiner HL, Guttman CR, De Jager PL. HLA B*44: protective effects in MS susceptibility and MRI outcome measures. *Neurology*. 2010 Aug 17;75(7):634-40. PMC2931768. doi: 10.1212/WNL.0b013e3181ed9c9c
99. **Khoury SJ**, Healy BC, Kivisäkk P, Viglietta V, Egorova S, Guttman CR, Wedgwood JF, Hafler DA, Weiner HL, Buckle G, Cook S, Reddy S. A randomized controlled double-masked trial of albuterol add-on therapy in patients with multiple sclerosis. *Arch Neurol*. 2010 Sep;67(9):1055-61. PMC2954052. doi: 10.1001/archneurol.2010.222
100. **Khoury SJ**, Bakshi R. Cerebral Pseudoatrophy or Real Atrophy After Therapy in Multiple Sclerosis. *Ann Neurol*. 2010;68(6):778-79. doi: 10.1002/ana.22254
101. Xia Z, Chibnik LB, Glanz BI, Liguori M, Shuman JM, Tran D, **Khoury SJ**, Chitnis T, Holyoak T, Weiner WL, Guttman CRG, De Jager PL. A Putative Alzheimer's Disease Risk Allele in PCK1 Influences Brain Atrophy in Multiple Sclerosis. *PLoS ONE*. 2010 Nov 5(11): e14169. PMC2994939. doi: 10.1371/journal.pone.0014169
102. Starossom SC, Imitola J, Wang Y, Cao L, **Khoury SJ**. Subventricular Zone Microglia Transcriptional Networks. *Brain Behav Immun*. 2011 Jul; 25 (5) 991–999. PMC3109092. doi: 10.1016/j.bbi.2010.11.002
103. Imitola J, Côté D, Rasmussen S, Xie XS, Liu Y, Chitnis T, Sidman RL, Lin CP, **Khoury SJ**. Multimodal coherent anti-stokes raman scattering microscopy reveals microglia-associated myelin and axonal dysfunction in multiple sclerosis-like lesions in mice. *J Biomed Optics*. 2011 Feb; 16(2):021109/1-021109/11. doi: 10.1117/1.3533312
104. Rasmussen S, Imitola I, Ayuso-Sacido A, Wang Y, Starossom S, Kivisäkk P, Zhu B, Meyer M, Bronson RT, Manuel Garcia-Verdugo J, **Khoury SJ**. Reversible Neural Stem Cell Niche Dysfunction In A Model Of Multiple Sclerosis. *Ann Neurol*. 2011 May;69 (5):878–891. PMC3082630. doi: 10.1002/ana.22299
105. Gross R, Healy BC, Cepok S, Chitnis T, **Khoury SJ**, Hemmer B, Weiner HL, Hafler DA, De Jager PL. Population structure and HLA DRB1*1501 in the response of subjects with multiple sclerosis to first-line treatments, *J Neuroimmunol* 2011Apr; 233(1-2):168-174. doi: 10.1016/j.jneuroim.2010.10.038
106. Liguori M, Healy BC, Glanz BI, **Khoury SJ**, Moscufo N, Weiner HL, De Jager PL, Guttman CR. HLA (A-B-C and -DRB1) alleles and brain MRI changes in multiple sclerosis: a longitudinal study, *Genes Immun*. 2011 Apr; 12(3):183-190. doi: 10.1038/gene.2010.58
107. Liguori M, Meier DS, Hildenbrand P, Healy BC, Chitnis T, Baruch NF, **Khoury SJ**, Weiner HL, Bakshi R, Barkhof F, Guttman CRG. One-year activity on subtraction MRI predicts subsequent 4-year activity and progression in multiple sclerosis. *J Neurol Neurosurg Psych*. 2011; 82(10): 1125-31. doi: 10.1136/jnnp.2011.242115
108. Bassil R, Zhu B, Lahoud Y, Riella LV, Yagita H, Elyaman W, **Khoury SJ**. Notch Ligand Delta-Like 4 Blockade Alleviates Experimental Autoimmune Encephalomyelitis by Promoting Regulatory T Cell Development. *J Immunol* 2011 Sept;187(5):2322-8. PMC3311114. doi: 10.4049/jimmunol.1100725
109. Zhu B, Kennedy JK, Wang Y, Sandoval-Garcia C, Cao L, Xiao S, Wu C, Elyaman W, **Khoury SJ**. Plasticity of Ly-6Chi Myeloid Cells in T Cell Regulation. *J Immunol* 2011 Sept;187(5):2418-32. PMC3159773. doi: 10.4049/jimmunol.1100403
110. Elyaman W, **Khoury SJ**, Scott DW, De Groot AS. Potential application of Tregitopes as immunomodulating agents in multiple sclerosis. *Neurology Research International*. 2011;2011:256460.
111. Zhu B, Trikudanathan S, Zozulya AL, Sandoval-Garcia C, Kennedy JK, Atochina O, Norberg T, Castagner B, Seeberger P, Fabry Z, Harn D, **Khoury SJ**, Guleria I. Immune modulation by Lacto-N-

- fucopentaose III in experimental autoimmune encephalomyelitis. *Clin Immunol* 2012 Mar; 142(3):351–361. PMC3288504. doi: 10.1016/j.clim.2011.12.006
112. Holland CM, Charil A, Csapo I, Liptak Z, Ichise M, **Khouri SJ**, Bakshi R, Weiner HL, Guttmann CR: The Relationship between Normal Cerebral Perfusion Patterns and White Matter Lesion Distribution in 1,249 Patients with Multiple Sclerosis. *J Neuroimaging*. 2011 Apr;22(2):129-136. doi: 10.1111/j.1552-6569.2011.00585.x
 113. Stankiewicz JM, Glanz BI, Healy BC, Arora A, Neema M, Benedict RHB, Guss ZD, Tauhid S, Buckle GJ, Houtchens MK, **Khouri SJ**, Weiner HL, Guttmann CRG, Bakshi. Brain MRI lesion load at 1.5T and 3T versus clinical status in multiple sclerosis R. *J Neuroimaging*. 2011 Apr;21(2):e50-6. doi: 10.1111/j.1552-6569.2009.00449.x
 114. Moodie J, Healy BC, Buckle GJ, Gauthier SA, Glanz BI, Arora A, Ceccarelli A, Tauhid S, Han X-M, Venkataraman A, Chitnis T, **Khouri SJ**, Guttmann CRG, Weiner HL, Neema M, Bakshi R. Magnetic resonance disease severity scale (MRDSS) for patients with multiple sclerosis: A longitudinal study. *J Neurol Sci* 2012 Apr;315(1-2):49-54. PMC3319060. doi: 10.1016/j.jns.2011.11.040
 115. Elyaman W, Bassil R, Bradshaw EM, Orent W, Lahoud Y, Zhu B, Radtke F, Yagita H and **Khouri SJ**. Notch and Smad3 Signaling Cooperate in the Induction of IL- 9-Producing T Cells. *Immunity* 2012 Apr; 36(4), 623–634. PMC3572366. doi: 10.1016/j.immuni.2012.01.020
 116. Ceccarelli A, Jackson JS, Tauhid S, Arora A, Gorky J, Dell’oglio E, Bakshi A, Chitnis T, **Khouri SJ**, Weiner HL, Guttmann CR, Bakshi R, Neema M. The Impact of Lesion In-Painting and Registration Methods on Voxel-Based Morphometry in Detecting Regional Cerebral Gray Matter Atrophy in Multiple Sclerosis. *Am J Neuroradiol* 2012, Mar;33: 1579-1585. PMC3425668. doi: 10.3174/ajnr.A3083
 117. Starossom SC, Mascanfroni ID, Imitola J, Cao L, Raddassi K, Hernandez SF, Bassil R, Croci DO, Cerliani JP, Wang Y, Elyaman W, **Khouri SJ***, Rabinovich GA* (*Co-Senior). Galectin-1 Deactivates Classically-Activated Microglia and Protects from Inflammation-Induced Neurodegeneration. *Immunity*, 2012 Aug; 37(2), 1-15. PMC3428471. doi: 10.1016/j.immuni.2012.05.023
 118. Purwar R, Schlapbach C, Xiao S, Kang HS, Elyaman W, Jiang X, Jetten AM, **Khouri SJ**, Fuhlbrigge RC, Kuchroo VK, Clark RA, Kupper TS. Robust tumor immunity to melanoma mediated by interleukin-9-producing T cells. *Nature Med* 2012 Aug; 38,1248-1254. PMC3518666. doi: 10.1038/nm.2856
 119. Ottoboni L, Keenan BT, Tamayo P, Kuchroo M, Mesirov JP, Buckle GJ, **Khouri SJ**, Hafler DA, Weiner HL, De Jager PL. An RNA Profile Identifies Two Subsets of Multiple Sclerosis Patients Differing in Disease Activity. *Sci Transl Med* 2012 Sept; 4, 153ra131. PMC3753678. doi: 10.1126/scitranslmed.3004186
 120. Ghandi R, Healy B, Gholipour T, Egorova S, Musallam A, Shuja M, Nejad P, Patel B, Hei H, **Khouri S**, Quintana F, Kivisakk P, Chitnis T, Weiner HL. Circulating microRNAs as biomarkers for disease staging in multiple sclerosis. *Ann Neurol*, 2013 Mar; 73:729–740. doi: 10.1002/ana.23880
 121. Messina S, Vargas-Lowy D, Musallam A, Healy B, Kivisakk P, Ghandi R, Bove R, Gholipour T, **Khouri S**, Weiner HL, Chitnis T. Increased leptin and A-FABP levels in relapsing and progressive forms of MS. *BMC Neurol*. 2013;13(1):172. PMC3829106. doi:10.1186/1471-2377-13-172.
 122. Vargas-Lowy D, Kivisakk P, Gandhi R, Raddassi K , Soltany P, Gorman MP, **Khouri SJ**, Chitnis T. Increased Th17 response to myelin peptides in pediatric MS. *Clin Immunol* 2013 Jan; 146, 176–184. doi: 10.1016/j.clim.2012.12.008
 123. Yamout B, Alroughani R, Al-Jumah M, **Khouri S**, Abou Zeid N, Dahdaleh M, Alsharoqi I, Inshasi J, Hashem S, Zakaria M, ElKallab K, Alsaadi T, Tawfeek T, Bohlega S. Consensus guidelines for the diagnosis and treatment of multiple sclerosis. *Curr Med Res Opin* 2013 Apr 29, 1–11. doi: 10.1185/03007995.2013.787979

124. Zhu B, Buttrick T, Bassil R, Zhu C, Olah M, Wu C, Xiao S, Orent W, Elyaman W, **Khoury SJ**. Interleukin 4 and retinoic acid synergistically induce regulatory dendritic cells expressing Aldh1a2. *J Immunol* 2013 Aug; 191, 3139–3151. PMC3773474. doi: 10.4049/jimmunol.1300329
125. Bove R, Musallam A, Healy BC, Houtchens M, Glanz BI, **Khoury S**, Guttman CR, De Jager PL and Chitnis T. No sex-specific difference in disease trajectory in multiple sclerosis patients before and after age 50. *BMC Neurology* 2013, 13:73. PMC3707791. doi: 10.1186/1471-2377-13-73
126. Kivisäkk P, Healy BC, Francois K, Ghandi R, Gholipour T, Egorova S, Sevdalinova V, Quintana F, Chitnis T, Weiner HL, **Khoury SJ**. Evaluation of circulating osteopontin levels in an unselected cohort of patients with multiple sclerosis: Relevance for biomarker development. *MS Journal* 2014 Apr; 20: 438-444. doi: 10.1177/1352458513503052
127. Yamout B, El-Salem K, Gebeily S, **Khoury S**, Abu Zaid N, Hamoulila K, Al Halabi MS, Al Uqla Z, Al Hamadani H, Hamoodi H, Al Omar A, Dahdaleh M. Multiple Sclerosis in the Levant: A Regional Consensus Statement. *Int J Neurosc* 2014 Sep; 124(9): 635-641
128. Ottoboni L, Frolich IY, Lee M, Healy BC, Keenan BT, Xia Z, Chitnis T, Guttman CR, **Khoury SJ**, Weiner HL, Hafler DA, De Jager PL. Clinical relevance and functional consequences of the TNFRSF1A multiple sclerosis locus. *Neurology* 2013 Nov;81:1891–1899. PMC3843384. doi: 10.1212/01.wnl.0000436612.66328.8a
129. Kivisäkk P, Francois K, Mbianda J, Ghandi R, Weiner HL, and **Khoury SJ**. Effect of natalizumab treatment on circulating plasmacytoid dendritic cells: a cross-sectional observational study in patients with multiple sclerosis. *PLoS ONE* 2014;9(7) p. e103716. PMC4116240. doi: 10.1371/journal.pone.0103716
130. Bassil R, Orent W, Olah M, Kurdi AT, Frangieh M, Buttrick T, **Khoury SJ**, and Elyaman W. BCL6 Controls Th9 Cell Development by Repressing *Il9* Transcription. *J Immunol* 2014 Jul; 193:198-207. PMC4130220. doi: 10.4049/jimmunol.1303184
131. Mouhieddine TH, Darwisha H, Fawaz L, Yamout B, Tamim H, **Khoury SJ**. Risk factors for multiple sclerosis and associations with anti-EBV antibody titers. *Clin Immunol* 2015 May;158: 59–66. doi: 10.1016/j.clim.2015.03.011
132. Yamout B, Zeineddine M, **Khoury SJ**. Safety and Efficacy of Reduced Fingolimod Dosage Treatment. *J Neuroimmunol* 2015, 285:13-15. doi: 10.1016/j.jneuroim.2015.05.012
133. Yamout B, Alroughani R, Al-Jumah M, Goueider R, Dahdaleh M, Inshasi J, Hashem S, Alsharoqi I, Sahraian M, **Khoury S**, Alkawi Z, Koussa S, Zakaria M, Al Khaburi J, Alsaadi T, Bohlega S. Consensus recommendations for the diagnosis and treatment of multiple sclerosis: the Middle East North Africa Committee for Treatment and Research In Multiple Sclerosis (MENACTRIMS), *Current Medical Research and Opinion* 2015 Jul; 31: 1349-1361. doi: 10.1185/03007995.2015.1047750
134. Merzaban JS, Imitola J, Starossom SC, Zhu B, Wang Y, Lee J, Ali AJ, Olah M, Abuelela AF, **Khoury SJ**, Sackstein R. Cell Surface Glycan Engineering of Neural Stem Cells Augments Neurotropism and Improves Recovery in a Murine Model of Multiple Sclerosis. *Glycobiology* 2015 Dec; 25: 1392–1409. doi: 10.1093/glycob/cwv046
135. Darwish H, Zeinoun P, Ghusn P, Khoury B, Tamim H, **Khoury SJ**. Serum 25-hydroxyvitamin D predicts cognitive performance in adults. *Neuropsychiatry disease and treatment*. 2015Aug; 11:2217–2223. PMC4556248. doi: 10.2147/NDT.S87014
136. Yamout B, Zeineddine MM, Tamim H, **Khoury SJ**^[1]_{SEP} Safety and efficacy of fingolimod in clinical practice: The experience of an academic center in the Middle East. *J Neuroimmunol* 2015; 289:93–97. doi: 10.1016/j.jneuroim.2015.10.015
137. Alroughani R, Akhtar S, Ahmed SF, **Khoury SJ**, al-Hashel JY, Sahraian^[1]_{SEP}MA, Al-Jumah M, Zeineddine MM, Farhat S, Doumiati H, Yamout B. JC virus seroprevalence and seroconversion in multiple sclerosis cohort: A Middle-Eastern study. *J Neurol Sci* 2016; 360:61–65. doi: 10.1016/j.jns.2015.11.044

138. Orent W, McHenry AR, Rao DA, White C, Klein H-U, Bassil R, Srivastava G, Replogle JM, Raj T, Frangieh M, Cimpean M, Cuedon N, Chibnik L, **Khoury SJ**, Karlson EW, Brenner MB, De Jager P, Bradshaw EM, Elyaman W. Rheumatoid Arthritis-Associated RBPJ Polymorphism Alters Memory CD4+ T Cells. *Hum Mol Genet* 2016; 25(2):1-14. PMC4881829. doi: 10.1093/hmg/ddv474
139. Fawaz L, Mrad MF, Kazan JM, Sayegh S, Akika R, **Khoury SJ**. Comparative effect of 25(OH)D3 and 1,25(OH)2D3 on Th17 cell differentiation. *Clin Immunol* 2016; 166-167: 59-71. doi: 10.1016/j.clim.2016.02.011
140. Kurdi AT, Bassil R, Olah M, Wu C, Xiao S, Taga M, Frangieh M, Buttrick T, Orent W, Bradshaw EM, **Khoury SJ**, Elyaman W. Tiam1/Rac1 Complex Controls *Il17a* Transcription and Autoimmunity. *Nat Comm* 2016; 7:1-14. PMC5062600. doi: 10.1038/ncomms13048
141. Yamout BI, Abou Zeid N, Taha AJ, Zeineddine MM, **Khoury SJ**. Three cases of herpes zoster radiculitis in MS patients treated with natalizumab. *MSRD* 2016; 9: 122-124. doi: 10.1016/j.msard.2016.07.012
142. **Khoury SJ**, Rochon J, Ding L, Byron M, Ryker K, Tosta P, Gao W, Freedman MS, Arnold DL, Sayre HP, Smilek DE; On behalf of the ACCLAIM Study Group. ACCLAIM: A randomized trial of abatacept (CTLA4-Ig) for relapsing-remitting multiple sclerosis. *Mult Scler J* 2016 (5): 686-695. PMC5288398. doi: 10.1177/1352458516662727
143. Dagher MM, Atieh JA, Soubra MK, **Khoury SJ**, Tamim H, Kaafarani BR. Medical Research Volunteer Program (MRVP): innovative program promoting undergraduate research in the medical field. *BMC Med Educ.* 2016;16(1):160. PMC4893826. doi:10.1186/s12909-016-0670-9
144. Ayoubi El NK, Ghassan S, Said M, Allam J, Darwish H, **Khoury SJ**. Retinal measures correlate with cognitive and physical disability in early multiple sclerosis. *J Neurol.* August 2016:1-9. doi:10.1007/s00415-016-8271-4. doi: 10.1007/s00415-016-8271-4
145. Azcutia V, Bassil R, Herter JM, Engelbertsen D, Newton G, Autio A, Mayadas T, Lichtman AH, **Khoury SJ**, Parkos CA, Elyaman W, Luscinskas FW. Defects in CD4+ T cell LFA-1 integrin-dependent adhesion and proliferation protect CD47-/- mice from EAE. *J Leukocyte Biol* 2017; 101(2):493-505. doi: 10.1189/jlb.3A1215-546RR
146. Darwish H, Haddad R, Osman S, Ghassan S, Yamout B, Tamim H, **Khoury SJ**. Effect of Vitamin D Replacement on Cognition in Multiple Sclerosis Patients. *Sci Reports* 2017; 7:45926. PMC5379671. doi: 10.1038/srep45926
147. Mrad MF, El Ayoubi NK, Esmerian MO, Kazan JM, **Khoury SJ**. Effect of vitamin D replacement on immunological biomarkers in patients with multiple sclerosis. *Clin Immunol* 2017;181:9-15. doi: 10.1016/j.clim.2017.05.017
148. Buttrick T, **Khoury SJ**, Elyaman W. Opposite Functions of STAT3 and Smad3 in Regulating Tiam1 Expression in Th17 cells. *Small GTPases* 2017; Jun 13:1-7. doi: 10.1080/21541248.2017.1341365
149. Yamout BI, Said M, Hannoun S, Zeineddine M, Massouh J, **Khoury SJ**. Rebound syndrome after teriflunomide cessation in a patient with multiple sclerosis. *J Neurol Sci.* 2017;380:79-81. doi:10.1016/j.jns.2017.07.014
150. Bove R, Chitnis T, Cree BA, Tintoré M, Naegelin Y, Uitdehaag BMJ, Kappos L, **Khoury SJ**, Montalban X, Hauser SL, Weiner HL; on behalf of the SUMMIT Consortium (Serially Unified Multicenter Multiple Sclerosis Investigation): creating a repository of deeply phenotyped contemporary multiple sclerosis cohorts. *Multiple Sclerosis Journal.* August 2017:1352458517726657. PMC5821573. doi:10.1177/1352458517726657
151. Yamout BI, **Khoury SJ**, Ayyoub N, Doumiani H, Fakhreddine M, AhmedSF, Tamim H, Al-Hashel JY, Behbehani R, Alroughani R. Alternative diagnoses in patients referred to specialized centers for suspected MS. *Multiple Sclerosis and Related Disorders* 2017: 18:85-89. doi: 10.1016/j.msard.2017.09.016

152. Longbrake EE, Kantor D, Pawate S, Bradshaw MJ, von Geldern G, MD, Chahin S, Cross AH, Parks BJ, Rice M, **Khoury SJ**, Yamout B, Zeineddine M, Russell-Giller S, Caminero-Rodriguez A, Edwards K, Lathi E, VanderKodde D, Meador W, Berkovich R, Ge L, Bacon TE, Kister I. Effectiveness of alternative dose fingolimod for multiple sclerosis. *Neurology: Clinical Practice* 2018;8;1-6. PMC5914753. doi:10.1212
153. Buttrick T, Wang W, Yung C, Trieu K, Patel K, **Khoury SJ**, Ai X, Elyaman W. Foxo1 Promotes Th9 Cell Differentiation and Airway Allergy. *Sci Reports* 2018; 8:818. PMC5770389. doi: 10.1080/21541248.2017.1341365
154. Yamout BI, Sahraian MA, Ayoubi El N, Tamim H, Nicolas J, **Khoury SJ**, Zeineddine M. Efficacy and safety of natalizumab extended interval dosing. *Multiple Sclerosis and Related Disorders* 2018; Jul 5;24:113–6. doi: 10.1016/j.msard.2018.06.015
155. Yamout BI, El-Ayoubi NK, Nicolas J, El Kouzi Y, **Khoury SJ**, Zeineddine MM. Safety and Efficacy of Rituximab in Multiple Sclerosis: A Retrospective Observational Study. *J Immunol Res* 2018; 9084759. doi /10.1155/2018/9084759
156. Hannoun S, Baalbaki M, Haddad R, Saaybi S, El Ayoubi NK, Yamout BI, **Khoury SJ**, Hourani R. Gadolinium Effect on Thalamus and Whole Brain Tissue Segmentation in Multiple Sclerosis. *Neuroradiol* 2018; 60(11); 1167-1173. doi: 10.1007/s00234-018-2082-5
157. Regev K , Healy BC, Paul, A, Diaz-Cruz C, Mazzola MA, Raheja R, Glanz BI, Kivis'akk P, Chitnis T, Jagodic M, Piehl, F, Olsson F, Khademi M, Hauser S, Oksenberg J , **Khoury SJ**, Weiner HL , and Gandhi R. Identification of MS-specific serum miRNAs in an international multicenter study. *Neurol Neuroimmunol Neuroinflamm* 2018;5:e491. PMC6117192. doi:10.1212/NXI.0000000000000491
158. Said M, El Ayoubi N, Hannoun S, Haddad R, Saba L, Jalkh Y, Yamout BI, **Khoury SJ**. The Bayesian risk estimate at onset (BREMSO) correlates with cognitive and physical disability in patients with early multiple sclerosis. *Multiple Sclerosis and Related Disorders* 2018; 26:96-102. doi: 10.1016/j.msard.2018.09.003
159. Starossom SC, Garcia JC, Woelfle T, Romero-Suarez S, Olah M, Watanabe F, Cao L, Yeste A, Tukker JJ, Quintana FJ, Imitola J, Witzel F, Schmitz D, Morkel M, Paul F, Infante-Duarte C, **Khoury SJ**. Chi3l3 induces oligodendrogenesis in an experimental model of autoimmune neuroinflammation. *Nat Comm* 2019; 10:1-14. doi: 10.1038/s41467-018-08140-7
160. Hannoun S, Issa R, El Ayoubi NK, Haddad R, Baalbaki M, Yamout BI, **Khoury SJ**, Hourani R. Gadoterate Meglumine Administration in Multiple Sclerosis has no Effect on the Dentate Nucleus and the Globus Pallidus Signal Intensities. *Acad Radiol* 2019;26(10);e284-e289 . doi: 10.1016/j.acra.2018.11.010
161. Glatigny S, Höllbacher B, Motley SJ, Tan C, Hundhausen C, Buckner JH, Smilek D, **Khoury SJ**, Ding L, Qin T, Pardo J, Nepom GT, Turka LA, Harris KM, Campbell DJ, Bettelli E. Abatacept Targets T Follicular Helper and Regulatory T Cells, Disrupting Molecular Pathways That Regulate Their Proliferation and Maintenance. *J Immunol* 2019; 202 (5);ji1801425. doi: 10.4049/jimmunol.1801425
162. Alroughani R, Akhtar S, Zeineddine M, EL Kouzi Y, El Ayoubi NK, Ahmed SF, Behbehani R, **Khoury SJ**, Al-Hashel JY, Yamout BI. Risk of relapses during pregnancy among multiple sclerosis patients. *Multiple Sclerosis and Related Disorders* 2019; 34:9-13. doi: 10.1016/j.msard.2019.06.007
163. Tamim H., Zeineddine S., Tabbara F., **Khoury S.**, Akiki Z., Khansa S., Taher A. Structure and evaluation of a residency research program in a university hospital. *BMC Medical Education* (2019) 19:406-9. doi: 10.1186/s12909-019-1858-6
164. Chehabeddine L., Al Saleh T., Baalbaki M., Saleh E., **Khoury SJ.**, Hannoun S. Cumulative administrations of gadolinium-based contrast agents: risks of accumulation and toxicity of linear vs macrocyclic agents. *Critical Reviews in Toxicology*, 2019; 49 (3): 262-269. doi: 10.1080/10408444.2019.1592109

165. Yamout B., Sahraian M., Bohlega S., Al-Jumah M., Goueider R., Dahdaleh M., Inshasi J., Hashem S., Alsharoqi I., **Khoury S.**, Alkhawajah M., Koussa S., Al Khaburi J., Almahdawi A., Alsaadi T., Slassi E., Daodi S., Zakaria M., Alroughani R. Consensus recommendations for the diagnosis and treatment of multiple sclerosis: 2019 revisions to the MENACTRIMS guidelines. *Multiple Sclerosis and Related Disorders* 2020; 37:9-13. Doi: 10.1016/j.msard.2019.06.007
166. Hala Darwish H., Farran N., Hannoun S., Tadros N., Yamout B., El Ayoubi NK., **Khoury SJ.** Serum vitamin D level is associated with speed of processing in multiple sclerosis patients. *J Steroid Biochem Mol Biol*, 2020; 200:105628. doi: 10.1016/j.jsbmb.2020.105628
167. El Najjar M., Eid C., **Khoury SJ.**, Hannoun S. “No evidence of disease activity”: Is it an aspirational therapeutic goal in multiple sclerosis? *Multiple Sclerosis and Related Disorders*, 2020; 40:101935. doi:10.1016/j.msard.2020.101935
168. Iacobaeus E, Arrambide G, Amato MP, et al. Aggressive multiple sclerosis (1): Towards a definition of the phenotype. *Mult Scler* 2020;25:135245852092536–14. doi: 10.71177/1352458520925369
169. Arrambide G, Iacobaeus E, Amato MP, et al. Aggressive multiple sclerosis (2): Treatment. *Mult Scler* 2020;25(3):135245852092459–19. doi: 10.1177/1352458520924595
170. Zeinoun P, Farran N, **Khoury SJ**, Darwish H. Development, psychometric properties, and pilot norms of the first Arabic indigenous memory test: The Verbal Memory Arabic Test (VMAT). *J Clin Exp Neuropsych* 2020;381(1):1-11. doi: 10.1080/13803395.2020.1773408
171. Jaafar N, Khoueiry M, **Khoury SJ**, Makki A. Rare Case of Spinal Neurosarcoidosis with Concomitant Epidural Lipomatosis. *Case Reports in Neurological Medicine* 2021;2021. doi: 10.1155/2021/5952724
172. Mrad MF, Saba ES, Nakib L, **Khoury SJ.** Exosomes from MS subjects express EBV-derived proteins and activate monocyte-derived macrophages. *Neurol Neuroimmunol Neuroinflamm* 2021; 8(4): e1004 doi: 10.1212/NXI.0000000000001004
173. El Ayoubi N, Bou Reslan S W, Baalbaki MM, Darwish H, **Khoury SJ.** Effect of fingolimod vs interferon treatment on OCT measurements and cognitive function in RRMS. *Multiple Sclerosis and Related Disorders*, 2021; 53: 103041. doi: 10.1016/j.msard.2021.103041
174. Saba ES, Karout M, Nasralla L, Kobeissy F, Darwish H, **Khoury SJ.** Long-term cognitive deficits after traumatic brain injury associated with microglia activation. *Clin Immunol* 2021; 230 (9); 108815. doi: 10.1016/j.clim.2021.108815
175. Kaplan TB, Gopal A, Block VJ, Suskind AM, Zhao C, Polgar-Turcsanyi M, Saraceno TJ, Gomez R, Santaniello A, SUMMIT Consortium, El Ayoubi N, Cree BAC, Hauser SL, Weiner H, Chitnis T, **Khoury S**, Bove R. Challenges to Longitudinal Characterization of Lower Urinary Tract Dysfunction in Multiple Sclerosis. *Multiple Sclerosis and Related Disorders* 2022; 62: 103793. doi: 10.1016/j.msard.2022.10379
176. Roos I, Malpas C, Leray E, Casey R, Horakova D, Havrdova E, Debouverie N, Patti F, De Seze J, Izquierdo G, Eichau S, Edan G, Prat A, Girard M, Ozakbas S, Grammond P, Zephir H, Ciron J, Maillart E, Moreau T, Amato MP, Labauge P, Alroughani R, Buzzard K, Skibina O, Terzi M, Laplaud DA, Berger E, Grand'Maison F, Lebrun-Frenay C, Cartechini E, Boz C, Lechner-Scott J, Clavelou P, Stankoff B, Prevost J, Kappos L, Pelletier J, Shaygannejad V, Yamout BI, **Khoury SJ**, et al. Disease reactivation after cessation of disease-modifying therapy in relapsing-remitting multiple sclerosis. *Neurol* 2022;99:e1926-e1944. doi: 10.1212/WNL.0000000000201029
177. Zhu C, Zhou Z, Roos I, Merlo D, Kalincik T, Ozakbas S, Skibina O, Kuhle J, Hodgkinson S, Boz C, Alroughani R, Lechner-Scott J, Barnett M, Izquierdo G, Prat A, Horakova D, Havrdova E, Macdonell R, Patti F, **Khoury SJ**, et al. Comparing switch to ocrelizumab, cladribine or natalizumab after fingolimod treatment cessation in multiple sclerosis. *J Neurol Neurosurg Psych* 2022; 0:1-8. doi: 10.1136/jnnp-2022-330104

178. Hitti E, Hadid D, **Khoury SJ**, Tamim H, Makki M, Karam C. Domestic Tethers: Gender Differences in Career Paths and Domestic Responsibilities of Top-Research Medical School Graduates. *PLOS ONE* 2022 17(4):e0267288 doi:10.1371/journal.pone.0267288
179. Dahham J, Rizk R, Hiligsmann M, Daccache C, **Khoury SJ**, Darwish H, Evers S, Kremer I. The Economic and societal burden of multiple sclerosis on lebanese society: a cost-of-illness and quality of life study protocol. *Expert Review of Pharmacoeconomics & Outcomes Research* 2022, 22:5, 869-876, doi: 10.1080/14737167.2022.2008242
180. El Ayoubi N, Sabbagh HM, Bou Rjeily N, Hannoun S, **Khoury SJ**. Rate Of Retinal Layer Thinning As A Biomarker For Conversion To Progressive Disease In Multiple Sclerosis. *Neurol Neuroimmunol Neuroinflamm* 2022; 9:e200030. doi:10.1212/NXI.0000000000200030
181. Sharmin S, Malpas CB, Roos I, Diouf I, Alroughani R, Ozakbas S, Izquierdo G, Eichau S, Horakova D, Havrdova EK, Patti F, Terzi M, Boz C, Yamout B, **Khoury SJ**, Onofrj M, Lugesesi A, Altintas A, Prat A, Girard M, Duquette P, José Sá M, La Spitaleri D, Sidhom Y, Gouider R, Mrabet S, Soysal A, Turkoglu R, Amato MP, Fragoso YD, Kalincik T. Early predictors of disability in paediatric multiple sclerosis: evidence from a multi-national registry. *J Neurol Neurosurg Psych* 2022;0:1–8. doi:10.1136/jnnp-2022-329713
182. El Ayoubi NK, Moussa H, Younes A, Haddad R, **Khoury SJ**. Use of retinal optical coherence tomography to differentiate suspected neuromyelitis optica spectrum disorder from multiple sclerosis: a cross-sectional study. *Multiple Sclerosis and Related Disorders* 2022; 68; 104160. doi: 10.1016/j.msard.2022.104160
183. Signori A, Lorscheider J, Vukusic S, Trojano JM, Iaffaldano P , Hillert J, Hyde R, Pellegrini F, Magyari M, Koch-Henriksen N, Soelberg Sørensen P, Spelman T, van der Walt A, Horakova D, Havrdova E, Girard M, Eichau S, Grand'Maison F, Gerlach O, Terzi M, Ozakbas S, Skibina O, Van Pesch V, Jose Sa M, Prevost J, Alroughani R, McCombe PA, Gouider R, Mrabet S, Castillo-Trivino T , Zhu C, de Gans C, Sánchez-Menoyo JL, Yamout B, **Khoury S**, Sormani MP , Kalincik T, Butzkueven H, on behalf of the Big MS Data Network. Heterogeneity on Long-Term Disability Trajectories in Secondary Progressive MS Patients – A Latent Class Analysis from Big MS Data Network. *J Neurol Neurosurg Psych* 2023;0:1–8. doi:10.1136/jnnp-2022-329987
184. Dahham J, Hiligsmann M, Kremer I, **Khoury SJ**, Darwish H, Hosseini H, Evers S, Rizk R. The societal costs of multiple sclerosis in Lebanon: a cross-sectional study. *Expert Review of Pharmacoeconomics & Outcomes Research* 2023; 23:4, 419-430. doi: 10.1080/14737167.2023.2184802
185. Daruwalla C, Shaygannejad V, Ozakbas S, Havrdova EK, Horakova D, Alroughani R, Boz C, Patti F, Onofrj M, Lugesesi A, Eichau S, Girard M, Prat A, Duquette P, Yamout B, **Khoury SJ**, Sajedi SA, et al. Early non-disabling relapses are important predictors of disability accumulation in people with relapsing-remitting multiple sclerosis. *Mult Scler* 2023 :135245852311519. doi:10.1177/13524585231151951
186. Moradi N, Sharmin S, Malpas CB, Shaygannejad V, Terzi M, Boz C, Yamout B, **Khoury SJ**, Turkoglu R, Karabudak R, Shalaby N, Soysal A, Altıntaş A, Inshasi J, Al-Harbi T, Alroughani R and Kalincik T. External validation of a clinical prediction model in multiple sclerosis. *Mult Scler* 2023, Vol. 29(2) 261–269. doi:10.1177/13524585221136036
187. Harding-Forrester S, Roos I, Nguyen AL, Malpas CB, Diouf I, Moradi N, Sharmin S, Izquierdo G, Eichau S, Patti F, Horakova D, Kubala Havrdova E, Prat A, Girard M, Duquette P, Grand'Maison F, Onofrj M, Lugesesi A, Grammond P, Ozakbas S, Amato MP, Gerlach O, Sola P, Ferraro D, Buzzard K, Skibina O, Lechner-Scott J, Alroughani R, Boz C, Van Pesch V, Cartechini E, Terzi M, Maimone D, Ramo-Tello C, Yamout B, **Khoury SJ**, La Spitaleri D, Sa MJ, Blanco Y, Granella F, Slee M, Butler E, Sidhom Y, Gouider R, Bergamaschi R, Karabudak R, Ampapa R, Sánchez-Menoyo JL, Prevost J, Castillo-Trivino T, McCombe PA, Macdonell R, Laureys G, Van Hijfte L, Oh J, Altintas A, de Gans K, Turkoglu R, van der Walt A, Butzkueven H, Vucic S, Barnett M, Cristiano E, Hodgkinson S, Iuliano G, Kappos L, Kuhle J, Shaygannejad V, Soysal A, Weinstock-Guttman B, Van Wijmeersch B, Kalincik T. Disability accrual in

- primary and secondary progressive multiple sclerosis. *J Neurol Neurosurg Psych* 2023 nnp-2022-330726. doi: 10.1136/jnnp-2022-330726
188. Kalincik T, Sharmin S, Roos I, Freedman MS, Atkins H, Burman J, Macdonell R, Grigg A, Torkildsen O, Bo L, Lehman AK, Havrdova K, Krasulova E, Treny M, Kozak T, van der Walt A, Butzkueven H, McCombe P, Skibina O, Lechner-Scott J, Willekens B, Cartechini E, Ozakbas S, Alroughani R, Kuhle J, Patti F, Duquette P, Lugaesi A, **Khoury SJ**, Turkoglu R, Hodgkinson S, John N, Maimone D, Sa MJ, van Pesch V, Gerlach O, Laureys G, Van Hinfte L, Karabudak R, Spitaleri D, Csepany T, Gouider R, Castillo-Trivino T, Taylor B, Sharrack B, Snowden JA, and the MSBase Study Group Collaborators; and the MSBase Study Group Authors. Comparative Effectiveness of Autologous Hematopoietic Stem Cell Transplant vs Fingolimod, Natalizumab, and Ocrelizumab in Highly Active Relapsing-Remitting Multiple Sclerosis. *JAMA Neurol* 2023 doi: 10.1001/jamaneurol.2023.1184
 189. Zhu C, Kalincik T, Horakova D, Zhou Z, Buzzard K, Skibina O, Alroughani R, Izquierdo G, Eichau S, Kuhle J, Patti F, Grand'Maison F, Hodgkinson S, Grammond P, Lechner-Scott J, Butler E, Prat A, Girard M, Duquette P, Macdonell RAL, Weinstock-Guttman B, Ozakbas S, Slee M, Sa MJ, Van Pesch V, Barnett M, Van Wijmeersch M, Gerlach O, Prevost J, Terzi M, Boz C, Laureys G, Van Hijfte L, Kermode AG, Garber J, Yamout B, **Khoury SJ**, Merlo D, Monif M, Jokubaitis V, van der Walt A, Butzkueven H, for the MSBase Study Group. Comparison Between Dimethyl Fumarate, Fingolimod, and Ocrelizumab After Natalizumab Cessation. *JAMA Neurol* 2023 doi:10.1001/jamaneurol.2023.1542
 190. Roos I, Hughes S, McDonnell G, Malpas CB, Sharmin S, Boz C, Alroughani R, Ozakbas S, Buzzard K, Skibina O, van der Walt A, Butzkueven H, Lechner-Scott J, Kuhle J, Terzi M, Laureys G, Van Hijfte L, John N, Grammond P, Grand'Maison F, Soysal A, Jensen AV, Rasmussen PV, Svendsen KB, Barzinji I, Nielsen HH, Sejbæk T, Prakash S, Stilund MLM, Weglewski A, Issa NM, Kant M, Sellebjerg F, Gray O, Magyari M, Kalincik T, et al. Rituximab vs Ocrelizumab in Relapsing-Remitting Multiple Sclerosis. *JAMA Neurol* 2023; doi:10.1001/jamaneurol.2023.1625
 191. Sharmin S, Roos I, Simpson-Yap S, Malpas C, Sánchez MM, Ozakbas S, Horakova D, Havrdova EK, Patti F, Alroughani R, Izquierdo G, Eichau S, Boz C, Zakaria M, Onofrij M, Lugaesi A, Weinstock-Guttman B, Prat A, Girard M, Duquette P, Terzi M, Amato MP, Karabudak R, Grand'Maison F, **Khoury SJ**, Grammond P, Lechner-Scott J, Buzzard K, Skibina O, van der Walt A, Butzkueven H, Turkoglu R, Altintas A, Maimone D, Kermode A, Shalaby N, V. Pesch V, Butler E, Sidhom Y, Gouider R, Mrabet S, Gerlach O, Soysal A, Barnett M, Kuhle J, Hughes S, Sa MJ, Hodgkinson S, Oreja-Guevara C, Ampapa R, Petersen T, Ramo-Tello C, Spitaleri D, McCombe P, Taylor B, Prevost J, Foschi M, Slee M, McGuigan C, Laureys G, V. Hijfte L, de Gans C, Solaro C, Oh J, Macdonell R, Aguera-Morales E, Bhim Singhal E, Gray O, Garber J, V. Wijmeersch B, Simu M, Triviño TC, Sanchez-Menoyo JL, Khurana D, Al-Asmi A, Al-Harbi T, Deri N, Fragoso Y, Lalive PH, Sinnige LGF, Shaw C, Shuey N, Csepany T, Sempere AP, Moore F, Decoo D, Willekens B, Gobbi C, Massey J, Hardy T, Parratt J, and Kalincik T, on behalf of the MSBase investigators. The risk of secondary progressive multiple sclerosis is geographically determined but modifiable. *Brain* 2023 Nov 2;146(11):4633-4644. doi: 10.1093/brain/awad218. PubMed PMID: 37369086; PubMed Central PMCID: PMC10629760.
 192. Chitnis T, Foley J, Ionete C, El Ayoubi NK, Saxena S, Gaitan-Walsh P, Lokhande H, Paul A, Saleh F, Weiner H, Qureshi F, Becich MJ, Rubio da Costa F, Gehman VM, Zhang F, Keshavan A, Jalaaliddini K, Ghoreyshi A, **Khoury SJ**. Clinical validation of a multi-protein, serum-based assay for disease activity assessments in multiple sclerosis. *Clin Immunol* 2023 doi:10.1016/j.clim.2023.109688
 193. Imitola J, Hollingsworth EW, Watanabe F, Olah M, Elyaman W, Starossom S, Kivissak P, **Khoury SJ**. Stat1 is an inducible transcriptional repressor of neural stem cells self-renewal program during neuroinflammation. *Front Cell Neurosci* 2023, vol 17:1156802. doi: 10.3389/fncel.2023.1156802
 194. Li Y, Saul A, Taylor B, Ponsonby AL, Simpson-Yap S, Blizzard L, Broadley S, Lechner-Scott J, Karabudak R, Patti F, Eichau S, Onofrij M, Ozakbas S, Horakova D, Kubala Havrdova E, Grand'Maison F, Alroughani R, Gerlach O, Amato MP, Altintas A, Girard M, Duquette P, Blanco Y, Ramo-Tello C, Laureys G, Kalincik T, **Khoury SJ**, Shaygannejad V, Etemadifar M, Singhal B, Mrabet S, Foschi M,

- Habek M, John N, Hughes S, McCombe P, Ampapa R, van der Walt A, Butzkueven H, de Gans K, McGuigan C, Oreja-Guevara C, Sa MJ, Petersen T, Al-Harbi T, Sempere AP, Van Wijmeersch B, Grigoriadis N, Prevost J, Gray O, Castillo-Triviño T, Macdonell R, Lugaresi A, Sajedi SA, van der Mei I. Examining the environmental risk factors of progressive-onset and relapsing-onset multiple sclerosis: recruitment challenges, potential bias, and statistical strategies. *J Neurol*. 2024, vol 271:472–485;. doi: 10.1007/s00415-023-11980-z.
195. Spelman T, Magyari M, Butzkueven H, Walt AV, Vukusic S, Trojano M, Iaffaldano P, Horáková D, Drahota J, Pellegrini F, Hyde R, Duquette P, Lechner-Scott J, Sajedi SA, Lalive P, Shaygannejad V, Ozakbas S, Eichau S, Alroughani R, Terzi M, Girard M, Kalincik T, Grand'Maison F, Skibina O, **Khoury SJ**, Yamout B, Sa MJ, Gerlach O, Blanco Y, Karabudak R, Oreja-Guevara C, Altintas A, Hughes S, McCombe P, Ampapa R, Gans K de, McGuigan C, Soysal A, Prevost J, John N, Inshasi J, Stawiarz L, Manouchehrinia A, Forsberg L, Sellebjerg F, Glaser A, Pontieri L, Joensen H, Rasmussen PV, Sejbaek T, Poulsen MB, Christensen JR, Kant M, Stilund M, Mathiesen H, Hillert J, Registry the DMR Italian MS Registry, Swedish MS Registry, MSBase Study Group, and OFSEP The Big MS Data Network: a collaboration of the Czech MS. Predictors of treatment switching in the Big Multiple Sclerosis Data Network. *Front Neurol* 2023, 14, 1274194. <https://doi.org/10.3389/fneur.2023.1274194>
 196. Signori A, Ponzano M, Kalincik T, Ozakbas S, Horakova D, Havrdova EK, Alroughani R, Patti F, Kuhle J, Izquierdo G, Eichau S, Yamout B, **Khoury SJ**, Karabudak R, Grammond P, Duquette P, Roos I, Butzkueven H, Walt A van der, Sormani MP. Emulating randomised clinical trials in relapsing-remitting multiple sclerosis with non-randomised real-world evidence: an application using data from the MSBase Registry. *J Neurol Neurosurg Psych* 2024, jnnp-2023-332603. <https://doi.org/10.1136/jnnp-2023-332603>
 197. Yamout B, Al-Jumah M, Sahraian MA, Almalik Y, Khaburi JA, Shalaby N, Aljarallah S, Bohlega S, Dahdaleh M, Almahdawi A, **Khoury SJ**, Koussa S, Slassi E, Daoudi S, Aref H, Mrabet S, Zeineddine M, Zakaria M, Inshasi J, Gouider R, Alroughani, R. Consensus Recommendations for Diagnosis and Treatment of Multiple Sclerosis: 2023 revision of The MENACTRIMS Guidelines. *Multiple Sclerosis and Related Disorders*, 2024, 105435. <https://doi.org/10.1016/j.msard.2024.105435>
 198. Sharmin S, Roos I, Malpas CB, Iaffaldano I, Simone M, Filippi M, Havrdova EK, Ozakbas S, Morra VB, Alroughani R, Zaffaroni M, Patti F, Eichau S, Salemi G, Di Sapio A, Inglese M, Portaccio E, Trojano M, Amato MP, Kalincik T, Horakova D, Prat A, Girard M, Duquette P, Boz B, Pozzilli C, Cocco E, Gallo P, Yamout B, **Khoury SJ**, Lugaresi A, et al. Disease-modifying therapies in managing disability worsening in paediatric-onset multiple sclerosis: a longitudinal analysis of a global and national registry. *The Lancet Child & Adolescent Health* 2024. In Press [https://doi.org/10.1016/s2352-4642\(24\)00047-6](https://doi.org/10.1016/s2352-4642(24)00047-6)
 199. Spelman T, Simoneau G, Hyde R, Kuhelj R, Alroughani R, Ozakbas S, Karabudak R, Yamout BI, **Khoury SJ**, Terzi M, Boz K, Horakova D, Havrdova EK, Weinstock-Guttman B, Patti F, Altintas A, Mrabet S, Gouider R, Inshasi J, Shaygannejad V, Eichau S, Ward WL, Butzkueven H, MSBase Investigators. Comparative Effectiveness of Natalizumab, Fingolimod, and Injectable Therapies in Pediatric-Onset Multiple Sclerosis: A Registry-Based Study. *Neurology* 102, 2024, e208114. <https://doi.org/10.1212/wnl.0000000000208114>
 200. Li Y, Saul A, Taylor B, Ponsonby A-L, Simpson-Yap S, Blizzard L, Broadley S, Lechner-Scott J on behalf of Ausimmune/AusLong Investigators Group. Karabudak R, Patti F, Eichau S, Onofrij M, Ozakbas S, Horakova D, Havrdova EK, Grand'Maison F, Alroughani R, Gerlach O, Amato MP, Altintas A, Girard M, Duquette P, Blanco Y, Ramo-Tello C, Laureys G, Kalincik T, **Khoury SJ**, et al. on behalf of MSBase Examining the environmental risk factors of progressive-onset and relapsing-onset multiple sclerosis: recruitment challenges, potential bias, and statistical strategies. *J. Neurol*. 271, 1 (2024), 472–485. doi:10.1007/s00415-023-11980-z
 201. Dekeyser C, Hautekeete M, Cambron M, Pesch VV, Patti F, Kuhle J, **Khoury S**, Scott JL, Gerlach O, Lugaresi A, Maimone D, Surcinelli A, Grammond P, Kalincik T, Habek M, Willekens B, Macdonell R, Lalive P, Csepany T, Butzkueven H, Boz C, Tomassini V, Foschi M, Sánchez-Menoyo JL, Altintas

- A, Mrabet S, Iuliano G, Sa MJ, Alroughani R, Karabudak R, Aguera-Morales E, Gray O, Gans K de, Walt A van der, McCombe PA, Deri N, Garber J, Al-Asmi A, Skibina O, Duquette P, Cartechini E, Spitaleri D, Gouider R, Soysal A, Hijfte LV, Slee M, Amato MP, Buzzard K, Laureys G. Routine CSF parameters as predictors of disease course in multiple sclerosis: an MSBase cohort study. *J Neurol Neurosurg Psychiatry* 2024;0:1–11. doi:10.1136/jnnp-2023-333307
202. Spelman T, Eichau S, Alroughani R, Ozakbas S, **Khoury SJ**, Patti F, Havrdova EK, Boz C, Terzi M, Kuhle J, Grammond P, Lechner-Scott J, Gray O, Amato MP, Laureys G, Shaygannejad V, Hyde R, Wang H, Bozin I, Belviso N, Quan C, Zeng F, van der Walt A, Butzkueven H, and the MSBase Study Group. Comparative effectiveness of dimethyl fumarate versus non-specific immunosuppressants: real-world evidence from MSBase. *MSJ-ECT* 2024;10(2). doi: 10.1177/20552173241247182
 203. Dahham J, Hiligsmann M, Kremer I, **Khoury SJ**, Darwish H, Hosseini H, Evers S, Hallit S, Rizk R. Health-related Quality of Life and Utilities among Lebanese Patients with Multiple Sclerosis: A Cross-Sectional Study. *MSARD* 2024 Jun; 86:105635. doi:10.1016/j.msard.2024.105635
 204. Zeineddine M, Al-Roughani R, Ahmed SF, **Khoury SJ**, El-Ayoubi N, Al-Mahdawi A, Al-Khabouri J, Al-Asmi A, Chentouf A, Inshasi J, Gouider R, Mrabet S, Shalaby N, Massouh J, Dimasi F, Boumediene F, Yamout B. Safety and Effectiveness of Disease Modifying Therapies after Switching from Natalizumab. *Mult Scler J* 2024 Jul;30(8):1026-1035, doi: 10.1177/13524585241261565
 205. Roos I, Sharmin S, Malpas C, Ozakbas S, Lechner-Scott J, Hodgkinson S, Alroughani R, Eichau Madueño S, Boz C, van der Walt A, Butzkueven H, Buzzard K, Skibina O, Foschi M, Grand'Maison F, John N, Grammond P, Terzi M, Prévost P, Barnett M, Laureys G, Van Hijfte L, Sanchez-Menoyo JL, Yolanda Blanco Y, Oh J, McCombe P, Ramo Tello C, Soysal, Alexandre Prat A, Duquette P, Yamout BI, **Khoury SJ**, van Pesch V, Macdonell R, Sá MJ, Slee M, Kuhle J, Maimone D, Spitaleri DLA, Willekens B, Al Asmi A, Tallantyre E, Robertson NP, Coles A, Brown JW and Kalincik. On behalf of the MSBase Study Group. Effectiveness of cladribine compared to fingolimod, natalizumab, ocrelizumab and alemtuzumab in relapsing-remitting multiple sclerosis. *Mult Scler J* 2024;0(0). doi:10.1177/13524585241267211
 206. Lal AP, Foong YC, Sanfilippo PG, Spelman T, Rath L, Levitz D, Fabis-Pedrini M, Foschi M, Habek M, Kalincik T, Roos I, Lechner-Scott J, John N, Soysal A, D'Amico E, Gouider R, Mrabet S, Gross-Paju K, Cárdenas-Robledo S, Moghadasi AN, Sa MJ, Gray O, Oh J, Reddel S, Ramanathan S, Al-Harbi T, Altintas A, Hardy TA, Ozakbas S, Alroughani R, Kermode AG, Surcinelli A, Laureys G, Eichau S, Prat A, Girard M, Duquette P, Hodgkinson S, Ramo-Tello C, Maimone D, McCombe P, Spitaleri D, Sanchez-Menoyo JL, Yetkin MF, Baghbanian SM, Karabudak R, Al-Asmi A, Jakob GB, **Khoury SJ**, Etemadifar M, Pesch V van, Buzzard K, Taylor B, Butzkueven H, Walt AV der. A multi-centre longitudinal study analysing multiple sclerosis disease-modifying therapy prescribing patterns during the COVID-19 pandemic. *J. Neurol.* 1–12 (2024). doi:10.1007/s00415-024-12518-7.
 207. El Ayoubi NK, Ismail A, Fahd F, Younes L, Chakra NA, **Khoury SJ**. Retinal optical coherence tomography measures in multiple sclerosis: a systematic review and meta-analysis. *Ann Clin Transl Neurol* 2024. doi: 10.1002/acn3.52165
 208. Kreiter D, Kalincik T, Hupperts R, Pattie F, Spitalerig D, Foschih M, Surcinellih A, Maimonej D, Yamout B, **Khoury SJ**, Lechner-Scott J, Ozakbas S, Gerlach O, on behalf of the MSBase Study Group. Effectiveness of disease-modifying treatment on spinal cord lesion formation in relapse-onset multiple sclerosis. An MSBase registry study. *CNS Drugs* 2024; 38: 921-930 doi:0.1007/s40263-024-01115-x
 209. De Brouwer E, Becker T, Werthen-Brabants L, Dewulf P, Iliadis D, Dekeyser C, Laureys G, Van Wijmeersch B, Popescu V, Dhaene T, Deschrijver D, Waegeman W, De Baets B, Stock M, Horakova D, Patti F, Izquierdo G, Eichau S, Girard M, Prat A, Lugaesi A, Grammond P, T Kalincik T, Alroughani R, Grand'Maison F, Skibina O, Terzi M, Lechner-Scott J, Gerlach O, **Khoury SJ**, Cartechini E, Van Pesch V, Sa MJ, Weinstock-Guttman B, Blanco Y, Ampapa R, Spitaleri D, Solaro C, Maimone D, Soysal A, Iuliano G, Gouider R, Castillo-Triviño T, Sanchez-Menoyo JL, Laureys G, van der Walt A, Oh J, Aguera-Morales E, Altintas A, Al-Asmi A, de Gans K, Fragoso Y, Csepany T, Hodgkinson S, Deri N,

- Al-Harbi T, Taylor B, Gray O, Lalive P, Rozsa C, McGuigan C, Kermode A, Sempere AP, Mihaela S, Simo M, Hardy T, Decoo D, Hughes S, Grigoriadis N, Sas A, Vella N, Moreau Y, Peeters L. Machine-learning-based prediction of disability progression in multiple sclerosis: An observational, international, multi-center study. *PLOS Digit Health* 2024; 3(7): e0000533. doi:10.1371/journal.pdig.0000533
210. Yaktine N, Fares S, **Khoury SJ**, Darwish H. Psychometric validation of the Arabic multiple sclerosis resiliency scale: Uncovering resilience factors in Lebanese MS patients for clinical and research advancements. *eNeurologicalSci* 2024; 34:100489. doi:10.1016/j.ensci.2023.100489
 211. Siriratnam P, Huda S, Van Der Walt A, Sanfilippo PG, SharminVS, Foong YC, Yeh WZ, Zhu C, **Khoury SJ**, Csepany T, Willekens B, et al. Prevalence of progression independent of relapse activity and relapse-associated worsening in patients with AQP4-IgG positive NMOSD. *Neurol* 2024;103:e209940. doi:10.1212/WNL.0000000000209940
 212. Siriratnam P, Sanfilippo P, van der Walt A, Sharmin S, Foong YC, Yeh WZ, Zhu C, **Khoury SJ**, Csepany T, Willekens B, Etemadifar M, Ozakbas S, Nytrova P, Altintas A, Al-Asmi A, Yamout B, Laureys G, Patti F, Simo M, Surcinelli A, Foschi M, McCombe PA, Alroughani R, Sánchez-Menoyo JL, Turkoglu R, Soysal A, Lechner Scott J, Kalincik T, Butzkueven H, Jokubaitis V, Huda S, Monif M. Predictors of relapse risk and treatment response in AQP4-IgG positive and seronegative NMOSD: A multicentre study. *J Neurol Neurosurg Psychiatry* 2024; 0:1–9. doi:10.1136/jnnp-2024-334090
 213. Machtoub D, Fares C, Sinan H, Al Hariri M, Nehme R, Chami J, Joukhdar R, Tcheroyan R, Adib S, **Khoury SJ**. Factors affecting fatigue progression in multiple sclerosis patients. *Scient Rep* (2024) 14:31682. doi:10.1038/s41598-024-80992-0
 214. D'hondt R, Dedja K, Aerts S, Van Wijmeersch B, Kalincik T, Reddel S, Havrdova EK, Lugaresi A, Weinstock-Guttman B, Mrabet S, Lalive P, Kermode AG, Ozakbas S, Patti F, Prat A, Tomassini V, Roos I, Alroughani R, Gerlach O, **Khoury SJ**, van Pesch V, Sa MS, Prevost J, Spitaleri D, McCombe P, Solaro C, van der Walt A, Butzkueven H, Laureys G, Sanchez-Menoyo JL, Gans K, Al-Asmi A, Deri N, Csepany T, Al-Harbi T, Carroll WM, Rozsa C, Singhal B, Hardy TA, Ramanathan S, Peeters L, Vens C, the MSBase Study Group. Explainable time-to-progression predictions in multiple sclerosis. *Computer Methods and Programs in Biomedicine* (2025). doi:10.1016/j.cmpb.2025.108624
 215. Toscano S, Spelman T, Ozakbas S, Alroughani R, Chisari CG, Fermo SL, Prat A, Girard M, Duquette P, Izquierdo G, Eichau S, Grammond P, Boz C, Kalincik T, Blanco Y, Buzzard K, Skibina O, Sa MJ, van der Walt A, Butzkueven H, Terzi M, Gerlach O, Grand'Maison F, Foschi M, Surcinelli A, Barnett M, Lugaresi A, Onofri M, Yamout B, **Khoury SJ**, et al. First-year treatment response predicts the following 5-year disease course in patients with Relapsing-Remitting Multiple Sclerosis. *Neurotherapeutics* (2025) 22(2):e00552. doi:10.1016/j.neurot.2025.e00552
 216. Müller J, Sharmin S, Lorscheider J, Horakova D, Havrdova EK, Eichau S, Patti F, Grammond P, Buzzard K, Skibina O, Prat A, Girard M, Grand'Maison F, Alroughani R, Lechner-Scott J, Spitaleri D, Barnett M, Cartechini E, Sa MJ, Gerlach O, van der Walt A, Butzkueven H, Prevost J, Castillo-Triviño T, Yamout B, **Khoury SJ**, Yaldizli O, Derfuss T, Granziera C, Kuhle J, Kappos L, Roos I, Kalincik T, for the MSBase Study Group. Treatment De-escalation in Relapsing-Remitting Multiple Sclerosis: An Observational Study. *CNS Drugs* (2025) 39:403–416. Doi: 10.1007/s40263-025-01164-w
 217. El Ayoubi NK, Ismail A, Sader G, Chakra NA, El Ahdab J, **Khoury SJ**. Retinal Optical Coherence Tomography longitudinal measures as prognostic biomarkers in multiple sclerosis: Systematic Review and Meta-analysis. *Neurol Neuroimmunol Neuroinflamm* (2025)12:e200416. doi:10.1212/NXI.0000000000200416
 218. El Ayoubi NK, Bal MN, Metri N, El Sammak S, Baalbaki M, **Khoury SJ**. Slower retinal thinning over a year predicts continuous NEDA status at follow up in Multiple Sclerosis: a longitudinal OCT study. *MSJ* (2025) 31 (7) doi: 10.1177/13524585251332499
 219. Müller JJ, Sharmin S, Lorscheider J, Ozakbas S, Karabudak R, Horakova D, Weinstock-Guttman B, Shaygannejad V, Etemadifar M, Alroughani R, Patti F, Eichau S, Prat A, Lugaresi A, Tomassini V,

- Kermode AG, Amato MP, Turkoglu R, Altintas A, Buzzard K, Soysal A, van derWalt A, Butzkueven H, Blanco Y, Gerlach O, **Khoury SJ**, Barnett M, John N, Lechner-Scott J, Foschi M, Surcinelli, van Pesch, Prevost J, Sa MJ, Maimone D, D'hooghe M, Hughes S, Hodgkinson S, McGuigan C, Cartechini E, Taylor B, Spitaleri , Slee M, McCombe P, Yamout B, Benkert P, Kuhle J, Ludwig Kappos L, Roos I, Kalincik T, PGCertBiostat and the MSBase Study Group. Standardized Definition of Progression Independent of Relapse Activity (PIRA) in Relapsing-Remitting Multiple Sclerosis. *JAMA Neurol* (2025). doi:10.1001/jamaneurol.2025.0495
220. El Ayoubi NK, Fahd F, Tamim H, Hannoun S, Bal M, El-Hallak E, **Khoury SJ**. Comparative effect of standard versus extended interval dosing of Rituximab or Ocrelizumab in multiple sclerosis. *Ann Clin Trans Neurol* (2025): 0:1-11 doi: 10.1002/acn3.70142
 221. Yamout B, Al-Roughani R, Ahmed SF, Al-Mahdawi A, **Khoury S**, El-Ayoubi N, Inshasi J, Al-Khabouri J, Al-Asmi A, Gouider R, Aljarallah S, Alkhawajah N, Al Malik Y, Abulaban A, Makkawi S, Khojah O, El-Hajj T, Massouh M, AlSalamat H, Dimassi H, Al-Hajje A, Salameh P, Boumediene F, Zeineddine M. The Comparative Effectiveness of Natalizumab and Anti-CD20 Monoclonal Antibodies in Relapsing-Remitting Multiple Sclerosis: A Real-World Propensity-Score Matched Study. *J Neurol Neurosurg Psychiatry* (2025)0:1–7. doi:10.1136/jnnp-2024-335704
 222. Jokubaitis VG, Al-Roughani, Altintas A, Eichau S, Hughes S, Willekens B, Horakova D, Havrdova EK, et al. Four years on: Pregnancy and birth outcomes reported in the MSBase pregnancy, neonatal outcomes, and Women's Health Registry (2020–2024). *MSJ* (2025). doi: 10.1177/13524585251349125
 223. Fissolo N, Schaedelin S, Villar LM, Lünemann JD, Correale J, Rejdak K, Schwab N, Vilaseca A, Held F, García-Merino A, Bittner S, Trojano M, Furlan R, Tumani H, Pérez-Miralles F, Rosenstein I, Galimberti D, Álvarez-Bravo G, Thouvenot E, Llüfriu S, **Khoury SJ**, Hoepner R, Martínez-Yélamos S, Hegen H, Drulovic J, Téllez-Lara N, Khalil M, Oechtering J, Pérez-Sempere Á, Rodríguez-Antigüedad A, Enrique-Martínez J, Strijbis E, Killestein J, Eichau S, Colombo E, Schaller-Nagengast J, Midaglia L, Sánchez-López AJ, Monreal E, Chan A, Paul F, Rovira ., Tintoré M, Lycke J, Zipp F, Hemmer B, Kuhle J, Montalban X, Comabella M, Group RS, Zettl UK, Falk S, Gutiérrez L, Gasior M, González JLV, Ferrer R, Quiroga-Varela A, Bachhuber F. Costa-Frossard L, 2025. Prognostic Factors for Multiple Sclerosis Symptoms in Radiologically Isolated Syndrome. *JAMA Neurol* (2025). 82. doi: 10.1001/jamaneurol.2025.1481
 224. Zhu C, Zhou C, Kalincik T, Roos I, Buzzard K, Skibina O, Alroughani R, Kuhle J, Girard M, Grammond P, Lechner-Scott J, Gerlach O, John N, McCombe P, Macdonell R, van Pesch V, Laureys G, Prevost J, Horakova D, Havrdova EK, Castillo-Trivino T, Ramo-Tello C, Blanco Y, Meca-Lallana JE, Lugaresi A, Tomassini V, Cartechini E, Amato MP, Spitaleri D, Patti F, Maimone D, Foschi M, Surcinelli A, D'Amico E, Yamout B, **Khoury SJ**, Sa MJ, Boz C, Ozakbas S, Weinstock-Guttman B, Merlo D, monif M, Jokubaitis VG, van der Walt A, Butzkueven H. Persistent progression independent of relapse activity in multiple sclerosis. *Brain Communications* (2025). doi: 10.1093/braincomms/fcaf306
 225. Roberts JI, Sharmin S, Horakova D, Havrdova EK, Ozakbas S, Lugaresi A, Tomassini V, Alroughani R, Buzzard K, Skibina O, Boz C, Turkoglu R, Maimone D, Yamout B, **Khoury SJ**, Spitaleri D, Lechner-Scott J, Girard M, Duquette P, Al-Asmi A, Ampapa R, Foschi M, Surcinelli A, Patti F, Van Pesch V, Ramo-Tello C, Sánchez-Menoyo JL, Altintas A, Grammond P, Cartechini E, Csepany T, Laureys G, Willekens B, Roos I, Kalincik T. Corticosteroid treatment of multiple sclerosis relapses is associated with lower disability worsening over 5 years. *J Neurol Neurosurg Psychiatry* (2025) 0:1–9. doi:10.1136/jnnp-2025-336343
 226. Pirmani A, De Brouwer E, Arany A, Oldenhof M, Passemiers A, Faes A, Kalincik T, Ozakbas S, Gouider R, Willekens B, Horakova D, Havrdova EK, Patti F, Prat A, Lugaresi A, Tomassini V, Grammond P, Cartechini E, Roos I, Boz C, Alroughani R, Amato MP, Buzzard K, Lechner-Scott J, Guimarães J, Solaro C, Gerlach O, Soysal A, Kuhle J, Sanchez-Menoyo JL, Spitaleri D, Csepany T, Van Wijmeersch B, Ampapa R, Prevost J, **Khoury SJ**, Van Pesch V, John N, Maimone D, Weinstock-Guttman B, Laureys

- G, et al. Personalized federated learning for predicting disability progression in multiple sclerosis using real-world routine clinical data. *NPJ digital medicine* (2025) doi: 10.1038/s41746-025-01788-8
227. Hannoun S, Fayad G, El Ayoubi NK, **Khoury SJ**. The effect of lesion filling on brain age estimation in multiple sclerosis. *BMC Med. Imaging* **25**, 356 (2025).doi: 10.1186/s12880-025-01897-6
 228. Foong YC, Merlo D, Gresle M, Buzzard K, Horakova D, Havrdova E, Kalincik T, Roos I, Hodgkinson S, Lechner-Scott J, Lugaresi A, Weinstock-Guttman B, Ozakbas S, Surcinelli A, M, Boz C, **Khoury SJ**, Yamout B, Laureys G, Blanco Y, Skibina O, Kuhle J, John N, Alroughani R, Prevost J, Van Pesch V, Tomassini V, Prat A, Girard M, Terzi M, Shaygannejad V, Ampapa R, Gray O, Etemadifar M, Guimarães J, McCombe PA, Gerlach O, Solaro C, Butzkueven H, Zhu C, van der Walt A, MSBase Study Group. Moderate–high efficacy disease-modifying therapies reduce relapse risk in late-onset multiple sclerosis. *J Neurol Neurosurg Psychiatry* 2025;0:1–9. doi:10.1136/jnnp-2025-336513
 229. Shipley J, Beadnall HN, Sanfilippo PG, Yeh WZ, Horakova D, Havrdova EK, Hradilek P, Kalincik T, Roos I, Prat A, Girard M, Rous Z, Pavelek Z, Gerlach OHH, Lechner-Scott J, Alroughani R, Ozakbas S, Peterka M, Buzzard K, Skibina OG, Maimone D, Foschi M, Surcinelli A, Karabudak R, Spitaleri DLA, Lugaresi A, Tomassini V, Gouider R, Mrabet S, Ferrando BR, Hodgkinson S, Štourač P, Guimarães J, John NA, Macdonell RAL, Meca-Lallana JE, Butzkueven H, van der Walt A, Jokubaitis, VG, Varchova M, Duquette P, Libertinova J, Izquierdo G, Eichau S, Patti F, Boz C, Barnett M, Ampapa R, Martinkova A, Ramo-Tello C, Grammond P, van Pesch V, D'Amico E, Weinstock-Guttman B, Shaygannejad V, Blanco Y, Kuhle J, Grand'Maison F, PrevostJ, Amato MP, Yamout B, **Khoury SJ**, etc, MSBase Investigator Group. Pregnancy-Related Disease Outcomes in Women With Moderate to Severe Multiple Sclerosis Disability. *JAMA Network Open* (2025) 8 (9), e2531581-e2531581
 230. Zibaoui RE, Ayoubi NKE, Khoury SJ. Predictors of Radiological and Clinical Disease Reactivation in Multiple Sclerosis Patients After Cessation of Natalizumab (Tysabri). *Brain and Behavior* (2025): 15 (9), e70904. doi:10.1002/brb3.70904
 231. Saba ES, Mrad MF, Nakib L, Dermesrobian V, Abboud J, **Khoury SJ**. Exosomal pHERV-W ENV as a dynamic biomarker for relapse prediction and prognosis in multiple sclerosis. *J Neuroinflamm* 2025;22(1):238. doi:10.1186/s12974-025-03567-4
 232. Lteif MR, Kalot J, El Jaouni M, **Khoury SJ**. E-health literacy levels of multiple sclerosis patients in Lebanon. *PloS one* 2025;20(10); e0335084. doi:10.1371/journal.pone.0335084
 233. Siriratnam P, Jokubaitis VG, Van Der Walt A, Sanfilippo PG, Zhu C, Etemadifar M, Ayse Altintas A, Abdullah Al-Asmi A, Laureys G, Meca-Lallana JE, Foschi M, Alroughani R, Gomez-Figueroa E, Habek M, **Khoury SJ**, Simo M, Singhal BSG, Jakob GB, Fabis-Pedrini MJ, Soysal A, Kermode AG, Shaygannejad V, Havrdova EK, Millan-Pascual J, Roos I, McCombe PA, Nytrova P, Boz C, Surcinelli A, Kalincik T, Patti F, Huda S, Butzkueven H, Monif M. Impact of Age at Onset on Relapse and Disability in AQP4-IgG NMOSD. *Neurol* 2025, In press
 234. Lizak N, Sharmin S, Horáková D, Havrdova EK, Eichau S, van der Walt A, Butzkueven H, Lechner-Scott J, Buzzard K, Skibina O, Gerlach O, Prat A, Girard M, Duquette P, Alroughani R, Patti F, Grand'Maison F, Sá MJ, Aguera-Morales E, Hodgkinson S, Grammond P, Kuhle J, Yamout B, **Khoury SJ**, Ozakbas S, Csepany T, John N, Laureys G, Terzi M, Amato MP, Boz C, Al-Asmi A, Cartechini E, Gouider R, Mrabet S, Roos I, Kalincik T. Managing reactivation of multiple sclerosis during treatment with natalizumab. *Mult Scler.* 2025 Dec 19:13524585251398682. doi: 10.1177/13524585251398682.
 235. Souissi A, Patti F, Spelman T, Chisari C, Gargouri A, John N, Kermode AG, Kalincik T, Butzkueven H, Sajedi SA, Lechner-Scott J, Roos I, Laureys G, Taylor B, Alroughani R, **Khoury SJ**, et al. Effect of late-onset on multiple sclerosis phenotype and outcome: evidence from a multi-national registry. *J Neurol.* 2026;273(2):114. doi: 10.1007/s00415-026-13632-4

- **Non-peer reviewed scientific or medical publications/materials in print or other media**

- *Proceedings of meetings or other non-peer reviewed research publications*

1. Sayegh MH, **Khoury SJ**, Hancock WW, Weiner HL, Carpenter CB. Induction of immunity and oral tolerance to alloantigen by polymorphic class II major histocompatibility complex allopeptides in the rat. *Transplant Proc* 1993;25(1 Pt 1):357-8.
2. Hohol MJ, **Khoury SJ**, Cook SL, Orav EL, Hafler DA, Weiner HL. Three-year open protocol continuation study of oral tolerization with myelin antigens in multiple sclerosis and design of a phase III pivotal trial. *Ann NY Acad Sci* 1996;778:243-50.
3. Fukaura H, Kent SC, Pietrusewicz MJ, **Khoury SJ**, Weiner HL, Hafler DA. Antigen-specific TGF-beta1 secretion with bovine myelin oral tolerization in multiple sclerosis. *Ann NY Acad Sci.* 1996;778:251-7.
4. Waaga AM, Murphy B, Chen W, **Khoury SJ**, Sayegh MH. Class II MHC allopeptide-specific T cell clones transfer delayed type hypersensitivity responses in vivo. *Transplant Proc* 1997;29(1-2):1008-9.
5. Kent SC, Fukaura H, Pietrusewicz MJ, **Khoury SJ**, Weiner HL, Hafler DA. Oral administration of myelin induces antigen-specific TGF-beta 1-secreting T cells in multiple sclerosis patients. *Ann NY Acad Sci* 1997;815:412-22.
6. Hafler DA, Kent SC, Pietrusewicz MJ, **Khoury SJ**, Weiner HL, Fukaura H. Oral administration of myelin induces antigen-specific TGF-beta 1-secreting T cells in multiple sclerosis patients. *Ann NY Acad Sci* 1997;835:120-31.
7. Chitnis T, Abdallah K.A. Issazadeh S, Chandraker A, Wells A.D., Turka L.A., Sayegh M.H., **Khoury SJ**. Transgenic expression of Bcl-xL in T cells enhances susceptibility to EAE. American Academy of Neurology 51st Annual Meeting 1999. Toronto, CA.
8. K.A. Abdallah, Chitnis T., Issazadeh S, Chandraker A, Wells A.D., Turka L.A., Sayegh M.H., **Khoury SJ**. Transgenic expression of Bcl-xL in T cells enhances susceptibility to EAE. FASEB, 1999. Washington D.C.
9. K. Kishimoto, Chitnis T, **Khoury SJ**, X.C. Li, T.B. Strom, A.D., Wells, L.A. Turka and M.H. Sayegh. Passive T cell death is not required for induction of transplantation tolerance to minor histocompatibility antigens mismatched allografts. ASTP, 2000. Chicago, IL
10. Frank M, Chitnis T, Abdallah KA, Sayegh MH, **Khoury SJ**. Tamoxifen ameliorates disease expression in experimental autoimmune encephalomyelitis (EAE). American Academy of Neurology 52nd Annual Meeting 2000. San Diego, CA.
11. Chitnis T, Najafian N, Abdallah KA, Sayegh MH, **Khoury SJ**. CD28 independent induction of experimental autoimmune encephalomyelitis. American Neurological Association Annual Meeting 2000. Boston, MA.
12. Chitnis T, Najafian N, Abdallah KA, Victor Dong, Hideo Yagita, Sayegh MH, **Khoury SJ**. CD28 independent induction of experimental autoimmune encephalomyelitis. Neuroimmunology Meeting 2001, Edinburgh, UK.
13. Chitnis T, Najafian N, Grusby, MJ, Sayegh MH, **Khoury SJ**. Effect of targeted disruption of STAT4 and STAT6 genes on the induction of experimental autoimmune encephalomyelitis. Federation of Clinical Immunologists 1st Meeting 2001, Boston, MA.
14. Chitnis T, Najafian N, Abdallah KA, Victor Dong, Hideo Yagita, Sayegh MH, **Khoury SJ**. CD28 independent induction of experimental autoimmune encephalomyelitis. American Academy of Arts and Sciences Research Council 5th Annual Retreat 2001. Cambridge, MA
15. Chitnis T, Najafian N, Grusby, MJ, Sayegh MH, **Khoury SJ**. Effect of targeted disruption of STAT4 and STAT6 genes on the induction of experimental autoimmune encephalomyelitis. American Academy of

Neurology 53rd Annual Meeting 2001, Philadelphia, PA.

16. Chitnis T, Najafian N, Abdallah KA, Sayegh MH, **Khoury SJ**. CD28 independent induction of experimental autoimmune encephalomyelitis. Mary Horrigan Connors Gender Biology Poster Session 2001, Brigham and Women's Hospital, Boston, MA.
17. Chitnis T, Benou C, Najafian N, Salama A.D., Grusby M.J., Sayegh M.H., **Khoury SJ** Blockade of T cell Costimulatory Pathways protects from Experimental Autoimmune Encephalomyelitis in Th2 deficient mice. American Academy of Neurology 54th Annual Meeting, 2002, Denver, CO.
18. Chitnis T, Imitola J., Benou C, **Khoury SJ** Apoptosis of Spinal Cord Neurons in Experimental Autoimmune Encephalomyelitis. American Academy of Neurology 54th Annual Meeting, 2002, Denver, CO.
19. Chitnis T, Imitola J, Elyaman W, Chawla P, Sharuk M, **Khoury SJ**. The role of the Wlds gene in axon-protection in experimental autoimmune encephalomyelitis. European Charcot Foundation Symposium. 2003 Lisbon, Portugal.
20. Chitnis T, Imitola J, Elyaman W, Chawla P, Sharuk M, **Khoury SJ**. The role of the Wlds gene in axon-protection in experimental autoimmune encephalomyelitis. Brigham and Women's Hospital Research Poster Council. 2003 Boston, MA.
21. Chitnis T, Salama A, Akiba H, Yagita, H, Sayegh MH, **Khoury SJ**. (2003) Blockade of the PD1 T cell costimulatory pathway exacerbates EAE in WT and CD28 deficient mice. American Academy of Neurology 55th Annual Meeting, 2003, Honolulu, HI.
22. Chitnis T, Imitola J, Chawla P, Sharuk M, **Khoury SJ**. (2003) The role of the Wlds phenotype in protection against Wallerian degeneration in EAE. American Academy of Neurology 55th Annual Meeting, 2003, Honolulu, HI.
23. Chitnis T, Imitola J, Elyaman W, Chawla P, Sharuk M, **Khoury SJ**. Role of CD200 in immune modulation and neural protection. International Society of Neuroimmunology meeting. 2004 Venice, Italy.
24. Chitnis T, Imitola J, Elyaman W, Chawla P, Sharuk M, **Khoury SJ**. Role of CD200 in immune modulation and neural protection. Federation of Clinical Immunology Societies meeting/International Immunology meeting. 2004 Montreal, Canada.
25. Chitnis T, Imitola J, Elyaman W, Chawla P, Sharuk M, **Khoury SJ**. Role of CD200 in axon protection in EAE. American Academy of Neurology 56th Annual Meeting. 2004 San Francisco, CA.
26. Elyaman W, Chitnis T, Najafian N, Sayegh MH, **Khoury SJ**, Memory cell induced disease in experimental autoimmune encephalomyelitis. American Academy of Neurology, 2005, Miami.
27. Elyaman W, Chitnis T, Imitola J, Sayegh MH, **Khoury SJ**, Memory cell induced disease in experimental autoimmune encephalomyelitis. Federation of Clinical Immunology Societies Meeting, 2005, Boston.
28. Gutierrez C, Chitnis T, Imitola J, Sayegh MH, **Khoury SJ**. IL-15 and IL-2 Fusion proteins attenuate experimental autoimmune encephalomyelitis. Federation of Clinical Immunology Societies Meeting, 2005, Boston.
29. Napoli S, **Khoury SJ**, Chitnis T. CNS demyelination and axonal loss in MS correlates with decreased expression of CD200, American Academy of Neurology Meeting, 2006, San Diego.
30. Chitnis T, Gauthier S, Buckle G, Glanz B, **Khoury SJ**, Weiner HL. Incidence of First MS Symptoms in Childhood: Experience from a Regional MS Center. American Academy of Neurology Meeting, 2007, Boston.
31. Bando Y, Zhu B, Albin M, Chitnis T, Sayegh MH, **Khoury SJ**. Expression of B7-H3 and B7-H4 in Experimental Autoimmune Encephalomyelitis, American Academy of Neurology Meeting, 2007, Boston.
32. Berger A, Healy BC, Egorova S, Duan Y, **Khoury SJ**, Weiner HL, Chitnis T, Guttman CR. MRI Findings

in Pediatric-Onset Multiple Sclerosis Patients: A Retrospective Study. American Academy of Neurology, 2008. Chicago. IL.

○ *Reviews, chapters, monographs and editorials*

1. **Khoury SJ**, Weiner HL, Hafler DA. T cell abnormalities in Multiple Sclerosis. In Stuart D. Cook (ed): *Handbook of Multiple Sclerosis*. Marcell Dekker Inc. NY. 1990;129-49.
2. **Khoury SJ**, Weiner HL. Therapy of multiple sclerosis with particular reference to recent clinical trials. *Curr Opin Neurol Neurosurg* 1991;4(2):191-6.
3. Weiner HL, Zhang ZJ, **Khoury SJ**, Miller A, al-Sabbagh A, Brod SA, Lider O, Higgins P, Sobel R, Nussenblatt PB, et al. Antigen-driven peripheral immune tolerance. Suppression of organ-specific autoimmune diseases by oral administration of autoantigens. *Ann N Y Acad Sci*. 1991 Dec 30;636:227-32. Review.
4. Freidman A, al-Sabbagh A, Santos LM, Fishman-Lobell J, Polanski M, Das MP, **Khoury SJ**, Weiner HL. Oral tolerance: a biologically relevant pathway to generate peripheral tolerance against external and self antigens. *Chem Immunol*. 1994;58:259-90. Review.
5. Weiner HL, Friedman A, Miller A, **Khoury SJ**, al-Sabbagh A, Santos L, Sayegh M, Nussenblatt RB, Trentham DE, Hafler DA. Oral tolerance: immunologic mechanisms and treatment of animal and human organ-specific autoimmune diseases by oral administration of autoantigens. *Annu Rev Immunol*. 1994;12:809-37. Review.
6. Weiner HL, Miller A, **Khoury SJ**, Zhang ZJ, al-Sabbagh A, Brod SA, Lider O, Higgins P, Sobel R, Matsui M, et al. Treatment of autoimmune diseases by oral tolerance to autoantigens. *Adv Exp Med Biol* 1995;371B:1217-23. Review.
7. Weiner HL, Hohol MJ, **Khoury SJ**, Dawson DM, Hafler DA. Therapy for multiple sclerosis. *Neurol Clin*. 1995 Feb;13(1):173-96. Review.
8. **Khoury SJ**, Weiner HL, Hafler DA. Immunological mechanisms in multiple sclerosis. In Stuart D. Cook (ed): *Handbook of Multiple Sclerosis*. 2nd edition. Marcell Dekker Inc. NY. 1996, pp 145-155.
9. Olek MJ, Hohol MJ, **Khoury SJ**, Dawson DM, Hafler DA, Weiner HL. Multiple sclerosis: therapeutic overview. In Stuart D. Cook (ed): *Handbook of Multiple Sclerosis*. 2nd edition. Marcell Dekker Inc. NY. 1996, pp 377-396.
10. Sayegh MH, **Khoury SJ**, Hancock WW, Weiner HL, Carpenter CB. Mechanisms of oral tolerance by MHC peptides. *Ann N Y Acad Sci*. 1996 Feb 13;778:338-45. Review.
11. **Khoury SJ**, Weiner HL. Multiple sclerosis: what have we learned from magnetic resonance imaging studies? *Arch Intern Med*. 1998 Mar 23;158(6):565-73. Review.
12. Lu L, **Khoury SJ**, Sayegh MH, Thompson AW. Dendritic cell tolerogenicity and prospects for dendritic cell-based therapy of allograft rejection and autoimmunity. In MT Lotze and AW Thompson (eds): *Dendritic Cells*. Academic Press. 1998,487-511
13. **Khoury S**, Sayegh MH, Turka LA. Blocking costimulatory signals to induce transplantation tolerance and prevent autoimmune disease. *Int Rev Immunol* 1999;18(3):185-99. Review.
14. Chitnis T, **Khoury SJ**. Neuroimmunology. In Bradley, Daroff, Fenichel, Marsden (eds): *Neurology in Clinical Practice*. Butterworth-Heinemann 1999, pp 805-822
15. Olek M, **Khoury SJ**. Multiple Sclerosis. In M.B. Goldman and M.C. Hatch (Eds) 1999: *Women and Health*. Academic Press 686-703
16. Chitnis T, **Khoury SJ**. (2000) Eds. Bradley W, Daroff R, Fenichel G. Neuroimmunology in *Neurology in Clinical Practice*. Butterworth-Heinemann. Chapter.
17. Makhoulouf K, **Khoury SJ**, Weiner HL: IL-12: immunological staging of Multiple Sclerosis and measure

- of response to therapy. In: L. Kappos, J. Kesselring, E. Rad, (Eds): *Multiple Sclerosis: tissue destruction and repair*, London: Martin Dunitz, 2001;219-24.
18. Najafian N, **Khoury SJ**, Sayegh MH. T cell costimulatory blockade as a novel immune intervention in autoimmune diseases. *Clin Dermatol* 2001;19(5):586-91.
 19. Karni A, Makhlof K, **Khoury SJ**, Weiner HL. Immunomodulation And Therapy In Multiple Sclerosis. In: O. Abramsky, D.A.S. Compston, A. Miller, G. Said, (Eds): *Brain Disease: Therapeutic Strategies and Repair*, UK: Martin Dunitz 2001:73-82.
 20. Makhlof K, Imitola J, **Khoury SJ**: Experimental Autoimmune Encephalomyelitis and Multiple Sclerosis: from bench to bedside. In: F. Dangond, Editor: *Disorders of myelin in the central and peripheral nervous system*, Boston: Butterworth-Heinemann 2002:139-54.
 21. Imitola J, Makhlof K, **Khoury SJ**: Role of Neural Stem and Oligodendrocyte Progenitor Cells in Demyelinating Diseases: Insights into Disease Mechanisms and Therapeutic Potential. In: F. Dangond, Editor: *Disorders of myelin in the central and peripheral nervous system*, Boston: Butterworth-Heinemann 2002:57-81.
 22. Makhlof K, Weiner HL, **Khoury SJ**. Potential of beta2-adrenoceptor agonists as add-on therapy for multiple sclerosis: focus on salbutamol (albuterol). *CNS Drugs* 2002;16(1):1-8. Review.
 23. Chitnis T, **Khoury SJ**. 20. Immunologic neuromuscular disorders. In: Shearer W, Editor: *Primer on Allergic and Immunologic Diseases*, 5th Edition. 2003 Feb;111(2 Suppl):659-68. Review.
 24. Chitnis T, **Khoury SJ**. Cytokine shifts and tolerance in experimental autoimmune encephalomyelitis. *Immunol Res* 2003;28(3):223-39. Review.
 25. Chitnis T, Khoury SJ. 20. Immunologic neuromuscular disorders. *J Allergy Clin Immunol* 2003;111(2 Suppl):S659-68. Review.
 26. Chitnis T, **Khoury SJ**. Neuroimmunology. In Bradley, Daroff, Fenichel, Marsden (eds): *Neurology in Clinical Practice*. 4th Edition. Butterworth-Heinemann. 2003: 46. Chapter.
 27. Makhlof K, **Khoury SJ**. Emerging disease-modifying therapies. In: J. Cohen, and R. Rudick, Editors: *Multiple Sclerosis Therapeutics*. Second Edition. Martin Dunitz 2003;515-522
 28. Makhlof K, **Khoury SJ**: Pregnancy and Multiple Sclerosis: therapeutic aspects. In: Noseworthy J, Editor, *Neurological Therapeutics: Principles and Practice*, London: Martin Dunitz 2003
 29. Chitnis T, **Khoury SJ**. Role of cytokines in MS and EAE. In: Herndon RM, editor. *Multiple sclerosis: immunology, pathology and physiology*. New York: Demos Medical Publishing; 2003. p. 145-58.
 30. Najafian N, **Khoury SJ**. T cell costimulatory pathways: blockade for autoimmunity. *Expert Opin Biol Ther*. 2003 Apr;3(2):227-36. Review.
 31. Yip S, Aboody KS, Burns M, Imitola J, Boockvar JA, Allport J, Park KI, Teng YD, Lachyankar M, McIntosh T, O'Rourke DM, **Khoury S**, Weissleder R, Black PM, Weiss W, Snyder EY. Neural stem cell biology may be well suited for improving brain tumor therapies. *Cancer J*. 2003 May-Jun;9(3):189-204. Review.
 32. Imitola J, Snyder EY, **Khoury SJ**. Genetic programs and responses of neural stem/progenitor cells during demyelination: potential insights into repair mechanisms in multiple sclerosis. *Physiol Genomics*. 2003 Aug 15;14(3):171-97. Review.
 33. Chitnis T, **Khoury SJ**. Role of co-stimulatory pathways in the pathogenesis of multiple sclerosis and experimental autoimmune encephalomyelitis. *J Allergy Clin Immunol*. 2003 Nov;112(5):837-49; quiz 850. Review.
 34. **Khoury SJ**, Sayegh MH. T cell costimulatory blockage: a novel therapeutic approach in autoimmune disease, allergy/asthma, and transplantation. In: Lichtenstein LM, Busse WW, Geha RS, editors. *Current*

- therapy in allergy, immunology and rheumatology. Sixth Edition. CV Mosby; 2003. p. 418-22.
35. **Khoury SJ**, Sayegh MH. The roles of the new negative T cell costimulatory pathways in regulating autoimmunity. *Immunity*. 2004 May;20(5):529-38. Review.
 36. Imitola J, Park KI, Teng YD, Nisim S, Lachyankar M, Ourednik J, Mueller FJ, Yiou R, Atala A, Sidman RL, Tuszynski M, **Khoury SJ**, Snyder EY. Stem cells: cross-talk and developmental programs. *Philos Trans R Soc Lond B Biol Sci*. 2004 May 29;359(1445):823-37. Review. PMC1693371
 37. Meier DS, Weiner HL, **Khoury SJ**, Guttman CR. Magnetic resonance imaging surrogates of multiple sclerosis pathology and their relationship to central nervous system atrophy. *J Neuroimaging*. 2004 Jul;14(3 Suppl):46S-53S. Review.
 38. Mueller FJ, McKercher SR, Imitola J, Loring JF, Yip S, **Khoury SJ**, Snyder EY. At the interface of the immune system and the nervous system: how neuroinflammation modulates the fate of neural progenitors in vivo. *Ernst Schering Res Found Workshop*. 2005;(53):83-114. Review.
 39. Chitnis T, Imitola J, **Khoury SJ**. Therapeutic strategies to prevent neurodegeneration and promote regeneration in multiple sclerosis. *Curr Drug Targets Immune Endocr Metabol Disord*. 2005 Mar;5(1):11-26. Review.
 40. Imitola J, Chitnis T, **Khoury SJ**. Cytokines in multiple sclerosis: from bench to bedside. *Pharmacol Ther*. 2005 May;106(2):163-77. Review.
 41. Imitola J, Chitnis T, **Khoury SJ**. Insights into the molecular pathogenesis of progression in multiple sclerosis: potential implications for future therapies. *Arch Neurol*. 2006 Jan;63(1):25-33. Review
 42. Chitnis T, **Khoury SJ**. Eds. Cohen J and Rudick R. T cell based Therapies in Multiple Sclerosis; in *Multiple Sclerosis Therapeutics* 3rd Edition. 2007. Chapter.
 43. Vigiotta V, **Khoury SJ**. Modulating Co-Stimulation. *Neurotherapeutics*. 2007 Oct;4(4):666-75. Review.
 44. Imitola J, **Khoury SJ**. Neural stem cells and the future treatment of neurological diseases: raising the standard. *Methods Mol Biol*. 2008;438:9-16.
 45. Chitnis T, **Khoury SJ**. Eds. Bradley W, Daroff R, Fenichel G. Neuroimmunology; in *Neurology in Clinical Practice*. Butterworth-Heinemann. 2008. Chapter.
 46. Villoslada P, Moreno B, Melero I, Pablos JL, Martino G, Uccelli A, Montalban X, Avila J, Rivest S, Acarin L, Appel S, **Khoury SJ**, McGeer P, Ferrer I, Delgado M, Obeso J, Schwartz M. Immunotherapy for neurological diseases. *Clin Immunol*. 2008 Sep;128(3):294-305. Review.
 47. **Khoury S**, Bakshi R. Cerebral pseudoatrophy or real atrophy after therapy in multiple sclerosis, *Ann Neurol* 2010; 68:778-779
 48. Chitnis T, **Khoury SJ**. Eds. Bradley W, Daroff R, Fenichel G. Neuroimmunology; in *Neurology in Clinical Practice*. Butterworth-Heinemann. 2011. Chapter.
 49. Houtchens M, Lublin FD, Miller AE, **Khoury SJ**. Eds. Bradley W, Daroff R, Fenichel G. Multiple Sclerosis and Other Inflammatory Demyelinating Diseases of the Central Nervous System; in *Neurology in Clinical Practice*. Butterworth-Heinemann. 2011. Chapter.
 50. Chitnis T, **Khoury SJ**. Eds. Cohen J and Rudick R. T cell-based Therapies in Multiple Sclerosis in *Multiple Sclerosis Therapeutics* 3rd Edition. 2011. Chapter.
 51. **SJ Khoury**. Th17 and Treg balance in systemic sclerosis. *Clin Immunol*. 2011; 139: 231–232
 52. Comabella M, **Khoury SJ**. Immunopathogenesis of multiple sclerosis. *Clin Immunol*. 2012 ;142(1):2-8
 53. Okada H, **Khoury SJ**. Type17 T-cells in Central Nervous System Autoimmunity and Tumors. *J Clin Immunol* 2012; 32:802–808

54. Stankiewicz JM, Khoury SJ. Eds. Stankiewicz JM and Weiner HL. Medication treatment in Multiple Sclerosis; in *Multiple Sclerosis: Diagnosis and Therapy*. 2012. Chapter
55. Comabella M, Khoury SJ. Eds. Stankiewicz JM and Weiner HL. Biomarkers; in *Multiple Sclerosis: Diagnosis and Therapy*. 2012. Chapter
56. Cousens LP, Najafian N, Mingozi F, Elyaman W, Mazer B, Moise L, Messitt T, Su Y, Sayegh M, High K, **Khoury SJ**, Scott DW, De Groot AS. In vitro and in vivo studies of IgG Derived Treg epitopes (Tregitopes): A Promising New Tool for Tolerance Induction and Treatment of Autoimmunity. *J Clin Immunol* 2013 Jan; 33 Suppl 1:S43-9
57. Chitnis T, **Khoury SJ**. Sex Influences in autoimmune disease. *Clin Immunol* 2013 Nov; 149: 169
58. **Khoury SJ**, and Bazarbachi A. A regional need for healthy growth. *Nat Middle East* 2013 Nov; 214
59. Chitnis T, **Khoury SJ**. Eds. Bradley W, Daroff R, Fenichel G. Neuroimmunology; in *Neurology in Clinical Practice*. Butterworth-Heinneman. 2015. Chapter
60. Bou Fakhredin R, Saade C, Kerek R, El-Jamal L, Khoury SJ, El-Merhi F. Imaging in multiple sclerosis: A new spin on lesions. *J Med Imaging Radiat Oncol*. July 2016. doi:10.1111/1754-9485.12498.
61. El Ayoubi N, **Khoury SJ**. Blood Biomarkers as Outcome Measures in Inflammatory Neurologic Diseases. *Neurotherapeutics* 2017;14(1): 135-147. doi: 10.1007/s13311-016-0486-7
62. Chitnis T, **Khoury SJ**. Eds. Bradley W, Daroff R, Fenichel G. Neuroimmunology; in *Neurology in Clinical Practice*. Butterworth-Heinneman. 2020. Chapter
63. Kivisäkk P., **Khoury SJ**. Co-signaling Molecules in Neurological Diseases. In: Azuma M., Yagita H. (eds) Co-signal Molecules in T Cell Activation. *Advances in Experimental Medicine and Biology*, vol 1189. Springer, Singapore. 2019. doi: 10.1007/978-981-32-9717-3_9
64. **Khoury SJ**. Progressive Multiple Sclerosis. *Ann Neurol* 2020;88 (3): 436-437. doi:10.1002/ana.25802
65. Amor S, Baker D, **Khoury SJ**, Schmierer K, Giovanonni G. SARS-CoV-2 and Multiple Sclerosis: Not all immune depleting DMTs are equal or bad. *Ann Neurol* 2020; 87(6): 794-797. doi:10.1002/ana.25770
66. Chaaban L, Safwan N, Hussein M, El-Sammak S, **Khoury S**, Hannoun S. Central vein sign: a putative marker for Multiple Sclerosis. *Acta Neurol Scand* 2022;145:279-287. doi:10.1111/ane.13553
67. Hannoun S, Abou-Mrad T, Naja K, **Khoury S**. Central Vein Sign and Paramagnetic Rim Sign: from Radiologically Isolated Syndrome to Multiple Sclerosis. *Eur J Neurol* 2023. doi: 10.1111/ene.15922
68. Rodin R, Chitnis T, **Khoury SJ**. Eds. Bradley W, Daroff R, Fenichel G. Neuroimmunology; in *Neurology in Clinical Practice 9th Edition*. Butterworth-Heinneman. 2024. Chapter
69. Saab G, **Khoury SJ**. Eds. Gershwin ME, Tsokos GC, Diamond B. *The Rose and MACKay textbook of autoimmune diseases 7th Edition*. 2024. Multiple Sclerosis. Chapter
70. **Khoury SJ**. Eds. Gershwin ME, Tsokos GC, Diamond B. *The Rose and MACKay textbook of autoimmune diseases 7th Edition*. 2024. Autoimmune disorders of the neurologic system. Section Editor
71. Khayat-Khoei M, Houtchens M, **Khoury SJ**. Eds. Troisi R, Rexrode K, Cozier Y, Goldman MB. *Women and Health 3rd Edition*. 2024. Multiple sclerosis. Chapter

